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Sheaf cohomology

Autor: **Andreu Cotanda Oliver**

Tutor: **Guillermo Peñafort Sanchis**

Facultat de Ciències Matemàtiques

Grau en Matemàtiques

Introduction

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PART I

Category theory

Introduction

Why categories? Historical introduction?

Chapter 1

Categories, morphisms and duality

1.1 Categories

Definition 1.1.1 (Category). A **category** C consists of:

- a collection $\mathrm{Ob}(\mathsf{C})$ of **objects**.
- for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ a collection $\mathrm{Hom}_{\mathsf{C}}(X, Y)$ of **morphisms from** X **to** Y.
- for each object $X \in \mathrm{Ob}(\mathsf{C})$ a morphisms $\mathbb{1}_X \in \mathrm{Hom}_{\mathsf{C}}(X, X)$ called the **identity** morphism.
- for each pair of morphisms $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$ a morphism $g \circ f \in \mathrm{Hom}_{\mathsf{C}}(X, Z)$ called their **composite**.

satisfying the following axioms:

- Composition is associative: for each quadruple $X, Y, Z, W \in \mathrm{Ob}(\mathsf{C})$ of objects, if $f \in \mathrm{Hom}_{\mathsf{C}}(Y, Z)$, $g \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ and $h \in \mathrm{Hom}_{\mathsf{C}}(W, X)$ then
$$(f \circ g) \circ h = f \circ (g \circ h)$$
- Composition satiesfies the unit laws: for each pair of objects $X, Y \in \mathrm{Ob}(\mathsf{C})$ if $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ then
$$\mathbb{1}_Y \circ f = f = f \circ \mathbb{1}_X$$

Notation. To indicate that X is an object of a category C we write $X \in \mathsf{C}$ instead of $X \in \mathrm{Ob}(\mathsf{C})$.

To indicate that $f \in \mathrm{Hom}_{\mathsf{C}}(X, Y)$ is a morphisms from X to Y we write $f : X \to Y$ or $X \xrightarrow{f} Y$.

Example 1.1.1.

- The category Set has the collection of all sets as objects and functions as morphisms.
- The category Grp has groups as objects and group homomorphisms as morphisms.
- The category Ab has abelian groups as objects and group homomorphisms as morphisms. It is a full subcategory of Grp.
- The category Ring has rings as objects and ring homomorphisms as mor-

phisms.[a]
- The category Rng has (not necessarilly unital) rings as objects and ring homomorphisms as morphism.
- The category Top has topological spaces as objects and continuous functions as morphisms.
- For a fixed ring R, the category $\mathsf{Mod}(R)$ has (left) R-modules as objects and R-linear maps as morphisms.
- For a fixed field k, the category Vect_k has k-vector spaces as objects and k-linear maps as morphisms.

[a] We adopt the convetion that all rings have an unit and all ring homomorphisms preserve it.

Examples of non-concrete categories?

Example 1.1.2 (Discrete category). Given a category C we can define the **discrete category** C_d considering the same collection of objects as C but only the identity morphisms between them.

Definition 1.1.2 (Subcategory). Given a category C, a **subcategory** D consists of a subcollection of objects and a subcollection of morphisms in C such that

- If $f : X \to Y$ is a morphism in D then X, Y are objects in D.
- If $f : X \to Y$ and $g : Y \to Z$ are morphisms in D, the composition $gf : X \to Z$ is in D.
- If X is an object in D then the identity map $\mathbb{1}_X$ is in D.

We say that D is a **full subcategory** if for all objects X, Y in D every morphism $f : X \to Y$ in C is also in D.

Example 1.1.3.

- Ab is a full subcategory of Grp.
- The category of finite sets FinSet is a full subcategory of Set.
- Ring is a subcategory of Rng that is not full.
- If k is a field, Vect_k is a full subcategory of $\mathsf{Mod}(k)$.

Definition 1.1.3 (Product category). Let C, D be two categories. The **product category** $\mathsf{C} \times \mathsf{D}$ is the category consisting of

- As objects: ordered pairs (C, D), where C is an object in C and D is an object in D, i.e:
$$\mathrm{Ob}_{\mathsf{C}\times\mathsf{D}} = \mathrm{Ob}_{\mathsf{C}} \times \mathrm{Ob}_{\mathsf{D}}$$
- As morphisms: ordered pairs $(C \xrightarrow{f} C', D \xrightarrow{g} D')$, where $C \xrightarrow{f} C'$ is a morphism in C and $D \xrightarrow{g} D'$ is a morphism in D, i.e: for all objects $(C, D), (C', D')$ in $\mathsf{C} \times \mathsf{D}$
$$\mathrm{Hom}_{\mathsf{C}\times\mathsf{D}}((C, D), (C', D')) = \mathrm{Hom}_{\mathsf{C}}(C, C') \times \mathrm{Hom}_{\mathsf{D}}(D, D')$$
- Composition is defined *componentwise*: if $(C \xrightarrow{f} C', D \xrightarrow{g} D')$ and $(C' \xrightarrow{f'} C'', D' \xrightarrow{g'} D'')$ are two morphisms in $\mathsf{C} \times \mathsf{D}$ them
$$(f', g') \circ (f, g) := (f' \circ f, g' \circ g) : C \times D \to C'' \times D''$$

Why do we need locally small categories? Explain size issues briefly

Definition 1.1.4 (Locally small category)**.** A category C is **locally small** if for every pair of objects $X, Y \in \mathsf{C}$ the collection $\mathrm{Hom}(X, Y)$ is a set.

Example 1.1.4. The categories Set, Grp, Ring, Top and $R - \mathsf{Mod}$ are locally small.

Lemma 1.1.1. Let C, D be two locally small categories. The product category $\mathsf{C} \times \mathsf{D}$ is locally small.

Proof. Let $C, C' \in \mathsf{C}$ and $D, D' \in \mathsf{D}$. Since C and D are locally small $\mathrm{Hom}_{\mathsf{C}}(C, C')$ and $\mathrm{Hom}_{\mathsf{D}}(D, D')$ are both sets. Therefore so is

$$\mathrm{Hom}_{\mathsf{C}}(C, C') \times \mathrm{Hom}_{\mathsf{D}}(D, D') = \mathrm{Hom}_{\mathsf{C}\times\mathsf{D}}((C, D), (C', D'))$$

□

Definition 1.1.5 (Small category)**.** A locally small category C is **small** if its collection of objects $\mathrm{Ob}(\mathsf{C})$ is a set.

Example 1.1.5 (Category of open sets)**.** Let X be a topological space.
The category $\mathsf{Op}(X)$ consisting of

- As objects: the open subsets of X.
- As morphisms: inclussion of open sets: $U \to V$ if and only if $U \subseteq V$

is small.

Lemma 1.1.2. Let C, D be two small categories. The product category $\mathsf{C} \times \mathsf{D}$ is small.

Proof. Since C and D are small, Ob_{C} and Ob_{D} are both sets. Therefore so is

$$\mathrm{Ob}_{\mathsf{C}} \times \mathrm{Ob}_{\mathsf{D}} = \mathrm{Ob}_{\mathsf{C}\times\mathsf{D}}$$

□

1.1.1 Duality

Definition 1.1.6 (Opposite category)**.** Let C be a category. The **opposite category** C^{op} consists of:

- The same objects as C: $\mathrm{Ob}(\mathsf{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathsf{C})$.
- For each pair of objects $X, Y \in \mathsf{C}^{\mathrm{op}}$: $\mathrm{Hom}_{\mathsf{C}^{\mathrm{op}}}(X, Y) = \mathrm{Hom}_{\mathsf{C}}(Y, X)$.
- If $f^{\mathrm{op}} : X \to Y$, $g^{\mathrm{op}} : Y \to Z$ are morphisms in C^{op} their composition is defined by
$$f^{\mathrm{op}} \circ g^{\mathrm{op}} = (g \circ f)^{\mathrm{op}}$$
where $f : Y \to X$ and $g : Z \to Y$ are the associated morphisms in C.

Example 1.1.6. Consider a preordered set $(P, \leq)$ viewed as a small category where a unique morphism $x \to y$ exists if and only if $x \leq y$.
The opposite category P^{op} corresponds to the same set P equipped with the reversed order $\geq$. That is, a morphism $x \to y$ exists in P^{op} if and only if $y \leq x$ in P.

Remark 1.1.1. Let $f^{\mathrm{op}} : Y \to X$ and $g^{\mathrm{op}} : Y \to X$ be two morphisms in the oppsite category C^{op}. Then $f^{\mathrm{op}} = g^{\mathrm{op}}$ in C^{op} if and only if $f = g$ in C.

Definition 1.1.7 (Dual statement)**.** A categorical statement P^* is defined as the **dual** of a statement P in a category C if P^* holds in the opposite category C^{op} if and only if P holds in C.

Remark 1.1.2. Note that by definition for any category C we have that $(\mathsf{C}^{\mathrm{op}})^{\mathrm{op}} = \mathsf{C}$. Therefore if a statement is true for all categories C, its dual is true for all categories C^{op}. Since C^{op} is an arbitrary category the dual statement is also true for all categories.
In practice, this means that proving an statement directly proves its dual for free.

1.2 Types of morphisms

Definition 1.2.1 (Monomorphism and epimorphism)**.** Let $f : X \to Y$ be a morphism in a category C. We say that f is a

- **Monomorphism** (or is **monic**) if for every pair of parallel morphisms $g, h : Z \to X$ in C, the equality $f \circ g = f \circ h$ implies $g = h$.
- **Epimorphism** (or is **epic**) if for every pair of parallel morphisms $g, h : Y \to Z$ in C, the equality $g \circ f = h \circ f$ implies $g = h$.

Example 1.2.1.

- In Set, an monomorphism (resp. epimorphism) is an injective (resp. surjective) function.
- In Grp, an monomorphism (resp. epimorphism) is an injective (resp. surjective) group homomorphism.
- In Top, an monomorphism (resp. epimorphism) is an injective (resp. surjective) continuous map.

Lemma 1.2.1 (Monomorphisms and epimorphisms are dual)**.** Let $f : X \to Y$ be a morphism in a category C. Consider the associated morphism $f^{\mathrm{op}} : Y \to X$ in the opposite category C^{op}.
Then f is a monomorphism (resp. epimorphism) in C if and only if f^{op} is an epimorphism (resp. epimorphism) in C^{op}.

Proof. Consider of pair of morphisms $g, h : Z \rightrightarrows X$ in C and their opposites $g^{\mathrm{op}}, h^{\mathrm{op}} : X \rightrightarrows Z$ in C^{op}.

Note that, by definition of composition in C^{op} we have

$$g^{\mathrm{op}} \circ f^{\mathrm{op}} = (f \circ g)^{\mathrm{op}}$$

Therefore we have that $f \circ g = f \circ h$ implies $g = h$ if and only if $(f \circ g)^{\mathrm{op}} = (f \circ h)^{\mathrm{op}}$ implies $g = h$ if and only if $g^{\mathrm{op}} \circ f^{\mathrm{op}} = h^{\mathrm{op}} \circ f^{\mathrm{op}}$ implies $g^{\mathrm{op}} = h^{\mathrm{op}}$. □

Definition 1.2.2 (Isomorphism)**.** A morphism $f : X \to Y$ in a category C is an **isomorphism** if there exists another morphism $g : Y \to X$ in C such that

$$g \circ f = \mathbb{1}_X \quad \text{and} \quad f \circ g = \mathbb{1}_Y$$

The morphism g is called the **inverse** of f and is denoted by f^{-1}.

Example 1.2.2.

- In Set, an isomorphism is a bijection.

- In Grp, an isomorphism is a bijective group homomorphism.
- In Top, an isomorphism is a homeomorphism.

Proposition 1.2.1. Let $f : X \to Y$ be an isomorphism. Then f is both a monomorphism and epimorphism.

Proof. Suppose $f : X \to Y$ is an isomorphism in a category C. Then there exists a morphism $f^{-1} : Y \to X$ in C such that

$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$

Consider a pair of morphisms $g, h : Z \rightrightarrows X$ such that $f \circ g = f \circ h$. Precomposing with f^{-1} we get

$$f^{-1} \circ (f \circ g) = f^{-1} \circ (f \circ h) \implies (f^{-1} \circ f) \circ g = (f^{-1} \circ f) \circ h \implies g = h$$

Therefore f is a monomorphism.

Analogously, if we consider pair of morphism $g', h' : Y \rightrightarrows Z$ such that $g' \circ f = h' \circ f$ this time by postcomposing with f^{-1} we get $g' = h'$, i.e: f is a epimorphism. □

Example 1.2.3. In general, a morphism $f : X \to Y$ that is both a monomorphism and epimorphism is not an isomorphism.
Indeed, in the category Top of topological spaces consider a set with more than one point X endowed with the discrete X_d and trivial X_t topology.
Consider the identity map $f : X_d \to X_t$, which is continuous. Since f is injective we have that it is a monomorphism and, since it is surjective, it is an epimorphism. However since there does not exists a continuous function $g : X_t \to X_d$, f cannot be a isomorphism.

Proposition 1.2.2. Let $f : X \to Y$ and $g : Y \to Z$ be isomorphisms in a category C. Then:

1. The inverse f^{-1} of f is unique.
2. The identity $\mathbb{1}_X$ is an isomorphism for every $X \in \mathsf{C}$.
3. $g \circ f$ is an isomorphism with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
4. If $g \circ f$ is an isomorphism and f is an isomorphism, then g is an isomorphism. Analogously if $g \circ f$ and g are isomorphisms, then f is an isomorphism.

Proof.

1. Suppose that $h_1, h_2 : Y \to X$ are inverses of f. We have
$$h_1 = h_1 \circ \mathbb{1}_Y = h_1 \circ (f \circ h_2) = (h_1 \circ f) \circ h_2 = \mathbb{1}_X \circ h_2 = h_2$$
2. For all $X \in \mathsf{C}$ by definition
$$\mathbb{1}_X \circ \mathbb{1}_X = \mathbb{1}_X$$
i.e: $\mathbb{1}_X$ is its own left and right inverse.
3. We only need to check that $f^{-1} \circ g^{-1}$ is both a left and right inverse of $g \circ f$. On the one hand
$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ f = \mathbb{1}_X$$
On the other
$$(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ g^{-1} = \mathbb{1}_Z$$

4. Let $h \coloneqq (g \circ f)^{-1} : Z \to X$. We are going to check that $f \circ h : Z \to Y$ is the inverse of g. On the one hand

$$g \circ (f \circ h) = (g \circ f) \circ h = \mathbb{1}_Z$$

On the other, since $h \circ (g \circ f) = \mathbb{1}_X$ precomposing with f and postcomposing with f^{-1} we have

$$\begin{aligned} f \circ (h \circ g \circ f) \circ f^{-1} = f \circ \mathbb{1}_X \circ f^{-1} = \mathbb{1}_Y &\implies (f \circ h) \circ g \circ (f \circ f^{-1}) = \mathbb{1}_Y \\ &\implies (f \circ h) \circ g = \mathbb{1}_Y \end{aligned}$$

□

Proposition 1.2.3. Let D be a subcategory of a category C. If $f : X \to Y$ is an isomorphism in D then f is also an isomorphism in C.

Proof. Since $f : X \to Y$ is an isomorphism in D there exists an inverse $f^{-1} : Y \to X \in \mathrm{Hom}_D(Y, X)$. Therefore since $\mathrm{Hom}_{\mathsf{D}}(Y, X) \subseteq \mathrm{Hom}_{\mathsf{C}}(Y, X)$, f^{-1} is an inverse of f in C. □

Proposition 1.2.4. Let D be a full subcategory of a category C. If $f : X \to Y$ is an isomorphism in C and $X, Y \in \mathsf{D}$ then f is also an isomorphism in D.

Proof. Since $f : X \to Y$ is an isomorphism in C there exists an inverse $f^{-1} : Y \to X \in \mathrm{Hom}_C(Y, X)$.

Since D is a full subcategory of D and $X, Y \in \mathsf{D}$ we have that $f^1 \in \mathrm{Hom}_{\mathsf{D}}(Y, X)$. Therefore f admits an inverse in D. □

1.3 Special objects

Definition 1.3.1 (Initial and terminal objects)**.** An object $I \in \mathsf{C}$ is **initial** if for every object $X \in \mathsf{C}$ there is a unique morphism $I \to X$.
Dually, an object $T \in \mathsf{C}$ is **terminal** if for every object $X \in \mathsf{C}$ there is a unique morphism $X \to T$.

Proposition 1.3.1 (Uniqueness of initial and terminal objects)**.** Let X, Y be two initial (resp. terminal) objects in a category C. Then there exists a unique isomorphism $X \xrightarrow{\sim} Y$ in C.

Proof. Since X is initial in C there exists a unique morphism $f : X \to Y$. Analogously there exist a unique morphism $g : Y \to X$.

Note that $g \circ f : X \to X$. Therefore since X is initial $g \circ f = \mathbb{1}_X$. Analogously since $f \circ g : Y \to Y$ we have $f \circ g = \mathbb{1}_Y$.

Therefore f and g are inverses of each other, i.e: $X \cong Y$. Moreover they are unique by definition. □

Definition 1.3.2 (Zero object)**.** An object $0 \in \mathsf{C}$ is a **zero** object if it is both initial and terminal.
Note that for any two objects $X, Y \in \mathsf{C}$ there exist maps $X \to 0 \to Y$. We call this composite map the **zero map** between X and Y and denote it $X \xrightarrow{0} Y$.

Corollary 1.3.1 (Uniqueness of zero objects)**.** Let $0, \widetilde{0}$ be two zero objects in a category C. Then there exists a unique isomorphism $0 \xrightarrow{\sim} \widetilde{0}$ in C.

Proof. Since 0 and $\widetilde{0}$ are in particular initial we can apply Proposition 1.3.1. □

Chapter 2

Functors and natural transformations

2.1 Functors

Definition 2.1.1 (Functor). Let C, D be two categories. A (covariant) **functor** $F : \mathsf{C} \to \mathsf{D}$ consists of:

- for each object $X \in \mathsf{C}$, an object $F(X) \in \mathsf{D}$.
- for each morphism $f : X \to Y$ in C, a morphism $F(f) : F(X) \to F(Y)$ in D,

such that:

- $F(\mathbb{1}_X) = \mathbb{1}_{F(X)}$ for every $X \in \mathsf{C}$.
- $F(g \circ f) = F(g) \circ F(f)$ for every composable pair of morphisms f, g in C.

A functor is **contravariant** if it reverses the order of composition: $F(g \circ f) = F(f) \circ F(g)$. This is equivalent to a covariant functor $F : \mathsf{C}^{\mathrm{op}} \to \mathsf{D}$.
Diagramatically, a covariant functor is represent by

$$\begin{array}{rccc} F: & \mathsf{C} & \longrightarrow & \mathsf{D} \\ & X & & FX \\ & f\downarrow & \longmapsto & \downarrow Ff \\ & Y & & FY \end{array}$$

And a contravariant functor by

$$\begin{array}{rccc} F: & \mathsf{C} & \longrightarrow & \mathsf{D} \\ & X & & FX \\ & f\downarrow & \longmapsto & \uparrow Ff \\ & Y & & FY \end{array}$$

Remark 2.1.1. We will disfavour the covariant/contravariant distintinction. Therefore all functors will be covariant (unless otherwise specified) and contravariant functors will be covariant functors from the opposite category.

Example 2.1.1 (Identity functor). Let C be a category. There exists an identity functor

$$\begin{array}{rccc} \mathbb{1}: & \mathsf{C} & \longrightarrow & \mathsf{C} \\ & X & & X \\ & f\downarrow & \longmapsto & \downarrow f \\ & Y & & Y \end{array}$$

Example 2.1.2 (Constant functor). Let C be a category. For every object $C \in \mathsf{C}$ there exists a constant functor

$$\begin{array}{rccc} \mathrm{ct}_C : & \mathsf{C} & \longrightarrow & \mathsf{C} \\ & X & & C \\ & f \downarrow & \longmapsto & \downarrow \mathbb{1}_C \\ & Y & & C \end{array}$$

Example 2.1.3. Let C be a For every object $C \in \mathsf{C}$ there exists a *covariant* Hom functor

$$\begin{array}{rccc} \mathrm{Hom}(C,-) : & \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & X & & \mathrm{Hom}(C,X) \\ & f \downarrow & \longmapsto & \downarrow f\circ \\ & Y & & \mathrm{Hom}(C,Y) \end{array}$$

and a *contravariant* Hom functor

$$\begin{array}{rccc} \mathrm{Hom}(-,C) : & \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & X & & \mathrm{Hom}(C,X) \\ & f \downarrow & \longmapsto & \uparrow \circ f \\ & Y & & \mathrm{Hom}(C,Y) \end{array}$$

We can combine this two into a *bifunctor*

$$\begin{array}{rccc} \mathrm{Hom}(-,-) : & \mathsf{C}^{\mathrm{op}} \times \mathsf{C} & \longrightarrow & \mathsf{Set} \\ & (X,Y) & & \mathrm{Hom}(X,Y) \\ & (f,g) \downarrow & \longmapsto & \downarrow g\circ - \circ f \\ & (Z,W) & & \mathrm{Hom}(Z,W) \end{array}$$

Example 2.1.4. Let D be a subcategory of a category C. There exist an inclussion functor

$$\begin{array}{rccc} \iota : & \mathsf{D} & \hookrightarrow & \mathsf{C} \\ & X & & X \\ & f \downarrow & \longmapsto & \downarrow f \\ & Y & & Y \end{array}$$

Example 2.1.5 (Preimage functor). Let X, Y be two topological spaces. Consider their categories of open sets $\mathsf{Op}(X), \mathsf{Op}(Y)$.
For every continuous map $f : X \to Y$ we have a preimage functor

$$\begin{array}{rccc} f^{-1} : & \mathsf{Op}(Y) & \longrightarrow & \mathsf{Op}(X) \\ & U & & f^{-1}(U) \\ & \downarrow & \longmapsto & \downarrow \\ & V & & f^{-1}(V) \end{array}$$

Proposition 2.1.1 (Functors preseve isomorphisms). Let C, D be two categories and $F : \mathsf{C} \to \mathsf{D}$ a functor between them. If f is an isomorphism in C then Ff is an isomorphism in D.

Proof. Let $f : X \to Y$ be a isomorphism in C. Then there exists an inverse $f^{-1} : Y \to X$

in C such that

$$f^{-1} \circ f = \mathbb{1}_X \quad f \circ f^{-1} = \mathbb{1}_Y$$

Applying F to this equalities we have

$$F(f^{-1} \circ f) = F(\mathbb{1}_X) \implies F(f^{-1}) \circ F(f) = \mathbb{1}_{F(X)}$$
$$F(f \circ f^{-1}) = F(\mathbb{1}_Y) \implies F(f) \circ F(f^{-1}) = \mathbb{1}_{F(Y)}$$

i.e: $F(f)$ has an inverse $F(f^{-1})$ in D. □

Definition 2.1.2 (Functor types). A functor $F : \mathsf{C} \to \mathsf{D}$ is:

- **faithful** if for every pair of objects $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$ is injective.
- **full** if for every pair $X, Y \in \mathsf{C}$, the map $\mathrm{Hom}_{\mathsf{C}}(X, Y) \to \mathrm{Hom}_{\mathsf{D}}(F(X), F(Y))$ is surjective.
- **fully faithful** if it is both full and faithful.
- **essentially surjective** if for every object $Y \in \mathsf{D}$ there exists an object $X \in \mathsf{C}$ such that $F(X) \cong Y$.

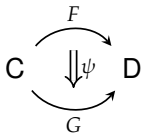
Example 2.1.6. A subcategory D of a category C is full if and only if the inclussion functor $\iota : \mathsf{D} \to \mathsf{C}$ is full.

2.2 Natural transformations

Definition 2.2.1. Let $F, G : \mathsf{C} \to \mathsf{D}$ be two functors. A **natural transformation** $\psi : F \implies G$ consists of a family of morphisms $\psi_X : F(X) \to G(X)$ in D for each object $X \in \mathsf{C}$, called the **components** of ψ, such that for every morphism $f : X \to Y$ in C, the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{\psi_X} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow[\psi_Y]{} & G(Y) \end{array}$$

Diagramatically, a natural transformations is represented by



If each component ψ_X is an isomorphism in D, we say that ψ is a **natural isomorphism**, and we write $F \cong G$.

Definition 2.2.2 (Composition of natural transformations).

- **Vertical composition**: Given natural transformations $\eta : F \implies G$ and $\varepsilon : G \implies H$, their vertical composite $\varepsilon \cdot \eta : F \implies H$ is defined componentwise by $(\varepsilon \cdot \eta)_X = \varepsilon_X \circ \eta_X$.
- **Horizontal composition**: Given natural transformations $\eta : F \implies G$ (where $F, G : \mathsf{C} \to \mathsf{D}$) and $\mu : H \implies K$ (where $H, K : \mathsf{D} \to \mathsf{E}$), their horizontal composite $\mu * \eta : H \circ F \implies K \circ G$ is defined by $(\mu * \eta)_X = \mu_{G(X)} \circ H(\eta_X) = K(\eta_X) \circ \mu_{F(X)}$.

Definition 2.2.3 (Equivalent categories)**.** Two categories C, D are **equivalent** if there exist functors $F : \mathsf{C} \to \mathsf{D}$ and $G : \mathsf{D} \to \mathsf{C}$ such that $G \circ F \cong \mathbb{1}_{\mathsf{C}}$ and $F \circ G \cong \mathbb{1}_{\mathsf{D}}$. In this case, F (and G) is called an **equivalence of categories**.

Theorem 2.2.1 ([Rie, Theorem 1.5.9])**.** A functor $F : \mathsf{C} \to \mathsf{D}$ defining an equivalence of categories is fully faithful and essentially surjective. Assuming the axiom of choice, any functor with these properties defines an equivalence of categories.

Chapter 3

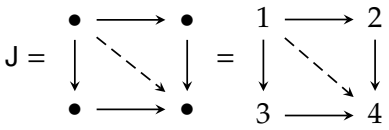
Limits and colimits

3.1 Diagrams, cones and cocones

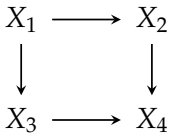
Definition 3.1.1 (Diagram). Let J be a small category.
A **diagram** X in a category C with shape J is a functor $X : \mathsf{J} \to \mathsf{C}$. For every $j \in \mathsf{J}$ we write $X_j := Xj$.

Example 3.1.1 (Constant diagram). A diagram $X : \mathsf{J} \to \mathsf{C}$ is **constant** if it is a constant functor const_c.
In this case the data of the diagram corresponds to the data given by a single object $c \in \mathsf{C}$.

Example 3.1.2. Consider the category

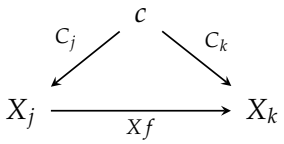


A diagram with shape J is a commutative square



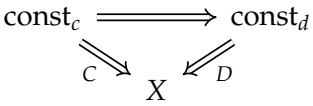
3.1.1 Cones and cocones

Definition 3.1.2 (Cone). A **cone** C over a diagram $X : \mathsf{J} \to \mathsf{C}$ with *apex* c is a natural transformation $C : \mathrm{const}_c \Rightarrow X$ from the constant diagram to X. The components $\{C_j : c \to X_j\}_{j \in \mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cone C over $X : \mathsf{J} \to \mathsf{C}$ with *apex* c is a collection of morphisms $C_j : c \to X_j$ for every object $j \in \mathsf{J}$ such that for every morphism $f : j \to k$ in J the following diagram in C commutes



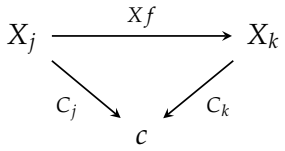
If $C : \mathrm{const}_c \Rightarrow X$ and $D : \mathrm{const}_d \Rightarrow X$ are two cones over X a morphism of cones is a natural transformation $\mathrm{const}_c \Rightarrow \mathrm{const}_d$ making the following diagram

commute

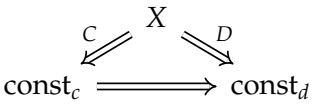


Therefore for any diagram $X : \mathsf{J} \to \mathsf{C}$ we have a category of cones $\Delta \downarrow X$.

Definition 3.1.3 (Cocone). A **cocone** C over a diagram $X : \mathsf{J} \to \mathsf{C}$ with *nadir* c is a natural transformation $C : X \Rightarrow \text{const}_c$ from the constant diagram to X. The components $\{C_j : X_j \to c\}_{j \in \mathsf{J}}$ are called the *legs* of the cone.
Less abstractly, a cocone C over $X : \mathsf{J} \to \mathsf{C}$ with *nadir* c is a collection of morphisms $C_j : X_j \to c$ for every object $j \in \mathsf{J}$ such that for every morphism $f : j \to k$ in J the following diagram in C commutes



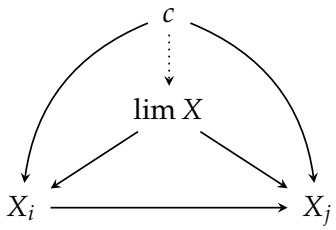
If $C : X \Rightarrow \text{const}_c$ and $D : X \Rightarrow \text{const}_d$ are two cocones over X a morphism of cocones is a natural transformation $\text{const}_c \Rightarrow \text{const}_d$ making the following diagram commute



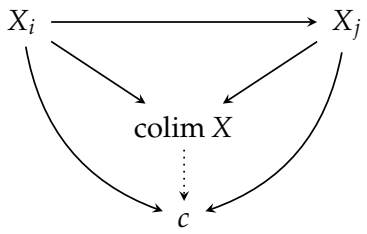
Therefore for any diagram $X : \mathsf{J} \to \mathsf{C}$ we have a category of cocones $X \downarrow \Delta$.

3.2 Limits and colimits

Definition 3.2.1 (Limit). A **limit** of a diagram $X : \mathsf{J} \to \mathsf{C}$ is a terminal object in the category of cones $\Delta \downarrow X$.
Less abstractly, a limit is a cone $\lim X : \text{const}_{\lim X} \Rightarrow X$ such that for any other cone $C : \text{const}_c \Rightarrow X$ there exists a unique morphism $c \to \lim X$ making the following diagram commute for all $i, j \in \mathsf{J}$



Definition 3.2.2 (Colimit). A **colimit** of a diagram $X : \mathsf{J} \to \mathsf{C}$, denoted $\text{colim}\, X$, is a terminal object in the category of cocones $X \downarrow \Delta$.
Less abstractly, a colimit is a cocone $\text{colim}\, X : X \Rightarrow \text{const}_{\text{colim}\, X}$ such that for any other cocone $C : X \Rightarrow \text{const}_c$ there exists a unique morphism $\text{colim}\, X \to c$ making the following diagram commute for all $i, j \in \mathsf{J}$



Proposition 3.2.1 (Uniqueness of limits and colimits). If a diagram $X : \mathsf{J} \to \mathsf{C}$ has a limit/colimit, it is unique up to unique isomorphism.

Proof. Direct from Proposition 1.3.1. □

Definition 3.2.3 (Functors and limits). Let $F : \mathsf{C} \to \mathsf{D}$ be a functor and $X : \mathsf{J} \to \mathsf{C}$ a diagram. We say that

- F **preserves** limits if whenever (L, φ) is the limit of X in C, $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D.
- F **reflects** limits if whenever (L, φ) is a cone over X such that $(F(L), F(\varphi))$ is the limit of $F \circ X$ in D then (L, φ) is the limit of X in C.

Replacing cones with cocones and limits with colimits we get dual statements.

3.2.1 Examples of limits and colimits

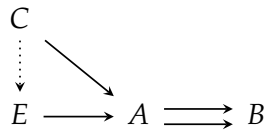
Definition 3.2.4 (Product). Let J be a discrete small category. This means that diagram $X : \mathsf{J} \to \mathsf{C}$ consists only of a set of objects $\{X_j\}_{j \in \mathsf{J}}$. We define the **product** of $\{X_j\}_{j \in \mathsf{J}}$, denoted $\prod_{j \in \mathsf{J}} X_j$, as the limit of the diagram X.
Less abstractly, a **product** is an object $\prod_{j \in \mathsf{J}} X_j \in \mathsf{C}$ together with maps $\prod_{j \in \mathsf{J}} X_j \to X_j$ for every $j \in \mathsf{J}$ such that, for any map $Y \to X_j$ there exists a unique morphism $Y \to \prod_{j \in \mathsf{J}} X_j$ making the following diagram commute

$$Y \dashrightarrow \prod_{j \in \mathsf{J}} X_j \longrightarrow X_j$$

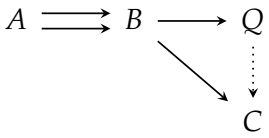
Definition 3.2.5 (Coproduct). Let J be a discrete small category. This means that diagram $X : \mathsf{J} \to \mathsf{C}$ consists only of a set of objects $\{X_j\}_{j \in \mathsf{J}}$. We define the **coproduct** of $\{X_j\}_{j \in \mathsf{J}}$, denoted $\coprod_{j \in \mathsf{J}} X_j$, as the colimit of the diagram X.
Less abstractly, a **coproduct** is an object $\coprod_{j \in \mathsf{J}} X_j \in \mathsf{C}$ together with maps $X_j \to \coprod_{j \in \mathsf{J}} X_j$ for every $j \in \mathsf{J}$ such that, for any map $X_j \to Y$ a unique morphism $\coprod_{j \in \mathsf{J}} X_j \to Y$ making the following diagram commute

$$Y \dashleftarrow \prod_{j \in \mathsf{J}} X_j \longleftarrow X_j$$

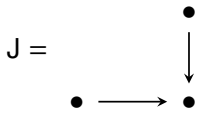
Definition 3.2.6 (Equaliser). Consider the *parallel pair* category $\mathsf{J} = \bullet \rightrightarrows \bullet$. A diagram $X : \mathsf{J} \to \mathsf{C}$ is a pair of morphisms $f, g : A \rightrightarrows B$, making a cone with apex C over X is a morphism $a : C \to A$ such that $fa = ga$.
The **equaliser** of f and g is the limit of this diagram. Les abstractly, it is a morphism $\mathrm{eq} : E \to A$ such that for any morphism $C \to A$ there exists a unique map $C \to E$ the following diagram commute



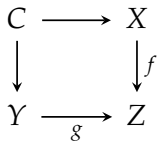
Definition 3.2.7 (Coequaliser)**.** Consider the *parallel pair* category $\mathsf{J} = \bullet \rightrightarrows \bullet$. A diagram $X : \mathsf{J} \to \mathsf{C}$ is a pair of morphisms $f, g : A \rightrightarrows B$, making a cocone with nadir C over X is a morphism $a : A \to C$ such that $af = ag$.
The **coequaliser** of f and g is the limit of this diagram. Less abstractly, it is a morphism $\mathrm{coeq} : B \to Q$ such that for any morphism $Y \to C$ there exists a unique map $Q \to C$ the following diagram commute



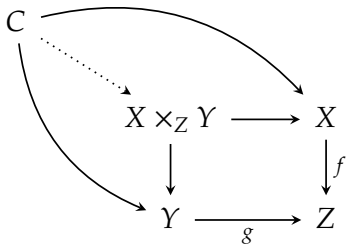
Definition 3.2.8 (Pullback)**.** Consider the category



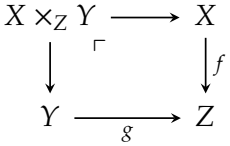
A diagram $X : \mathsf{J} \to \mathsf{C}$ is a pair of morphism $X \xrightarrow{f} Z \xleftarrow{g} Y$, making a cone with apex C over X a pair of morphisms $C \to X, C \to Y$ making the following diagram commute



A **pullback** (or **fibre product**) of two morphisms $f : X \to Z$ and $g : Y \to Z$ is the limit of the diagram $X \xrightarrow{f} Z \xleftarrow{g} Y$. It is denoted by $X \times_Z Y$.
Less abstractly, it is a pair of morphisms $X \times_Z Y \to X, X \times_Z Y \to Y$ such that for any pair of morphisms $C \to X, C \to Y$ making the square commute, there exists a unique morphism $C \to X \times_Z Y$ such that

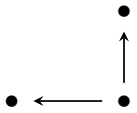


Pushout diagrams are often denoted by

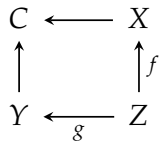


Definition 3.2.9 (Pushout)**.** Consider the category

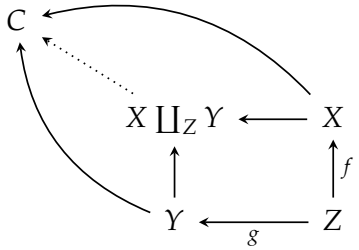
J =



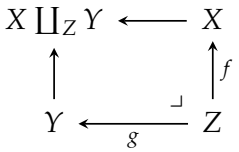
A diagram $X : \mathsf{J} \to \mathsf{C}$ is a pair of morphism $X \xleftarrow{f} Z \xrightarrow{g} Y$, making a cocone with nadir C over X a pair of morphisms $X \to C, Y \to C$ making the following diagram commute



A **pushout** of two morphisms $f : Z \to X, g : Z \to Y$ is the colimit of the diagram $X \xleftarrow{f} Z \xrightarrow{g} Y$. It is denoted by $X \coprod_Z Y$.
Less abstractly, it is a pair of morphisms $X \to X \coprod_Z Y, Y \to X \coprod_Z Y$ such that for any pair of morphisms $X \to C, Y \to C$ making the square commute, there exists a unique morphism $X \coprod_Z Y \to C$ such that



Pushout diagrams are often denoted by



Definition 3.2.10 (Directed set)**.** A **directed set** is a preordered set $(I, \leq)$ such that for every $i, j \in I$, there exists $k \in I$ with $i \leq k$ and $j \leq k$.
We can view this as a category following Example 1.1.6.

Definition 3.2.11 (Inverse limit)**.** An **inverse limit** (or **projective limit**) is a limit of a diagram over a directed set I. We write it as $\varprojlim_{i \in \mathsf{I}} X_i$.

Definition 3.2.12 (Direct limit)**.** A **direct limit** (or **inductive limit**) is a colimit of a diagram over a directed set I. We write it as $\varinjlim_{i \in \mathsf{I}} X_i$.

Chapter 4

More on functors

4.1 Representable functors

Definition 4.1.1. A functor $F : \mathsf{C} \to \mathsf{Set}$ is **representable** if there exists an object $X \in \mathsf{C}$ such that

$$F \cong \mathrm{Hom}_{\mathsf{C}}(X, -)$$

where $\mathrm{Hom}_{\mathsf{C}}(X, -)$ is the covariant Hom functor.

Definition 4.1.2 (Concrete category)**.** A category C is **concrete** if there exists a faithful functor $U : \mathsf{C} \to \mathsf{Set}$. Such a functor is often called a **forgetful functor**.

Lemma 4.1.1 (Yoneda - [Rie, Theorem 2.2.4])**.** Let C be a locally small category, $X \in \mathsf{C}$ an object, and $F : \mathsf{C} \to \mathsf{Set}$ a functor. There is a bijection:

$$\Phi : \mathrm{Hom}(\mathrm{Hom}(X, -), F) \xrightarrow{\sim} F(X)$$

given by $\alpha \mapsto \alpha_X(\mathbb{1}_X)$. This bijection is natural in both X and F.

4.2 Adjunctions

Definition 4.2.1 (Adjoint functors)**.** Let C, D be two locally small categories and let $L : \mathsf{C} \to \mathsf{D}$ and $R : \mathsf{D} \to \mathsf{C}$ be two functors. We say that L is **left adjoint** to R, and R is **right adjoint** to L, denoted $L \dashv R$, if there is a bijection:

$$\phi_{X,Y} : \mathrm{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{C}}(X, R(Y)) \tag{4.2.1}$$

which is natural in both $X \in \mathsf{C}$ and $Y \in \mathsf{D}$.

Definition 4.2.2 (Unit and counit)**.** The adjunction (4.2.1) is uniquely determined by two natural transformations:

- The **unit** $\eta : \mathbb{1}_{\mathsf{C}} \implies R \circ L$, where $\eta_X = \phi_{X,L(X)}(\mathbb{1}_{L(X)})$.
- The **counit** $\varepsilon : L \circ R \implies \mathbb{1}_{\mathsf{D}}$, where $\varepsilon_Y = \phi^{-1}_{R(Y),Y}(\mathbb{1}_{R(Y)})$.

These satisfy the **triangular identities**: $R(\varepsilon_Y) \circ \eta_{R(Y)} = \mathbb{1}_{R(Y)}$ and $\varepsilon_{L(X)} \circ L(\eta_X) = \mathbb{1}_{L(X)}$.

Remark 4.2.1. Left and right adjoints are dual concepts.

Proposition 4.2.1 (Adjoints are unique)**.** The right adjoint (resp. left adjoint) to a functor, provided it exists, is unique up to natural isomorphism

Proof. Let $L : \mathsf{C} \to \mathsf{D}$ be a functor.

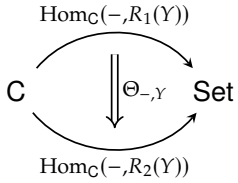
Let $R_1, R_2 : \mathsf{D} \to \mathsf{C}$ be two functors which are right adjoint to L. This means that for all $X \in \mathsf{C}$ and $Y \in \mathsf{D}$ we have bijections

$$\phi_{X,Y} : \mathrm{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{C}}(X, R_1(Y))$$
$$\psi_{X,Y} : \mathrm{Hom}_{\mathsf{D}}(L(X), Y) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Using this we can define a bijection

$$\Theta_{X,Y} = \psi_{X,Y} \circ \phi_{X,Y}^{-1} : \mathrm{Hom}_{\mathsf{C}}(X, R_1(Y)) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{C}}(X, R_2(Y))$$

Since ψ and ϕ are natural in X so is Θ. Therefore for a fixed $Y \in \mathsf{D}$ we have a natural isomorphism



□

Proposition 4.2.2. Right adjoints preserve all limits that exist in D.
Dually left adjoints preserve all colimits that exist in C.

4.3 Functor categories

Definition 4.3.1 (Functor category)**.** Let C and D be categories. The **functor category** $[C, D]$ (also denoted D^{C}) has:

- **Objects**: functors $F : \mathsf{C} \to \mathsf{D}$.
- **Morphisms**: natural transformations $\eta : F \implies G$.

Composition and identities are defined componentwise.

Lemma 4.3.1. Let C be a small category. Then

- If D is a locally small category then the functor category $[\mathsf{C}, \mathsf{D}]$ is locally small.
- If D is a small category then the functor category $[\mathsf{C}, \mathsf{D}]$ is small.

Lemma 4.3.2. Let C and D be two categories and $[\mathsf{C}, \mathsf{D}]$ be a functor category. If D has limits (resp. colimits) of shape J then $[\mathsf{C}, \mathsf{D}]$ has limits (resp. colimits) of shape J. Furthermore, they are computed objectwise.

Corollary 4.3.1 (Yoneda embedding - [Rie, Corollary 2.2.8])**.** Let C be a locally small category. The functor $Y : \mathsf{C}^{\mathrm{op}} \to [\mathsf{C}, \mathsf{Set}]$ given by $X \mapsto \mathrm{Hom}(X, -)$ is fully faithful. This allows us to view any locally small category as a full subcategory of a functor category.

4.4 Higher categories

We can go a step further.

Definition 4.4.1. Let $F : \mathsf{C} \to \mathsf{D}$, $G : \mathsf{C} \to \mathsf{D}$ be two functors. The **natural transformations category** $\mathsf{Nat}(F, G)$ constists of:

- **Objects**: natural transformations $F \Rightarrow G$.
- **Morphisms**:

PART II

Homological algebra

Chapter 5

Additive and abelian categories

5.1 Additive categories

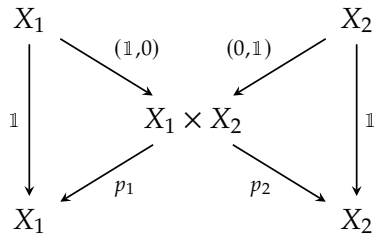
Definition 5.1.1 (Additive category). A category A is **additive** if it verifies:

- **A.1)** There exists a zero object $0 \in \mathsf{A}$.
- **A.2)** A has finite categorical products.
- **A.3)** For every pair of objects $X, Y \in \mathsf{A}$ the set $\mathrm{Hom}_{\mathsf{A}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.

Lemma 5.1.1. Finite products and coproducts coincide in an additive category.

Proof. Consider $X_1, X_2 \in \mathsf{A}$. We only need to check that $X_1 \times X_2 \cong X_1 \amalg X_2$ since the case for a finite collection of elements follows directly by induction. To do this, we will see that $X_1 \times X_2$ satisfies the universal property of the coproduct.

Consider the following diagram



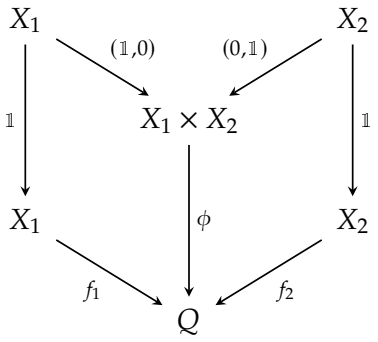
Note that for any pair of morphisms $(\alpha_1, \beta_1), (\alpha_2, \beta_2) : R \to X_1 \times X_2$ we have that

$$(\alpha_1, \beta_1) + (\alpha_2, \beta_2) = (\alpha_1 + \alpha_2, \beta_1, \beta_2)$$

Therefore if p_1, p_2 are two projections on $X_1 \times X_2$ whose sum is the identity we have that

$$(\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2 = \mathbb{1}_{X_1 \times X_2}$$

Consider now two morphisms $f_i : X_i \to Q$. Then its sum $\phi = f_1 \circ p_1 + f_2 \circ p_2$ produces the following diagram



Furthermore, this is unique. Indeed, consider $\phi' : X_1 \times X_2 \to Q$ another morphism filling the diagram. Then

$$\begin{aligned}
\phi &= \phi \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi \circ (\mathbb{1}, 0) \circ p_1 + \phi \circ (0, \mathbb{1}) \circ p_2 \\
&= f_1 \circ p_1 + f_2 \circ p_2 \\
&= \phi' \circ (\mathbb{1}, 0) + \phi' \circ (0, \mathbb{1}) \\
&= \phi' \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi' \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi'
\end{aligned}$$

Therefore $X_1 \times X_2$ verifies the universal property of $X_1 \amalg X_2$. □

Notation. If $\{X_i\}_{i=1}^n$ is a finite set of objects in an additive category we will denote its product (and therefore by Lemma 5.1.1 its coproduct) by $\bigoplus_{i=1}^n X_i$ and call it its **direct sum**.

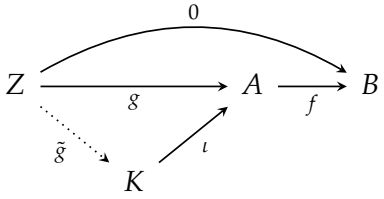
Lemma 5.1.2. A morphism $\varphi : A \to B$ in an additive category is a monomorphism if and only if for every morphism $\xi : Z \to A$

$$\varphi \circ \xi = 0 \implies \xi = 0$$

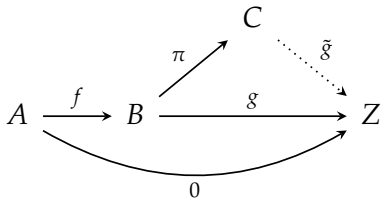
Dually φ is an epimorphism if and only if for every morphism $\xi : B \to Z$

$$\xi \circ \varphi = 0 \implies \xi = 0$$

Definition 5.1.2 (Kernel)**.** Let $f : A \to B$ be a morphism in an additive category. A morphism $\iota : K \to A$ is a **kernel** of f if $f \circ \iota = 0$ and for every morphism $g : Z \to A$ such that $f \circ g = 0$ there exists a unique $\tilde{g} : Z \to K$ making the following diagram commute



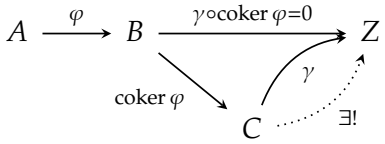
Definition 5.1.3 (Cokernel). Let $f : A \to B$ be a morphism in an additive category. A morphism $\pi : B \to C$ is a **cokernel** of f if $\pi \circ f = 0$ and for every morphism $g : B \to Z$ such that $\pi \circ g = 0$ there exists a unique $\tilde{g} : C \to Z$ making the following diagram commute



Lemma 5.1.3. In an additive category kernels are monomorphisms and cokernels are epimorphisms.

Proof. We will only show that cokernels are epimorphisms. The statement about kernels follows analogously.

Let $\varphi : A \to B$ be a morphism and let $\operatorname{coker} \varphi : B \to C$ be its cokernel. Consider another morphism $\gamma : C \to Z$ such that $\gamma \circ \operatorname{coker} \varphi = 0$. Since $(\gamma \circ \operatorname{coker} \varphi) \circ \varphi = 0$ by definition of the cokernel we have



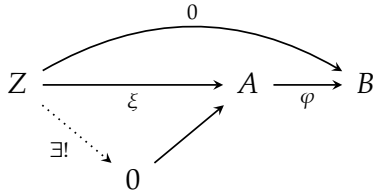
Therefore since $\gamma \circ \operatorname{coker} \varphi = 0 = 0 \circ \operatorname{coker} \varphi$ uniqueness forces $\gamma = 0$ and thus by Lemma 5.1.2 φ is an epimorphism. □

Lemma 5.1.4. Let $\varphi : A \to B$ be a morphism in an additive category. Then φ is a monomorphism if and only if $0 \to A$ is its kernel and φ is an epimorphism if and only if $B \to 0$ is its cokernel.

Proof. We will do the kernel one.

$\Longrightarrow$ Suppose φ is a monomorphism.
If $\xi : Z \to A$ is a morphism such that $\varphi \circ \xi = 0$ then by Lemma 5.1.2 $\xi = 0$. In particular ξ factors uniquely through $0 \to A$, i.e: $0 \to A$ is the kernel of φ.

$\Longleftarrow$ Suppose now that $0 \to A$ is the kernel of φ.
Consider a morphism $\xi : Z \to A$ such that $\varphi \circ \xi = 0$. By the kernel properties we have the following diagram



Therefore by uniqueness $\xi = 0$, making φ a monomorphism by Lemma 5.1.2.

□

5.1.1 Additive functors

Definition 5.1.4 (Additive functor)**.** Let A, B be two additive categories. A functor $F : \mathsf{A} \to \mathsf{B}$ is **additive** if it preserves the abelian group structure of all Hom-sets.

5.2 Abelian categories

Definition 5.2.1 (Abelian category)**.** An additive category A is **abelian** if it verifies

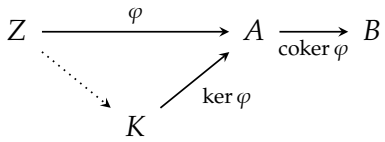
- **A.4)** Kernels and cokernels exist in A.
- **A.5)** Every monomorphism is the kernel of some morphism and every epimorphism is the cokernel of some morphism.

Lemma 5.2.1. In an abelian category A every kernel is the kernel of its cokernel and every cokernel is the cokernel of its kernel. In formulas

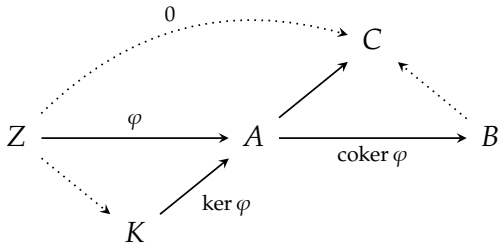
$$\ker(\operatorname{coker}\varphi) = \ker\varphi \qquad \operatorname{coker}(\ker\varphi) = \operatorname{coker}\varphi$$

Proof. Lets check cokernels this time.

Let $\operatorname{coker}\varphi : A \to B$ be the cokernel of a morphism $\varphi : Z \to A$. Since A is abelian φ has a kernel $\ker\varphi : K \to A$. Note that $Z \to A \to B$ is 0, therefore by the kernel universal property we have



Consider a morphism $A \to C$ such that $(K \to A \to C) = 0$. Now using the cokernel universal property we obtain

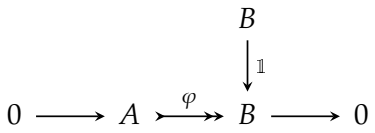


Therefore $\operatorname{coker}\varphi$ satisfies the universal property of the cokernel of $\ker\varphi : K \to A$. □

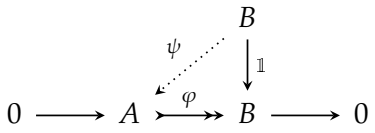
Lemma 5.2.2. Let $\varphi : A \to B$ be a morphism in an abelian category A. If φ is both a monomorphism and epimorphism then φ is an isomorphism.

Proof. By Lemma 5.1.4 the kernel of φ is $0 \to A$ and its cokernel is $B \to 0$. Furthermore by Lemma 5.2.1 φ is the cokernel of $0 \to A$ and the kernel of $B \to 0$.

Consider the following diagram

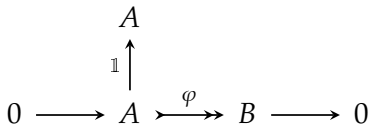


Since $(B \to B \to 0) = 0$ and φ is the kernel of $B \to 0$ we have

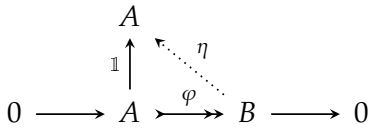


Therefore since $\varphi\psi = \mathbb{1}$ we have shown that φ has a right-inverse.

Consider now the diagram



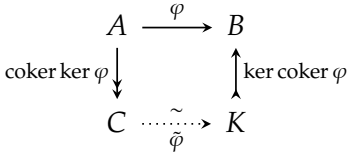
Since $(0 \to A \to A) = 0$ and φ is the cokernel of $0 \to A$ we have



Therefore since $\eta\varphi = \mathbb{1}$ we have shown that φ has a left-inverse. □

The factorization property

Theorem 5.2.1 (Factorization property). Every morphism $\varphi : A \to B$ in an abelian category A can be decomposed as



Before proving this statement, we need to consider some auxiliary lemmas.

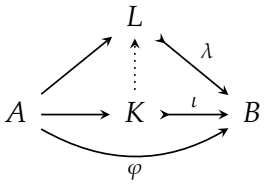
Lemma 5.2.3. Let $\varphi : A \to B$ be a morphism in an abelian category and let $\iota : K \to B$ be the kernel of its cokernel. Then

- ι is a monomorphism.
- φ factors through ι.
- ι is initial with these properties.

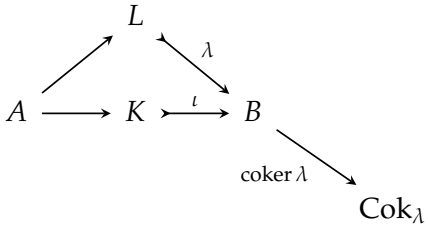
Proof.

- By Lemma 5.1.3 ι is a monomorphism.
- Since $A \to B \to \operatorname{coker}\varphi$ is the zero morphism by the universal property of kernels there is an induced morphism $K \to A$.

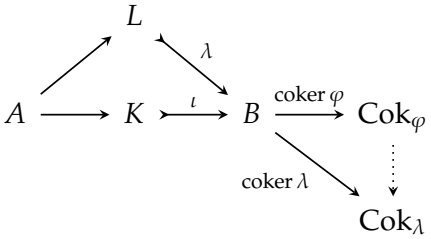
– We want to see that if $\lambda : L \rightarrowtail B$ is a monomorphism that factors through φ then ι factors uniquely through λ



Consider coker λ



Since φ factors through λ we have that $A \to B \to \mathrm{Cok}_\lambda = 0$. Therefore by the universal property of the cokernel we have an induced map



Therefore since $K \to \mathrm{Cok}_\varphi = 0$ we have that $K \to B \xrightarrow{\mathrm{coker}\,\lambda} \mathrm{Cok}_\lambda = 0$.
Finally since by Lemma 5.2.1 $\lambda : L \to B$ is the kernel of its cokernel there exists a unique morphism $K \to L$ making the diagram commute.

□

Definition 5.2.2. Let $\varphi : A \to B$ be a morphism in an abelian category. The **image** of φ is

$$\mathrm{im}\,\varphi = \ker \mathrm{coker}\,\varphi$$

and the **coimage** of φ is

$$\mathrm{coim}\,\varphi = \mathrm{coker} \ker \varphi$$

Notation. If $\varphi : A \rightarrowtail B$ is a monomorphism we denote by ${}^{B}/_{A}$ the image of coker φ.

In this way the target of coim φ is ${}^{B}/_{\ker \varphi}$.

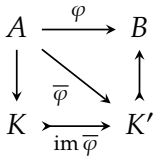
Lemma 5.2.4. Let $\varphi : A \to B$ be a morphism in an abelian category with $\mathrm{im}\,\varphi : K \to B$ and $\mathrm{coim}\,\varphi : A \to C$. Then the induced morphisms $A \to K$ and $B \to C$ are, respectively, an epimorphism and a monomorphism.

Proof. We will see that $\overline{\varphi} : A \to K$ is a epimorphism. The other statement follows analogously.

Consider the image of $\overline{\varphi}$

$$K' \overset{\operatorname{im}\overline{\varphi}}{\rightarrowtail} K \overset{\operatorname{coker}\overline{\varphi}}{\twoheadrightarrow} \operatorname{Cok}$$

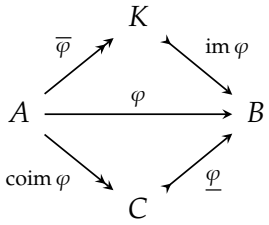
Since by Lemma 5.2.1 every epimorphism is the cokernel of its kernel we have that $\operatorname{coker}\overline{\varphi} = \operatorname{coker}\operatorname{im}\overline{\varphi}$. Moreover since $K' \rightarrowtail K$ is a monomorphism through which $\overline{\varphi}$ factors we have



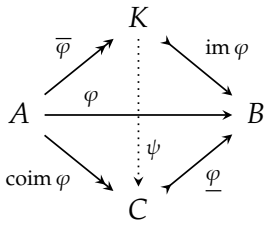
Since $\operatorname{im}\varphi$ is initial among this kind of morphisms by Lemma 5.2.3 necessarily $\operatorname{im}\overline{\varphi}$ is an isomorphism.

Therefore $0 = \operatorname{coker}\operatorname{im}\overline{\varphi} = \operatorname{coker}\overline{\varphi}$, proving that $\overline{\varphi}$ is an epimorphism by Lemma 5.1.4. □

To summarize, we have the following situation



Proof of Theorem 5.2.1. By the universal property of $\operatorname{im}\varphi$ and $\operatorname{coim}\varphi$ there exists a unique morphism $\psi : K \to C$ making the above diagram commute



Since $\varphi \circ \psi = \operatorname{im}\varphi$ is a monomorphism so is ψ. Since $\psi \circ \overline{\varphi} = \operatorname{coim}\varphi$ is a epimorphism so is $\overline{\psi}$. Therefore by Lemma 5.2.2 ψ is an isomorphism.

Taking $\tilde{\varphi} = \psi^{-1} : C \to K$ we are done. □

5.3 Exactness

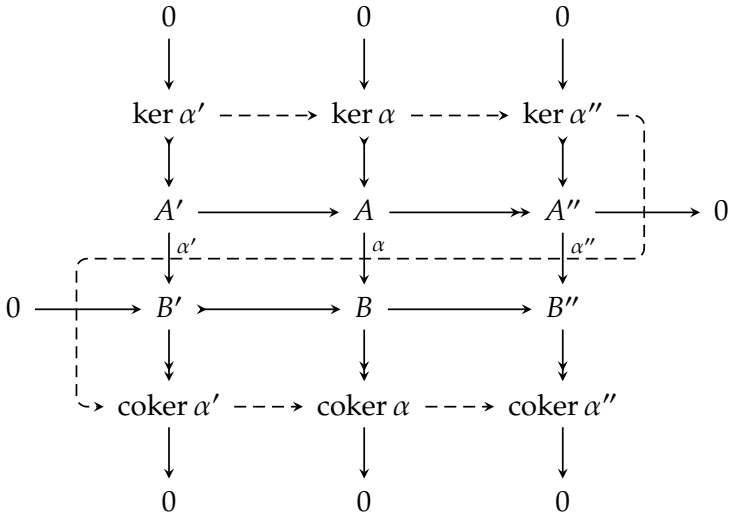
Definition 5.3.1 (Exact sequence)**.** An sequence of morphisms

$$X \overset{f}{\longrightarrow} Y \overset{g}{\longrightarrow} Z$$

in an abelian category A is **exact** at Y if $f \circ g = 0$ and $\operatorname{im} f = \ker g$.

Definition 5.3.2 (Short exact sequence). An sequence of morphisms
$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$
in an abelian-category A is **exact** is called a **short exact sequence**.
Exact complexes of the form $0 \to X \to Y \to Z$ (resp. $X \to Y \to Z \to 0$) are called left (resp. right) short exact sequences.

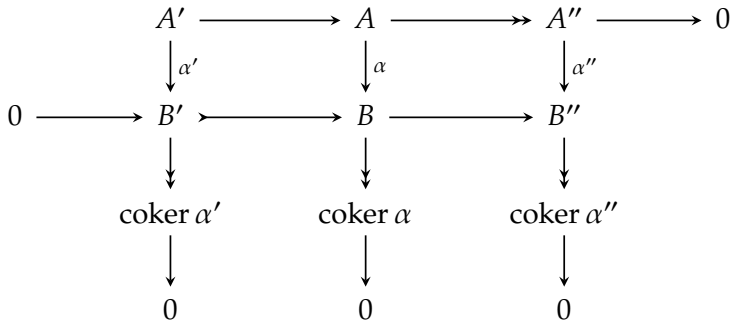
Remark 5.3.1. For any morphism $f : X \to Y$ in an A we get short exact sequences
$$0 \longrightarrow \ker f \longrightarrow X \longrightarrow \operatorname{im} f \longrightarrow 0$$
$$0 \longrightarrow \operatorname{im} f \longrightarrow Y \longrightarrow \operatorname{coker} f \longrightarrow 0$$
Therefore by Lemma 5.1.4 the sequence $0 \to X \xrightarrow{f} Y$ (resp. $X \xrightarrow{f} Y \to 0$) is exact if and only if f is a monomorphism (resp. epimorphism).

Theorem 5.3.1 (Snake lemma). In any abelian category, any diagram of the form


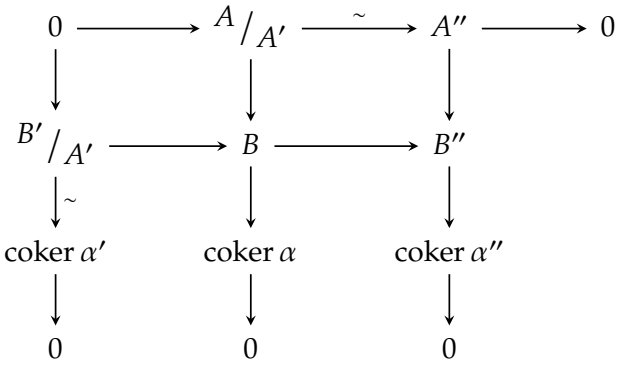
with exact rows and columns can be completed by dashed arrows making this sequence exact.

Proof. We will give a purely *arrow-theoretic* proof following [Wis].

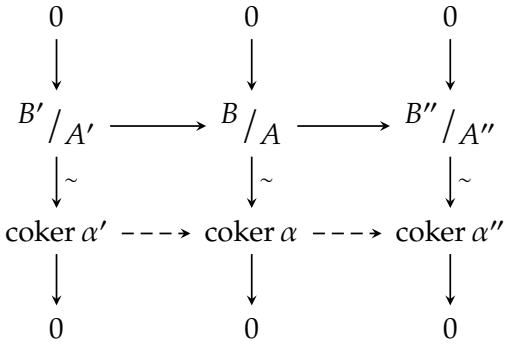
We will be using Lemma 5.1.3 and Lemma 5.2.2 throughout the proof without explicit mention.

Consider first the diagram


Take first cokernels by A'



And then take cokernels by A/A'



This produces an exact sequence $\operatorname{coker}\alpha' \to \operatorname{coker}\alpha \to \operatorname{coker}\alpha''$.

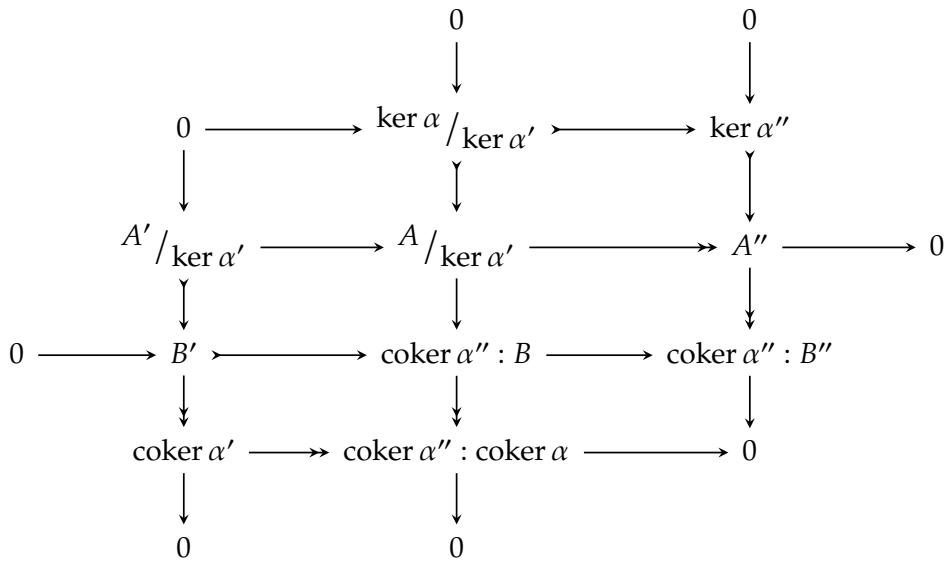
Taking kernels instead we get an exact sequence $\ker\alpha' \to \ker\alpha \to \ker\alpha''$.

It remains to produce a map $\delta : \ker\alpha'' \to \operatorname{coker}\alpha'$ and show that it induces an isomorphism

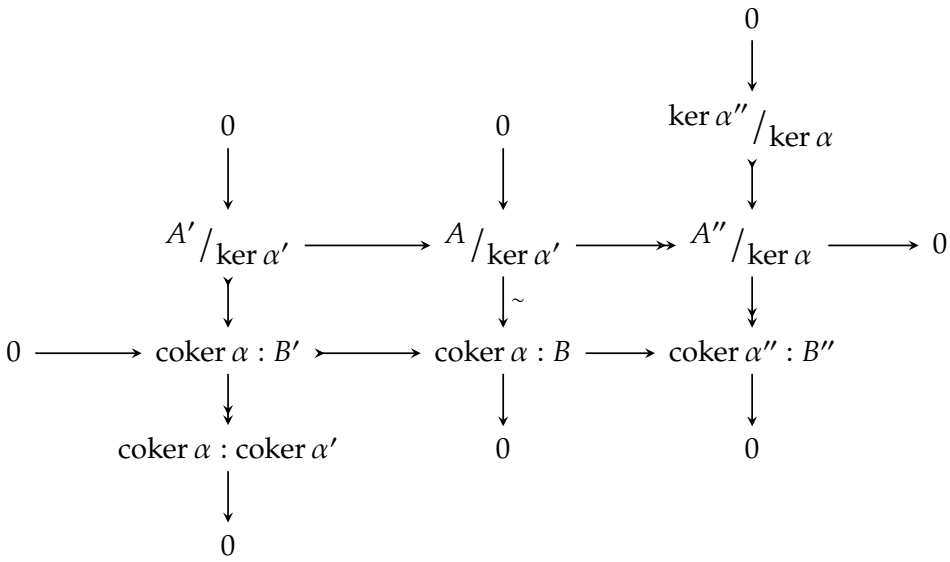
$$\operatorname{coker}(\ker\alpha \to \ker\alpha'') \cong \ker(\operatorname{coker}\alpha \to \operatorname{coker}\alpha')$$

To simplify the notation, we will denote $\ker(\operatorname{coker}\alpha \to \operatorname{coker}\alpha'')$ by $\operatorname{coker}\alpha : \operatorname{coker}\alpha''$.

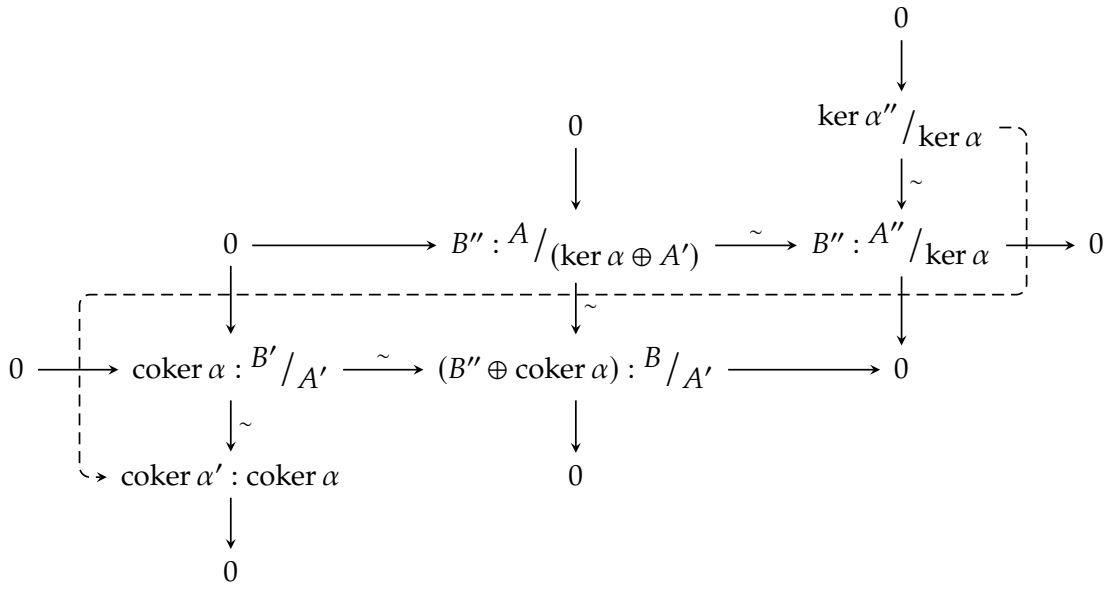
First, take cokernels by $\ker\alpha'$ and kernels into $\operatorname{coker}\alpha''$ in the full diagram



Now, take cokernels by $\ker \alpha / \ker \alpha'$ and kernels into $\operatorname{coker} \alpha : \operatorname{coker} \alpha''$



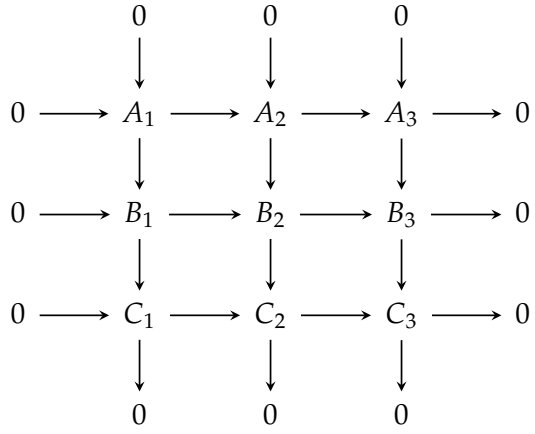
Finally, taking cokernels by $A' / \ker \alpha'$ and kernels into $\operatorname{coker} \alpha'' : B''$ we get



□

Theorem 5.3.2 (Freyd-Mitchell embedding). Let A be a small abelian category. Then there exists a ring R (with unit, not necessarily commutative) and a fully faithful exact functor $F : \mathsf{A} \to \mathsf{Mod}(R)$.
This means that A and $\mathsf{Mod}(R)$ are equivalent categories.

Theorem 5.3.3 (Nine lemma). Consider the following diagram in an abelian category



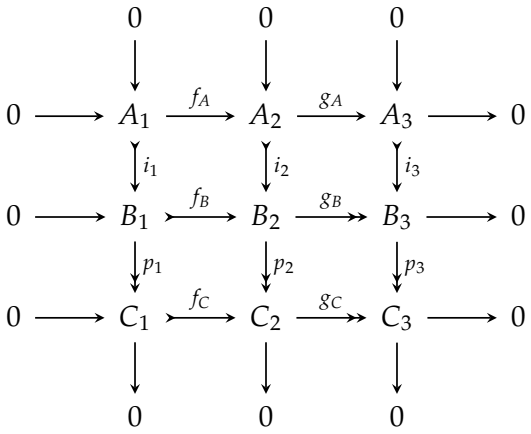
Suppose that all the columns are exact. Then

1. If the two bottom rows are exact, so is the first row.
2. If the two top rows are exact, so is the last row.
3. If the top and bottom rows are exact and $A_2 \to C_2$ is the zero morphism, the middle row is exact.

By symmetry, exchainging "row" and "column" gives three more true statements.

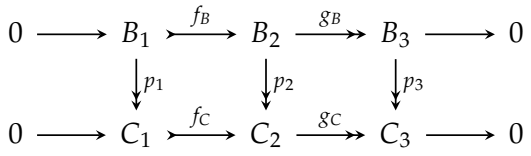
Proof. We will only show the first statement. The other two follow analogously considering the corresponding diagram.

We have the following

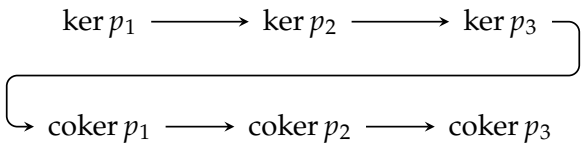


By Remark 5.3.1 we need to check that f_A is a monomorphism and g_A is an epimorphism.

Applying the snake lemma to



we get a exact sequence



Since the columns $0 \to A_k \xrightarrow{i_k} B_k \xrightarrow{p_k} C_k \to 0$ are exact we have

$$\ker p_k = A_k \quad \operatorname{coker} p_k = 0$$

i.e: the snake lemma sequence is

$$A_1 \xrightarrow{f_A} A_2 \xrightarrow{g_A} A_3 \longrightarrow 0$$

Since this sequence is exact we get that $\ker f_A = \operatorname{coker} g_A$ and that g_A is an epimorphism.

We still need to check that f_A is a monomorphism. To do this, by Lemma 5.1.2 let $x : X \to A_1$ be a morphism in A such that $f_A \circ x = 0$. We need to check that $x = 0$.

We have

$$\begin{aligned} 0 &= i_2 \circ (f_A \circ x) && \text{(Composition is associative)} \\ &= (i_2 \circ f_A) \circ x && \text{(Commutativity of the diagram)} \\ &= (f_B \circ i_1) \circ x \end{aligned}$$

Since $f_B \circ i_1$ is a monomorphism (both f_B and i_1 are monomorphisms) by Lemma 5.1.2 we have that $x = 0$. □

5.3.1 Exact functors

Definition 5.3.3 (Exact functor). Let A, B be two abelian categories.
An additive functor $F : \mathsf{A} \to \mathsf{B}$ is called left (resp. right) **exact** if for any short exact sequence in A

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z$$

(resp. $X \to Y \to Z \to 0$) the squence

$$0 \longrightarrow F(X) \longrightarrow F(Y) \longrightarrow F(Z)$$

(resp. $F(X) \to F(Y) \to F(Z) \to 0$) is exact.
If F is both left and right exact, then F is called **exact**.

Example 5.3.1. The functors $\mathrm{Hom}_{\mathsf{A}}(X, -)$ and $\mathrm{Hom}_{\mathsf{A}}(-, X)$ are both left-exact functors from A to Ab.

Proof. We will check the statement about $\mathrm{Hom}_{\mathsf{A}}(X, -)$, the other follows analogously.

Consider a short left-exact sequence

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

in A. We want to show that

$$0 \longrightarrow \mathrm{Hom}(X, A) \xrightarrow{f_*} \mathrm{Hom}(X, B) \xrightarrow{g_*} \mathrm{Hom}(X, C)$$

is exact in Ab.

By Remark 5.3.1, to show exacteness at $\mathrm{Hom}(X, A)$ we need to check that f_* is a monomorphism in Ab, i.e: f_* is injective.

Let $\alpha \in \mathrm{Hom}(X, A)$ such that $f_*(\alpha) = f \circ \alpha = 0$. Since $0 \to A \xrightarrow{f} B$ is exact by Remark 5.3.1 f is a monomorphism. Therefore $f \circ \alpha = 0 \iff \alpha = 0$, proving that f_* is injective.

To see exactness at $\mathrm{Hom}(X, B)$ we are going to check that $\mathrm{im}\, f_* = \ker g_*$:

- $\subseteq$ Let $\phi = f_*(\alpha)$ for some $\alpha \in \mathrm{Hom}(X, A)$. Applying g_* we get $g_*(\phi) = (g \circ f) \circ \alpha$, therefore by exactness of the original sequence $\phi \in \ker g_*$.
- $\supseteq$ Let $\beta \in \mathrm{Hom}(X, B)$ such that $g_*(\beta) = 0$. Since the original sequence is exact $\ker g = f$, therefore by the universal property of the kernel there exists $\alpha : X \to A$ such that $f \circ \alpha = \beta$.

□

The following is [Sch, Lemma 5.2.2].

Lemma 5.3.1. Let $F : \mathsf{A} \to \mathsf{B}$ be an additive functor.

1. The following are equivalent
   (a) F is left exact.

   (b) F commutes with kernels: for any morphism $f : X \to Y$, $F(\ker f) \xrightarrow{\sim} \ker F(f)$.
   (c) F commutes with finite limits.
   (d) For any short exact sequence $0 \to X \to Y \to Z \to 0$ in A the sequence $0 \to F(X) \to F(Y) \to F(Z)$ is exact in B.

2. The following are equivalent

   (a) F is exact.
   (b) For any exact sequence $X \to Y \to Z$ in A the sequence $F(X) \to F(Y) \to F(Z)$ is exact in B.
   (c) F commutes with finite limits and finite colimits.

Theorem 5.3.4. Let $L : \mathsf{A} \to \mathsf{B}$ and $R : \mathsf{B} \to \mathsf{A}$ be an adjoint pair of additive functors. Then L is right exact and R is right exact.

Proof. Let

$$\tau : \mathrm{Hom}_{\mathsf{B}}(LA, B) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{A}}(A, RB)$$

Consider a short exact sequence in B

$$0 \longrightarrow B' \longrightarrow B \longrightarrow B'' \longrightarrow 0$$

By naturality of τ for every object $A \in \mathsf{A}$ we have

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Hom}_{\mathsf{B}}(LA, B') & \longrightarrow & \mathrm{Hom}_{\mathsf{B}}(LA, B) & \longrightarrow & \mathrm{Hom}_{\mathsf{B}}(LA, B'') \\ & & \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \sim} \\ 0 & \longrightarrow & \mathrm{Hom}_{\mathsf{A}}(A, RB') & \longrightarrow & \mathrm{Hom}_{\mathsf{A}}(A, RB) & \longrightarrow & \mathrm{Hom}_{\mathsf{A}}(A, RB'') \end{array}$$

By left-exactness of the Hom functor the top row is exact. Therefore the bottom row must also be exact.

By the Yoneda lemma we have that

$$0 \longrightarrow RB' \longrightarrow RB \longrightarrow RB''$$

must be exact, meaning that R is left exact.

In particular $L^{\mathrm{op}} : A^{\mathrm{op}} \to B^{\mathrm{op}}$ is left exact, making L right exact. □

Chapter 6

Complexes and cohomology

6.1 Complexes and cohomology

Throughout this chapter we will be working over a fixed abelian category A.

Definition 6.1.1 (Complex). A sequence of objects and morphisms

$$\cdots \xrightarrow[d_{i+2}]{} C_{i+1} \xrightarrow[d_{i+1}]{} C_i \xrightarrow[d_i]{} C_{i-1} \xrightarrow[d_{i-1}]{} \cdots$$

in A is a **chain complex** if for every $i \in \mathbb{Z}$ $\quad d_i \circ d_{i+1} = 0$. We denote a chain complex as $(C_\bullet, d_\bullet)$ or simply as $C_\bullet$.
By reversing the order, we say that a sequence

$$\cdots \xrightarrow{d^{i-2}} C^{i-1} \xrightarrow{d^{i-1}} C^i \xrightarrow{d^i} C^{i+1} \xrightarrow{d^{i+1}} \cdots$$

is a **cochain complex** if for every $i \in \mathbb{Z}$ $\quad d^i \circ d^{i-1} = 0$. We denote a cochain complex as $(C^\bullet, d^\bullet)$ or simply as $C^\bullet$.

Remark 6.1.1. In purelly algebraic terms, chains are cochains are *equivalent*. We will adopt the cochain notation, but all proofs and definitions translate to the chain notation. Therefore from now onwards we will be refering to *cochains* as *chains* or *sequences*.

Definition 6.1.2 (Morphism of complexes). Let $C^\bullet, D^\bullet$ be two complexes in A. A **morphism of complexes** $f : C^\bullet \to D^\bullet$ is a collection of morphisms $\{f^i : C^i \to D^i\}_{i\in\mathbb{Z}}$ making, for every $i \in \mathbb{Z}$, the following diagram commute

$$\begin{array}{ccc} C^i & \xrightarrow{d^i_C} & C^{i+1} \\ {\scriptstyle f^i}\downarrow & & \downarrow{\scriptstyle f^{i+1}} \\ D^i & \xrightarrow{d^i_D} & D^{i+1} \end{array}$$

Definition 6.1.3. Given any abelian category A we can construct a **complex category** $\mathsf{Ch}(\mathsf{A})$ by taking $\mathrm{Ob}_{\mathsf{Ch}(\mathsf{A})}$ to be the collection of all complexes with values in A and, for every $C^\bullet, D^\bullet \in \mathrm{Ob}_{\mathsf{Ch}(\mathsf{A})}$, $\mathrm{Hom}_{\mathsf{Ch}(\mathsf{A})}(C^\bullet, D^\bullet)$ to be the collection of all morphisms of complexes between $C^\bullet$ and $D^\bullet$.

Remark 6.1.2. A morphism of complexes $f : C^\bullet \to D^\bullet$ is a monomorphism (resp. epimorphism) if for every $i \in \mathbb{Z}$ the morphism $f^i : C^i \to D^i$ is a monomorphism (resp. epimorphism) in A.

Theorem 6.1.1. If A is an abelian-category, so is its category of complexes $\mathsf{Ch}(\mathsf{A})$.

Proof. By taking addition *elementwise* and $0^\bullet$ as the zero object we can make $\mathsf{Ch}(\mathsf{A})$ into an additive category. Therefore we only need to prove that $\mathsf{Ch}(\mathsf{A})$ verifies the axioms of an abelian category.

Let $f : A^\bullet \to B^\bullet$ be a morphism of complexes.

Since A is abelian, for every $i \in \mathbb{Z}$ the map $f^i : A^i \to B^i$ admits a kernel $\ker f^i : K^i \rightarrowtail A^i$ and a cokernel $\operatorname{coker} f^i : B^i \twoheadrightarrow C^i$.

Consider the diagram

$$\begin{array}{ccc}
K^i & & K^{i+1} \\
{\scriptstyle \ker f^i}\downarrow & & \downarrow{\scriptstyle \ker f^{i+1}} \\
A^i & \xrightarrow{d^i_C} & A^{i+1} \\
{\scriptstyle f^i}\downarrow & & \downarrow{\scriptstyle f^{i+1}} \\
B^i & \xrightarrow{d^i_D} & B^{i+1} \\
{\scriptstyle \operatorname{coker} f^i}\downarrow & & \downarrow{\scriptstyle \operatorname{coker} f^{i+1}} \\
C^i & & C^{i+1}
\end{array}$$

Since the inner square is commutative we get unique morphisms $K^i \to K^{i+1}$ and $C^i \to C^{i+1}$ making the whole diagram commute, obtaining chain complexes $K^\bullet$ and $C^\bullet$ and morphisms of chains $\ker f^\bullet$, $\operatorname{coker} f^\bullet$. It is then easy to check that $\ker f^\bullet$ and $\operatorname{coker} f^\bullet$ verify the corresponding universal properties.

On the other hand, by Remark 6.1.2 the other axiom is easily verified *element by element*. □

Definition 6.1.4 (Bounded complexes). Let $C^\bullet$ be a complex. We say that $C^\bullet$ is

- **bounded below** if there exists $k \in \mathbb{Z}$ such that $\forall i \leq k \quad C^i = 0$.
- **bounded above** if there exists $k \in \mathbb{Z}$ such that $\forall i \geq k \quad C^i = 0$.
- **bounded** if it is both bounded below and above.

Notation. We denote by $\mathsf{Ch}^{-,+,b}(\mathsf{A}) \hookrightarrow \mathsf{Ch}(\mathsf{A})$ the full subcategory of the chain complex category consisting of bounded below, bounded above and bounded complexes respectively.

We also have variants $\mathsf{Ch}^{\geq 0,\leq 0}(\mathsf{A}) \hookrightarrow \mathsf{Ch}(\mathsf{A})$ of complexes with $\forall i \geq 0$ (resp $\forall i \leq 0$) $C^i = 0$.

By Theorem 6.1.1 all of this categories are abelian.

Remark 6.1.3. For all integers $i \in \mathbb{Z}$ there exists a fully faithfull functor

$$\iota_r : \mathsf{A} \to \mathsf{Ch}(\mathsf{A})$$

sending an object $A \in \mathsf{A}$ to a complex

$$\cdots \longrightarrow 0 \xrightarrow{d^{r-1}} A \xrightarrow{d^r} 0 \longrightarrow \cdots$$

and a morphism $\varphi : A \to B$ in A to a morphism of complexes

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & 0 & \xrightarrow{d^{r-1}} & A & \xrightarrow{d^r} & 0 & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow{\scriptstyle \varphi} & & \downarrow & & \\ \cdots & \longrightarrow & 0 & \xrightarrow{d^{r-1}} & B & \xrightarrow{d^r} & 0 & \longrightarrow & \cdots \end{array}$$

In this way, we can identify A as its *image* via $\iota = \iota_0$ in $\mathsf{Ch}(\mathsf{A})$.

Definition 6.1.5 (Shift functor). For any integer $i \in \mathbb{Z}$ we define the **shift** functor

$$\begin{aligned} \mathsf{Ch}(\mathsf{A}) &\to \mathsf{Ch}(\mathsf{A}) \\ M^\bullet &\mapsto M[i]^\bullet \end{aligned}$$

by $M[i]^k = M^{k+i}$ and $d^k_{M[i]^\bullet} = (-1)^i d^{k+i}_{M^\bullet}$.

Definition 6.1.6 (Exact complex). A complex $C^\bullet$ is **exact** at position i if the sequence

$$C^{i-1} \xrightarrow{d^{i-1}} C^i \xrightarrow{d^i} C^{i+1}$$

is exact.
If the complex is exact at every position, we say that it is **exact**.

6.2 Cohomology

Definition 6.2.1 (Cohomology). The **i-th cohomology** of a complex $C^\bullet$ is

$$\mathrm{H}^i(C^\bullet) = {\ker d^i} \,/_{\operatorname{im} d^{i-1}}$$

Remark 6.2.1. The definition of cohomology should be parsed as follows: let $I \to C^i$ be $\operatorname{im} d^{i-1}$ and $K \to C^i$ be $\ker d^i$. Since $C^\bullet$ is a complex $\operatorname{im} d^{i-1}$ factors through $\ker d^i$, giving a monomorphism $I \rightarrowtail K$. Then $\mathrm{H}^i(C^\bullet) = \operatorname{coker}(I \rightarrowtail K)$.

Proposition 6.2.1. For every integer i, the assignement $\mathrm{H}^i : M^\bullet \mapsto H^i(M^\bullet)$ taking a complex $M^\bullet$ to its i-th cohomology group is a functor $\mathsf{Ch}(\mathsf{A}) \to \mathsf{A}$.

Proof. We want to show that for all complexes $M^\bullet, N^\bullet \in \mathsf{Ch}(\mathsf{A})$ and $i \in \mathbb{Z}$, H^i induces in a natural way a homomorphism of abelian groups

$$\operatorname{Hom}_{\mathsf{Ch}(\mathsf{A})}(M^\bullet, N^\bullet) \to \operatorname{Hom}_{\mathsf{A}}(\mathrm{H}^i(M^\bullet), \mathrm{H}^i(N^\bullet))$$

Let $\alpha^\bullet : M^\bullet \to N^\bullet$ be a morphism of complexes. Let's consider the relevant part of the commutative diagram defining $\alpha^\bullet$:

$$\begin{array}{ccccc} M^{i-1} & \xrightarrow{d^{i-1}_{M^\bullet}} & M^i & \xrightarrow{d^i_{M^\bullet}} & M^{i+1} \\ {\scriptstyle \alpha^{i-1}}\downarrow & & {\scriptstyle \alpha^i}\downarrow & & {\scriptstyle \alpha^{i+1}}\downarrow \\ N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1} \end{array}$$

By the commutativity of the right square we have that $d^i_{N^\bullet} \circ \alpha^i = \alpha^{i+1} \circ d^i_{M^\bullet}$ is 0 on $\ker d^i$. Therefore the restriction of α^i to $\ker d^i_{M^\bullet}$ factors through $\ker d^i_{N^\bullet}$. Composing with the projection we get a morphism

$$\ker d^i_{M^\bullet} \to \mathrm{H}^i(N^\bullet)$$

Since this morphism is 0 when restricted to $\operatorname{im} d^{i-1}_{M^\bullet}$ by the commutativity of the left square we get that α^i induces a morphism

$$\mathrm{H}^i(M^\bullet) \to \mathrm{H}^i(N^\bullet)$$

in A as we wanted. □

Remark 6.2.2. By Remark 6.1.3 we can see cohomology as a functor $\mathrm{H}^i : \mathsf{Ch}(\mathsf{A}) \to \mathsf{Ch}(\mathsf{A})$ using ι_i. Note that for this particular complex $M^\bullet$ we have that

$$\mathrm{H}^\bullet(\mathrm{H}^\bullet(M^\bullet)) = \mathrm{H}^\bullet(M^\bullet)$$

Theorem 6.2.1 (Long Exact Sequence in Cohomology). Every short exact sequence

$$0 \longrightarrow A^\bullet \xrightarrow{f^\bullet} B^\bullet \xrightarrow{g^\bullet} C^\bullet \longrightarrow 0$$

in $\mathsf{Ch}(\mathsf{A})$ induces a long exact sequence in cohomology

$$\cdots \longrightarrow \mathrm{H}^{i-1}(B) \longrightarrow \mathrm{H}^{i-1}(C) \longrightarrow \mathrm{H}^i(A) \longrightarrow \mathrm{H}^i(B) \longrightarrow \mathrm{H}^i(C) \longrightarrow \mathrm{H}^{i+1}(A) \longrightarrow \mathrm{H}^{i+1}(B) \longrightarrow \cdots$$

Proof. For every integer i, applying the snake lemma to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^i & \xrightarrow{f^i} & B^i & \xrightarrow{g^i} & C^i & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle d^i_A} & & \downarrow{\scriptstyle d^i_B} & & \downarrow{\scriptstyle d^i_C} & & \\ 0 & \longrightarrow & A^{i+1} & \xrightarrow{f^{i+1}} & B^{i+1} & \xrightarrow{g^{i+1}} & C^{i+1} & \longrightarrow & 0 \end{array}$$

we get an exact sequence

$$\operatorname{coker} d^i_A \longrightarrow \operatorname{coker} d^i_B \longrightarrow \operatorname{coker} d^i_C \longrightarrow 0$$

Applying now the snake lemma to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A^{i+1} & \xrightarrow{f^{i+1}} & B^{i+1} & \xrightarrow{g^{i+1}} & C^{i+1} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle d^{i+1}_A} & & \downarrow{\scriptstyle d^{i+1}_B} & & \downarrow{\scriptstyle d^{i+1}_C} & & \\ 0 & \longrightarrow & A^{i+2} & \xrightarrow{f^{i+2}} & B^{i+2} & \xrightarrow{g^{i+2}} & C^{i+2} & \longrightarrow & 0 \end{array}$$

we get an exact sequence

$$0 \longrightarrow \ker d^{i+1}_A \longrightarrow \ker d^{i+1}_B \longrightarrow \ker d^{i+1}_C$$

Since $d^{i+1} \circ d^i = 0$ we have the following diagram

$$\begin{array}{ccccccc} & & \operatorname{coker} d_A^i & \longrightarrow & \operatorname{coker} d_B^i & \longrightarrow & \operatorname{coker} d_C^i & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \overline{d}_A^i} & & \downarrow{\scriptstyle \overline{d}_B^i} & & \downarrow{\scriptstyle \overline{d}_C^i} & & \\ 0 & \longrightarrow & \ker d_A^{i+1} & \longrightarrow & \ker d_B^{i+1} & \longrightarrow & \ker d_C^{i+1} & & \end{array}$$

Applying the snake lemma to this diagram we get an exact sequence

$$\begin{array}{ccccc} \ker \overline{d}_A^i & \longrightarrow & \ker \overline{d}_B^i & \longrightarrow & \ker \overline{d}_A^i \\ & & & \overset{\delta^i}{\searrow} & \\ \operatorname{coker} \overline{d}_A^i & \longrightarrow & \operatorname{coker} \overline{d}_B^i & \longrightarrow & \operatorname{coker} \overline{d}_C^i \end{array}$$

It only remains to prove that $\ker \overline{d}_A^i \cong \mathrm{H}^i(A)$ and $\operatorname{coker} \overline{d}_A^i \cong \mathrm{H}^{i+1}(A)$.

Consider the diagram

$$\begin{array}{ccccccccc} & & 0 & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \operatorname{im} d_A^{i-1} & = & \operatorname{im} d_A^{i-1} & \longrightarrow & 0 & & \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \ker d_A^i & \longrightarrow & A^i & \longrightarrow & \operatorname{im} d_A^i & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathrm{H}^i(A) & \longrightarrow & A^i/_{\operatorname{im} d_A^{i-1}} & \xrightarrow{\psi} & \operatorname{im} d_A^i & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & 0 & & \end{array}$$

Since the first two rows are exact by the nine lemma we have a short exact sequence

$$0 \longrightarrow \mathrm{H}^i(A) \longrightarrow A^i/_{\operatorname{im} d_A^{i-1}} \xrightarrow{\psi} \operatorname{im} d_A^i \longrightarrow 0$$

Therefore since $\overline{d}_A^i : A^i/_{\operatorname{im} d_A^{i-1}} \xrightarrow{\psi} \operatorname{im} d_A^i \xrightarrow{\iota} \ker d_A^{i+1}$ we have

$$\ker(\overline{d}_A^i) = \ker \psi \cong \mathrm{H}^i(A^\bullet)$$

On the other hand, note that $\operatorname{coker} \overline{d}_A^i = \operatorname{coker}(\iota \circ \psi) = \operatorname{coker} \iota$ and

$$\operatorname{coker} \iota = \ker d_A^{i+1}/_{\operatorname{im} \iota} = \ker d_A^{i+1}/_{\operatorname{im} d_A^i} \cong \mathrm{H}^{i+1}(A)$$

□

Triangles

Consider a short exact sequence of complexes

$$0 \longrightarrow A^\bullet \longrightarrow B^\bullet \longrightarrow C^\bullet \longrightarrow 0$$

Note that we can map $C^\bullet$ to $A^\bullet$ by sending all the complex $C^\bullet$ to 0 *within* $A^\bullet$

$$C^\bullet \xrightarrow{0} A^\bullet$$

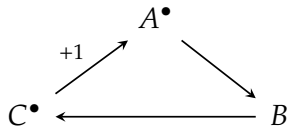
In this way, the exactness of the original sequence is equivalent to the exactness at the center of

$$C^\bullet \xrightarrow{0} A^\bullet \longrightarrow B^\bullet$$

$$A^\bullet \longrightarrow B^\bullet \longrightarrow C^\bullet$$

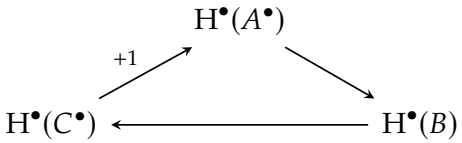
$$B^\bullet \longrightarrow C^\bullet \xrightarrow{0} A^\bullet$$

We can represent this using the *exact triangle*



The arrow marked by +1 in this case refers to the zero morphism. We use +1 to record that we can view this morphism from $C^\bullet$ to the shift $A[1]^\bullet$. Therefore, the triangle is shorthand for the diagram

Using Theorem 6.2.1 we get another *exact triangle*



However this time the +1 morphism is, generally, non-zero.

Chapter 7

Cones and homotopy

7.1 The mapping cone of a morphism

Definition 7.1.1 (Mapping cone). Let $\alpha^\bullet : L^\bullet \to M^\bullet$ be a morphism of complexes. We define the **mapping cone** $\mathrm{Cone}(\alpha)^\bullet$ as the complex with objects

$$\forall i \in \mathbb{Z} \quad \mathrm{Cone}(\alpha)^i = L[1]^i \oplus M^i = L^{i+1} \oplus M^i$$

and differentials

$$\forall i \in \mathbb{Z} \quad d^i_{\mathrm{Cone}(\alpha)^\bullet}(\ell, m) = (-d^{i+1}_{L^\bullet}(\ell), \alpha^{i+1}(\ell) + d^i_{M^\bullet}(m))$$

Pictorially

$$\begin{array}{ccccccc} \cdots & \longrightarrow & L^{i+1} & \xrightarrow{-d_L^{i+1}} & L^{i+1} & \longrightarrow & \cdots \\ & & \oplus & \searrow^{\alpha^{i+1}} & \oplus & & \\ \cdots & \longrightarrow & M^i & \xrightarrow[d_M^i]{} & M^{i+1} & \longrightarrow & \cdots \end{array}$$

Remark 7.1.1. We have morphisms of complexes $M^\bullet \to \mathrm{Cone}(\alpha)^\bullet$ and $\mathrm{Cone}(\alpha)^\bullet \to L[1]^\bullet$ induced by the morphisms $M^i \to L[1]^i \oplus M^i \to L[1]^i$.
Furthermore, since the we have short exact sequences

$$0 \longrightarrow M^i \longrightarrow L[1]^i \oplus M^i \longrightarrow L[1]^i \longrightarrow 0$$

the sequence of complexes

$$0 \longrightarrow M^\bullet \longrightarrow L[1]^\bullet \oplus M^\bullet \longrightarrow L[1]^\bullet \longrightarrow 0$$

is exact.

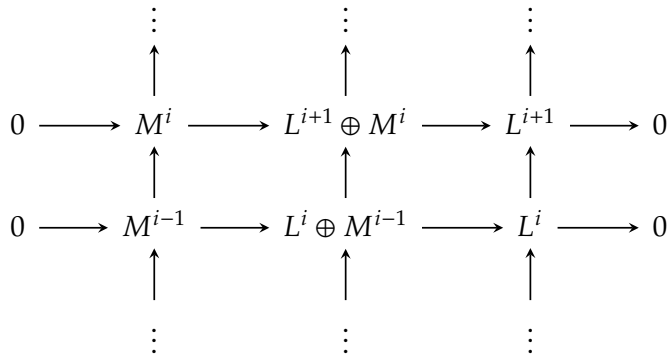
Proposition 7.1.1. There exists an exact triangle

$$\begin{array}{ccc} & \mathrm{H}^\bullet(M^\bullet) & \\ {\scriptstyle +1}\nearrow & & \searrow \\ \mathrm{H}^\bullet(L[1]^\bullet) & \xleftarrow{\quad\delta\quad} & \mathrm{H}^\bullet(\mathrm{Cone}(\alpha)^\bullet) \end{array}$$

where the connecting morphism δ is the morphism induced by $\alpha^\bullet$ in cohomology.

Proof. The existence of the triangle comes directly from Theorem 6.2.1, so we only need to check that δ agrees with the morphism induced by $\alpha^\bullet$.

By chasing the diagram



with a class $\mathrm{H}^{i-1}(L[1]^\bullet) = \mathrm{H}^i(L^\bullet)$ represented by $\ell \in L^i$ (such that $d^i_{L^\bullet}(\ell) = 0$) we find

$$\begin{array}{ccc} \alpha^i(\ell) & \longmapsto & (0, \alpha^i(\ell)) \\ & & \uparrow \\ & & (\ell, 0) \end{array} \longmapsto \ell$$

which verifies the claim. □

Corollary 7.1.1. Let $\alpha^\bullet : L^\bullet \to M^\bullet$ be a morphism of complexes. Then the induced morphism $\mathrm{H}^\bullet(L^\bullet) \to \mathrm{H}^\bullet(M^\bullet)$ is an isomorphism if and only if the mapping cone $\mathrm{Cone}(\alpha)^\bullet$ is an exact complex.

Proof. Using Proposition 7.1.1, consider the following segment of the long exact sequence

$$\cdots \longrightarrow \mathrm{H}^{i-1}(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \mathrm{H}^i(L^\bullet) \xrightarrow{\alpha^*} \mathrm{H}^i(M^\bullet) \longrightarrow \mathrm{H}^i(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \mathrm{H}^{i+1}(L^\bullet) \xrightarrow{\alpha^*} \mathrm{H}^{i+1}(M^\bullet) \longrightarrow \mathrm{H}^{i+1}(\mathrm{Cone}(\alpha)^\bullet) \longrightarrow \cdots$$

$\Longrightarrow$ Suppose α^* is an isomorphism. Since $\mathsf{Ch}(\mathsf{A})$ is abelian by Lemma 5.2.2 α^* is both a monomorphism and an epimorphism. Consider

$$\cdots \xrightarrow{\alpha^*} \mathrm{H}^i(M^\bullet) \xrightarrow{f} \mathrm{H}^i(\mathrm{Cone}(\alpha)^\bullet) \xrightarrow{g} \mathrm{H}^{i+1}(L^\bullet) \xrightarrow{\alpha^*} \cdots$$

Using that the first α^* is a surjective and exactness we have that

$$\ker f = \operatorname{im} \alpha^* = \mathrm{H}^i(M^\bullet)$$

i.e: f is the zero map.
Using that the last α^* is injective and exactness we have that

$$\ker g = \ker \alpha^* = \{0\}$$

i.e: g is the zero map.
Therefore we have that $\mathrm{H}^i \operatorname{Cone}(\alpha)^\bullet = 0$, meaning that the complex is exact.

$\Longleftarrow$ Suppose $\operatorname{Cone}(\alpha)^\bullet$ is an exact complex. Therefore since its cohomology vanishes at every index the sequence becomes

$$\cdots \longrightarrow 0 \longrightarrow \mathrm{H}^i(L^\bullet) \xrightarrow{\alpha^*} \mathrm{H}^i(M^\bullet) \longrightarrow 0 \longrightarrow \cdots$$

Therefore by exactness we have that α^* is an isomorphism.

□

7.2 Quasi-isomorphisms

Definition 7.2.1 (Quasi-isomorphism)**.** A morphism of complexes $\alpha^\bullet$ is a **quasi-isomorphism** (abbreviated q.i.s) if it induces an isomorphism in cohomology.

Remark 7.2.1. By Corollary 7.1.1, $\alpha^\bullet$ is a quasi-isomorphism if and only it its mapping cone $\operatorname{Cone}(\alpha)^\bullet$ is an exact complex.

Example 7.2.1 (Quasi-isomorphisms are not isomorphisms)**.** In $\mathsf{Ch}(\mathsf{Ab})$, consider the morphism of complexes

$$\begin{array}{ccccccccccc}
\cdots & \longrightarrow & 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\cdot 2} & \mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \pi} & & \downarrow & & \\
\cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{Z}/_{2\mathbb{Z}} & \longrightarrow & 0 & \longrightarrow & \cdots
\end{array}$$

where π is the obvious projection. Note that this is a quasi-isomorphism. However, it is not an isomorphism because there are no (non-trivial) group homomorphisms $\mathbb{Z}/_{2\mathbb{Z}} \to \mathbb{Z}$.

Lemma 7.2.1. Let A be an additive category and let $F : \mathsf{Ch}(\mathsf{A}) \to \mathsf{D}$ be an additive functor such that, for every quasi-isomorphism $\rho^\bullet$, $F(\rho^\bullet)$ is an isomorphism. Let $M^\bullet$ be an exact complex in $\mathsf{Ch}(\mathsf{A})$.

- The complex $F(M^\bullet)$ is a zero object in D.
- Let $\alpha^\bullet : L^\bullet \to N^\bullet$ be a morphism of complexes in $\mathsf{Ch}(\mathsf{A})$ that factors through an exact complex

$$\begin{array}{ccc}
L^\bullet & \xrightarrow{\alpha^\bullet} & N^\bullet \\
 & \searrow \quad \nearrow & \\
 & M^\bullet &
\end{array}$$

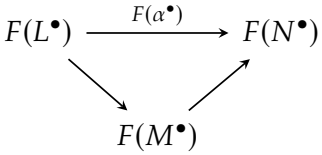
  Then $F(\alpha^\bullet)$ is the zero morphism.

Proof. Since $M^\bullet$ is exact, the zero morphism $M^\bullet \to M^\bullet$ is a quasi-isomorphism. Therefore we have

$$F(M^\bullet) \xrightarrow{F(0)} F(M^\bullet) \xrightarrow{F(0)^{-1}} F(M^\bullet), \qquad \mathbb{1}_{F(M^\bullet)}$$

Since F is additive $F(0) = 0$, making $\mathbb{1}_{F(M^\bullet)} = 0$.

For the other part, applying F to the diagram we get



Therefore since $F(M^\bullet)$ is a zero object in D we have that $F(\alpha^\bullet) = 0$. □

7.3 Homotopy

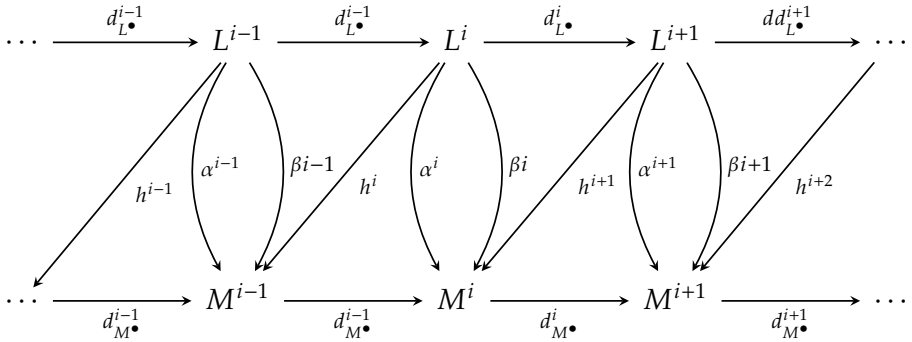
Definition 7.3.1. A **homotopy** h between two morphisms of complexes $\alpha^\bullet, \beta^\bullet : L^\bullet \to M^\bullet$ is a collection of morphisms

$$h^i : L^i \to M^{i-1}$$

such that

$$\forall i \in \mathbb{Z} \quad \beta^i - \alpha^i = d^{i-1}_{M^\bullet} \circ h^i + h^{i+1} \circ d^i_{L^\bullet}$$

Pictorially, we have the following **non-commutative** diagram



If there is an homotopy between $\alpha^\bullet$ and $\beta^\bullet$ we say that $\alpha^\bullet$ is **homotopic** to $\beta^\bullet$ and denote it by $\alpha^\bullet \sim \beta^\bullet$.

Definition 7.3.2. A morphism $\alpha^\bullet : L^\bullet \to M^\bullet$ is a **homotopy equivalence** if there exists a morphism $\beta^\bullet : M^\bullet \to L^\bullet$ such that $\alpha^\bullet \circ \beta^\bullet \sim \mathbb{1}_{M^\bullet}$ and $\beta^\bullet \circ \alpha^\bullet \sim \mathbb{1}_{N^\bullet}$. In this case, the complexes $L^\bullet$ and $M^\bullet$ are said to be **homotopy equivalent**.

Proposition 7.3.1. If $\alpha^\bullet, \beta^\bullet : L^\bullet \to M^\bullet$ are homotopic morphisms of complexes then $\alpha^\bullet, \beta^\bullet$ induce the same morphisms on cohomology $\mathrm{H}^\bullet(L^\bullet) \to \mathrm{H}^\bullet(M^\bullet)$.

Proof. Let $\overline{\ell} \in \mathrm{H}^i(L^\bullet)$. We have that $\overline{\ell}$ is represented by an element $\ell \in \ker d^i_{L^\bullet}$, and its images in $\mathrm{H}^i(M^\bullet)$ under α^*, β^* are represented by $\alpha^i(\ell), \beta^i(\ell)$.

Since $\alpha^\bullet, \beta^\bullet$ are homotopic there exist morphisms $h^i : L^i \to M^{i-1}$ such that

$$\beta^i(\ell) - \alpha^i(\ell) = d^{i-1}_{M^\bullet}(h^i(\ell)) + h^{i+1}(\cancelto{0}{d^i_{L^\bullet}(\ell)})$$

Therefore

$$\beta^i(\ell) - \alpha^i(\ell) \in \operatorname{im} d^i_{M^\bullet}$$

making $\beta^i(\ell) - \alpha^i(\ell)$ vanish in $\mathrm{H}^i(M^\bullet)$ as needed. □

Remark 7.3.1. If $\alpha^\bullet$ is homotopic to the identity by Corollary 7.1.1 the mapping cone of $\alpha^\bullet$ must be exact.

Corollary 7.3.1. Homotopy equivalent complexes have isomorphic cohomology.

Proof. If $\alpha^\bullet : L^\bullet \to M^\bullet$ and $\beta^\bullet : M^\bullet \to L^\bullet$ are morphisms of complexes such that $\beta^\bullet \circ \alpha^\bullet$ and $\alpha^\bullet \circ \beta^\bullet$ are homotopic to the identity then by Proposition 7.3.1 they induce inverse morphisms in cohomology. □

Lemma 7.3.1. Let $F : \mathsf{A} \to \mathsf{B}$ be an additive functor.

- If $\alpha^\bullet \sim \beta^\bullet$ are homotopic in $\mathsf{Ch}(\mathsf{A})$ then $\mathsf{Ch}(F)(\alpha^\bullet) \sim \mathsf{Ch}(F)(\beta^\bullet)$ in $\mathsf{Ch}(\mathsf{B})$.
- If $L^\bullet, M^\bullet$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{A})$ then $\mathsf{Ch}(F)(L^\bullet)$ and $\mathsf{Ch}(F)(M^\bullet)$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{B})$.

Proof. Note that the second assertion follows directly from the first.

To see the first, if h is a homotopy between $\alpha^\bullet, \beta^\bullet$ then

$$\forall i \in \mathbb{Z} \quad \beta^i - \alpha^i = d^{i-1}_{M^\bullet} \circ h^i + h^{i+1} \circ d^i_{L^\bullet}$$

Since F is additive

$$\forall i \in \mathbb{Z} \quad F(\beta^i) - F(\alpha^i) = F(d^{i-1}_{M^\bullet}) \circ F(h^i) + F(h^{i+1}) \circ F(d^i_{L^\bullet})$$

showing that the collection $\{F(h^i)\}_{i\in\mathbb{Z}}$ produces a homotopy between $F(\alpha^\bullet)$ and $F(\beta^\bullet)$. □

The following is direct from the previous statements.

Theorem 7.3.1. Let $F : \mathsf{A} \to \mathsf{B}$ be an additive between two abelian categories A, B. If $L^\bullet, M^\bullet$ are homotopy equivalent complexes in $\mathsf{Ch}(\mathsf{A})$ then the cohomology complexes

$$\mathrm{H}^\bullet(\mathsf{Ch}(F)(L^\bullet)) \cong \mathrm{H}^\bullet(\mathsf{Ch}(F)(M^\bullet))$$

are isomorphic.

Chapter 8

Injective and projective objects

8.1 Injective and projective objects

Definition 8.1.1.

- An object $I \in \mathsf{A}$ is **injective** if all the diagrams of the form

$$\begin{array}{ccc} X & \longrightarrow & I \\ \downarrow & & \\ Z & & \end{array}$$

  where $X \rightarrowtail Z$ is a monomorphism admit an extension

$$\begin{array}{ccc} X & \longrightarrow & I \\ \downarrow & \nearrow & \\ Z & & \end{array}$$

- An object $P \in \mathsf{A}$ is **projective** if all the diagrams of the form

$$\begin{array}{ccc} & & X \\ & & \downarrow \\ P & \longrightarrow & Z \end{array}$$

  where $X \twoheadrightarrow Z$ is an epimorphism admit an extension

$$\begin{array}{ccc} & \nearrow & X \\ & & \downarrow \\ P & \longrightarrow & Z \end{array}$$

Proposition 8.1.1.

- An object $I \in \mathsf{A}$ is injective if and only if the functor $\mathrm{Hom}_{\mathsf{A}}(-, I)$ is exact.
- An object $P \in \mathsf{A}$ is projective if and only if the functor $\mathrm{Hom}_{\mathsf{A}}(P, -)$ is exact.

Proof. By Example 5.3.1 we only need to check right exactness.

We will do the injective version this time:

$\Longrightarrow$ Suppose $I \in A$ is injective and consider a short exact sequence

$$0 \longrightarrow X \xrightarrow{f} Y \xrightarrow{g} Z \longrightarrow 0$$

Using Lemma 5.3.1, we want to see that

$$0 \longrightarrow \mathrm{Hom}(Z, I) \xrightarrow{g^*} \mathrm{Hom}(Y, I) \xrightarrow{f^*} \mathrm{Hom}(X, I)$$

is exact.
Since by Example 5.3.1 this sequence is exact at $\mathrm{Hom}(Z, I)$ and $\mathrm{Hom}(Y, I)$, we only need to check exactness at $\mathrm{Hom}(X, I)$, i.e: we want to see that f^* is surjective.
Consider a morphism $\alpha : X \to I$. Since f is a monomorphism and I is injective there exists a morphism $\beta : Y \to I$ such that $\beta \circ f = f^*(\beta) = \alpha$, making f^* surjective as wanted.

$\Longleftarrow$ Suppose $\mathrm{Hom}(-, I)$ is exact and consider a monomorphism $f : A \rightarrowtail B$ in A.
We have the following short exact sequence

$$0 \longrightarrow A \xrightarrow{f} B \longrightarrow \operatorname{coker} f \longrightarrow 0$$

Applying $\mathrm{Hom}(-, I)$ to this sequence we get

$$0 \longrightarrow \mathrm{Hom}(\operatorname{coker} f, I) \longrightarrow \mathrm{Hom}(B, I) \xrightarrow{f^*} \mathrm{Hom}(A, I) \longrightarrow 0$$

which is exact since $\mathrm{Hom}(-, I)$ is exact. In particular we have that f^* is surjective.
Therefore if $\gamma : A \to I$ is any morphism there exists $\beta \in \mathrm{Hom}(B, I)$ such that $\gamma = \beta \circ f$, which is exactly the universal property of injective objects.

□

We haven't proven existence of injective or projective objects. This is because in general they may not exist.

Since we will be interesed in using them, we introduce the following types of categories.

Definition 8.1.2. Let A be an abelian category. We say that:

- A has **enough injectives** if for every object $A \in \mathsf{A}$ there exists an injective object $I \in \mathsf{A}$ and a monomorphism $A \rightarrowtail I$.
- A has **enough projectives** if for every object $A \in \mathsf{A}$ there exists a projective object $P \in \mathsf{A}$ and an epimorphism $P \twoheadrightarrow A$.

Proposition 8.1.2. Let $R : \mathsf{B} \to \mathsf{A}$ be an additive functor with a left adjoint $L : \mathsf{A} \to \mathsf{B}$ that is exact. If $I \in \mathsf{B}$ is an injective object then $R(I)$ is also an injective object.
Dually, if an additive functor $L : \mathsf{A} \to \mathsf{B}$ is left adjoint to an exact functor $R : \mathsf{B} \to \mathsf{A}$ and $P \in \mathsf{A}$ is projective then $L(P)$ is projective.

Proof. We want to show that $\mathrm{Hom}_{\mathsf{A}}(-, R(I))$ is exact. By Example 5.3.1 we only need to check right exactness.

Let

$$\tau : \mathrm{Hom}_{\mathsf{B}}(L(A), B) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{A}}(A, R(B))$$

Given a monomorphism $f : A \rightarrowtail A'$ in A we have

$$\begin{array}{ccc} \mathrm{Hom}_{\mathsf{B}}(L(A'), I) & \xrightarrow{Lf^*} & \mathrm{Hom}_{\mathsf{B}}(L(A), I) \\ \downarrow\sim & & \downarrow\sim \\ \mathrm{Hom}_{\mathsf{A}}(A', R(I)) & \xrightarrow{f^*} & \mathrm{Hom}_{\mathsf{A}}(A, R(I)) \end{array}$$

commutes by naturality of τ. Since L is exact and I is injective Lf^* is a monomorphism. Therefore f^* is a monomorphism, proving that $R(I)$ is injective. □

8.2 Resolutions and homotopy

Definition 8.2.1.

- An **injective resolution** of an object $A \in \mathsf{A}$ is a quasi-isomorphism $\iota(\mathsf{A}) \to Q^\bullet$, where $Q^\bullet$ is a complex in $\mathsf{Ch}^{\geq 0}(I)$.
- A **projective resolution** of an object $A \in \mathsf{A}$ is a quasi-isomorphism $P^\bullet \to \iota(\mathsf{A})$, where $P^\bullet$ is a complex in $\mathsf{Ch}^{\leq 0}(P)$.

We will refer to this resolutions using $Q^\bullet, P^\bullet$, *forgetting* about the quasi-isomorphism.

Lemma 8.2.1. Let A be an abelian category.

- If A has enough injectives then every object $A \in \mathsf{A}$ admits an injective resolution.
- If A has enough projectives then every object $A \in \mathsf{A}$ admits a projective resolution.

Proof. We will do the enough injectives case.

Since A has enough injectives there exists an injective object $Q^0 \in \mathsf{A}$ and a monomorphism $d^0 : A \rightarrowtail Q^0$. Then there exists an injective object $Q^1 \in \mathsf{A}$ an a monomorphism $d^1 : \operatorname{coker} d^0 \rightarrowtail Q^0$ and so on. Pictorially

$$\begin{array}{ccccccc} A & \overset{d^0}{\rightarrowtail} & Q^0 & & & & \\ & & \downarrow & & & & \\ & & \operatorname{coker} d^0 & \rightarrowtail & Q^1 & & \\ & & & & \downarrow & & \\ & & & & \operatorname{coker} d^1 & \rightarrowtail & Q^2 \\ & & & & & & \downarrow \\ & & & & & & \vdots \end{array}$$

Therefore following this diagram we get an injective resolution

$$Q^\bullet = \cdots \longrightarrow 0 \longrightarrow Q^0 \longrightarrow Q^1 \longrightarrow Q^2 \longrightarrow \cdots$$

□

We will assume that A is an abelian-cateogory with enough injectives.

By *flipping* the corresponding arrows, all the results may be adapted to the projective case.

Lemma 8.2.2. Let A be an object in a A, let $M^\bullet$ be an injective resolution of A and let $Q^\bullet$ be a complex in $\mathsf{Ch}^{\geq 0}(I)$. Let $\varphi : \mathrm{H}^0(M^\bullet) = A \to \mathrm{H}^0(Q^\bullet)$ be an arbitrary morphism. Then

- There exists a morphism of complexes $\alpha : M^\bullet \to Q^\bullet$ inducing φ at the level of H^0.
- Different morphisms satisfying the above are homotopy equivalent.

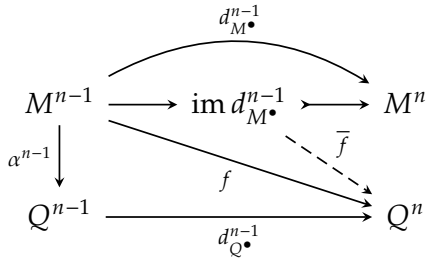
Proof. We have to define $\forall i \in \mathbb{Z} \quad \alpha^i : M^i \to Q^i$. Since for $i < 0$ we have $\alpha^i = 0$ we can truncate the complexes, obtaining

$$\begin{array}{ccccccccccccc}
0 & \longrightarrow & \mathrm{H}^0(M^\bullet) = A & \xrightarrow{\varepsilon} & M^0 & \xrightarrow{d^0_{M^\bullet}} & M^1 & \xrightarrow{d^1_{M^\bullet}} & M^2 & \xrightarrow{d^2_{M^\bullet}} & \cdots \\
 & & \downarrow{\scriptstyle \varphi} & & \vdots{\scriptstyle \alpha^1} & & \vdots{\scriptstyle \alpha^1} & & \vdots{\scriptstyle \alpha^2} & & \\
0 & \longrightarrow & \mathrm{H}^0(Q^\bullet) & \xrightarrow[\iota]{} & Q^0 & \xrightarrow[d^0_{Q^\bullet}]{} & Q^1 & \xrightarrow[d^1_{Q^\bullet}]{} & Q^2 & \xrightarrow[d^2_{Q^\bullet}]{} & \cdots
\end{array}$$

Note that the top complex is exact.

We will construct $\alpha^\bullet$ using induction:

- Case $n = 0$. Consider the following diagram
$$\begin{array}{ccc}
A & \xrightarrow{\varepsilon} & M^0 \\
\downarrow{\scriptstyle \varphi} & & \\
\mathrm{H}^0(Q^\bullet) & \xrightarrow[\iota]{} & Q^0
\end{array}$$
Since Q^0 is injective using its universal property with the morphism $\iota \circ \varphi$ we obtain morphism $\alpha^0 : M^0 \to Q^0$ making the diagram commute.
- Inductive step. Suppose we have already construced $\alpha^0, \ldots, \alpha^{n-1}$ such that
$$\forall k \in \{1, \ldots, n-1\} \quad \alpha^k \circ d^{k-1}_{M^\bullet} = d^k_{Q^\bullet} \circ \alpha^{k-1}$$
We want to construct $\alpha^n : M^n \to Q^n$.
Consider the diagram
$$\begin{array}{ccc}
M^{n-1} & \xrightarrow{d^{n-1}_{M^\bullet}} & M^n \\
{\scriptstyle \alpha^{n-1}}\downarrow & & \\
Q^{n-1} & \xrightarrow[d^{n-1}_{Q^\bullet}]{} & Q^n
\end{array}$$
Let $f = d^{n-1}_{Q^\bullet} \circ \alpha^{n-1} : M^{n-1} \to Q^n$. Note that
$$f \circ d^{n-2}_{M^\bullet} = d^{n-1}_{Q^\bullet} \circ \alpha^{n-1} \circ d^{n-2}_{M^\bullet} = \underbrace{d^{n-1}_{Q^\bullet} \circ d^{n-2}_{Q^\bullet}}_{0} \circ \alpha^{n-2} = 0$$
meaning that f vanishes on $\operatorname{im} d^{n-2}_{M^\bullet} = \ker d^{n-1}_{M^\bullet}$ by exactness of $M^\bullet$.
Therefore f induces a map $\overline{f} : \operatorname{im} d^{n-1}_{M^\bullet} \to Q^n$



Using that Q^n is injective we obtain a morphism $\alpha^n : M^n \to Q^n$.

Falta unicitat

□

Proposition 8.2.1. Any two injective resolutions (resp. projective) of an object A in an abelian category A are homotopy equivalent.

Corollary 8.2.1. Any two injective resolutions (resp. projective) of an object A in an abelian category A have the same cohomology.

Proposition 8.2.2. Let A_0, A_1 be two objects in an abelian category A. Let $Q_i^\bullet$ be an injective resolution of A_i, $i \in \{0,1\}$. Then every morphism $\varphi : A_0 \to A_1$ in A is induced by a morphism $\alpha^\bullet : I_0^\bullet \to I_1^\bullet$, uniquely determined up to homotopy. We call the morphism $\alpha^\bullet$ a *lifting* of φ.

Chapter 9

Derived functors

We follow [Wei94].

9.1 Delta functors

Definition 9.1.1. Let A, B be two abelian categories. A **cohomological δ-functor** is a collection of additive functors $T^n : \mathsf{A} \to \mathsf{B}$ for $n \in \mathbb{N}$ together with, for every short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

a morphism $\delta^n : T^n(C) \to T^{n+1}(A)$. We take the convention that $\forall n < 0 \quad T^n = 0$. Furthermore

1. For each short exact sequence as above there exists a long exact sequence

$$\begin{array}{l} \cdots \longrightarrow T^{n-1}(B) \longrightarrow T^{n-1}(C) \\ \qquad\qquad\qquad\qquad\qquad\qquad \xrightarrow{\delta^{n-1}} \\ T^n(A) \longrightarrow T^n(B) \longrightarrow T^n(C) \\ \qquad\qquad\qquad\qquad\qquad\qquad \xrightarrow{\delta^{n}} \\ T^{n+1}(A) \longrightarrow T^{n+1}(B) \longrightarrow \cdots \end{array}$$

   In particular T^0 is left exact.
2. For each morphism of short exact sequences

$$(0 \to A' \to B' \to C' \to 0) \mapsto (0 \to A \to B \to C \to 0)$$

   the δ^n give commutative diagrams

$$\begin{array}{ccc} T^n(C') & \xrightarrow{\delta^n} & T^{n+1}(A') \\ \downarrow & & \downarrow \\ T^n(C) & \xrightarrow{\delta^n} & T^{n+1}(A) \end{array}$$

Example 9.1.1. Cohomology is a cohomological δ-functor $\mathrm{H}^\bullet$ from $\mathsf{Ch}^{\geq 0}(\mathsf{A})$ to A.

Definition 9.1.2. A morphism $S \to T$ of cohomological δ-functors is a collection of natural transformations $S^n \to T^n$ that commute with δ.

Definition 9.1.3. A cohomological δ-functor is **universal** if given another cohomological δ-functor S and $f^0 : T^0 \to S^0$ there exists a unique morphism of δ-functors $T \to S$ extending f^0.

9.2 Derived functors

Definition 9.2.1. Let $F : \mathsf{A} \to \mathsf{B}$ be a left exact functor between two abelian categories A and B. If A has enough injectives we define the i-th **right derived functor** ($i \geq 0$) as

$$\forall A \in \mathsf{A} \quad R^iF(A) := \mathrm{H}^i(F(I^\bullet))$$

where $I^\bullet$ is an injective resolution of A.
Note that this is well-defined by Corollary 8.2.1.

Remark 9.2.1. Since

$$0 \longrightarrow F(A) \longrightarrow F(I^0) \longrightarrow F(I^1)$$

is exact we always have $R^0F(A) \cong F(A)$.

Lemma 9.2.1. Let $F : \mathsf{A} \to \mathsf{B}$ be a left exact functor between two abelian categories A and B. For every $i \in \mathbb{N}$ the right derived functor R^iF is an additive functor from A to B.

Theorem 9.2.1. Let $F : \mathsf{A} \to \mathsf{B}$ be a left exact functor between two abelian categories A and B. The right derived functors $R^\bullet F$ form a cohomological δ-functor.

Theorem 9.2.2. Let $F : \mathsf{A} \to \mathsf{B}$ be a left exact functor between two abelian categories A and B. Suppose that A has enough injectives. Then for any object $A \in \mathsf{A}$ the right derived functors $R^\bullet F$ form an universal δ-functor.

Definition 9.2.2. Let $F : \mathsf{A} \to \mathsf{B}$ be a left exact functor between two abelian categories A and B. An object $Q \in \mathsf{A}$ is called F-**acyclic** if

$$\forall i > 0 \quad R^iF(Q) = 0$$

Example 9.2.1. Every injective object $I \in \mathsf{A}$ is F-acyclic. Indeed, we can compute $R^iF(I)$ using the resolution $0 \to I \xrightarrow{\mathbb{1}} I \to 0$ which implies that for all $i > 0 \quad R^iF(I) = 0$.

Theorem 9.2.3. Let $F : \mathsf{A} \to \mathsf{B}$ be a left exact functor between two abelian categories A and B. If

$$0 \longrightarrow A \longrightarrow Q^0 \longrightarrow Q^1 \longrightarrow \cdots \longrightarrow Q^m \longrightarrow M \longrightarrow 0$$

is exact with Q^i F-acyclic then for all $i \geq m+2 \quad R^iF(A) \cong R^{i-m-1}F(A)$ and $R^{m+1}F(A)$ is the cokernel of $F(Q^m) \to F(M)$.
In particular if $A[0] \to Q^\bullet$ is an F-acyclic resolution of A then

$$R^iF(A) \cong \mathrm{H}^i(F(Q^\bullet))$$

Proof. We will prove this using induction.

In the case $m = 0$ we have the short exact sequence

$$0 \longrightarrow A \longrightarrow Q^0 \longrightarrow M \longrightarrow 0$$

Using the long exact sequence in cohomology we can extend this sequence to

$$\begin{array}{l} 0 \longrightarrow F(A) \longrightarrow F(Q^0) \longrightarrow F(M) \xrightarrow{\delta^0} \\ R^1F(A) \longrightarrow R^1F(Q^0) \longrightarrow R^1F(M) \xrightarrow{\delta^1} \\ R^2F(A) \longrightarrow R^2F(Q^0) \longrightarrow \cdots \end{array}$$

Since Q^0 is F-acyclic we have that $\forall i > 0 \quad R^iF(Q^0) = 0$. Therefore we have

1. For $i = 1$ the segment
$$F(Q^0) \longrightarrow F(M) \xrightarrow{\delta^0} R^1F(A) \longrightarrow 0$$
By exactness at $R^1F(A)$ we have
$$R^1F(A) \cong {}^{F(M)}\big/_{\operatorname{im} F(Q^0)} \cong \operatorname{coker}\left(F(Q^0) \to F(M)\right)$$

2. For $i \geq 2$ the segment
$$0 \longrightarrow R^{i-1}F(A) \xrightarrow{\delta^{i-1}} R^iF(A) \longrightarrow 0$$
By exactness we have that $R^{i-1}F(A) \cong R^iF(A)$.

This finishes the proof of the base case.

Let's suppose now that we have proven the result for $m-1$ and check it for m. Consider the exact sequence

$$0 \longrightarrow A \longrightarrow Q^0 \longrightarrow Q^1 \longrightarrow \cdots \longrightarrow Q^m \longrightarrow M \longrightarrow 0$$

Denote by $Z_0 = A$, $Z_{k+1} = \operatorname{coker}\left(Z_k \to Q^k\right)$ and $Z_{m+1} = M$. We can split the original sequence into $m+1$ short exact sequences

$$0 \to A \to Q^0 \to Z_1 \to 0 \tag{9.2.1}$$
$$\forall k \in \{1, \ldots, m-1\} \quad 0 \to Z_k \to Q^k \to Z_{k+1} \to 0 \tag{9.2.2}$$
$$0 \to Z_m \to Q^m \to M \to 0 \tag{9.2.3}$$

We now apply the inductive hypothesis to this sequences:

1. Using (9.2.1) for all $i \geq 2$ we get
$$R^iF(A) \cong R^{i-1}F(Z_1)$$

2. Using (9.2.2) for all $i \geq m+2$ we have
$$R^iF(A) \cong R^{i-(m+1)}F(Z_{m+1}) \cong R^{i-m-1}F(M)$$

3. If $i = m+1$ we get
$$R^{m+1}F(A) \cong R^1F(Z_m) \cong \operatorname{coker}\left(F(Q^m) \to F(M)\right)$$

This concludes the proof of the first part of the theorem.

To see the second part, let

$$0 \longrightarrow A \longrightarrow Q^0 \longrightarrow Q^1 \longrightarrow \cdots$$

be a F-acyclic resolution. We want to see that $R^iF(A) \cong H^i(F(Q^\bullet))$.

Fix $i \geq 1$ and truncate the sequence to form a exact sequence

$$0 \longrightarrow A \longrightarrow Q^0 \longrightarrow \cdots \longrightarrow Q^{i-1} \longrightarrow Z_i \longrightarrow 0$$

where

$$Z_i \coloneqq \ker\left(Q^i \to Q^{i+1}\right) \cong \operatorname{im}\left(Q^{i-1} \to Q^i\right)$$

Since F is left-exact applying it to the short exact sequence $0 \to Z_i \to Q^i \to Q^{i+1}$ we get the short exact sequence

$$0 \longrightarrow F(Z_i) \longrightarrow F(Q^i) \longrightarrow F(Q^{i+1})$$

Meaning that

$$F(Z_i) \cong \ker\left(F(Q^i) \to F(Q^{i+1})\right)$$

Using the result above (for $m = i - 1$) we have

$$\begin{aligned} R^iF(A) &\cong \operatorname{coker}\left(F(Q^{i-1}) \to F(Z_i)\right) \\ &\cong {F(Z_i)}\Big/{\operatorname{im}\left(F(Q^{i-1}) \to F(Z_i)\right)} \\ &\cong {\ker\left(F(Q^i) \to F(Q^{i+1})\right)}\Big/{\operatorname{im}\left(F(Q^{i-1}) \to F(Z_i)\right)} \\ &\cong \mathrm{H}^i(F(Q^\bullet)) \end{aligned}$$

□

PART III

Sheaves

Chapter 10

Sheaves

10.1 Presheaves

Definition 10.1.1 (Presheaf). Let X be a topological space and let $\mathsf{Op}(X)$ be its posetal category of open sets. Let C be a category. The category of C-valued **presheaves** over X is the functor category

$$\mathsf{PSh}_X(\mathsf{C}) := \mathsf{Fun}(\mathsf{Op}(X)^{\mathrm{op}}, \mathsf{C})$$

The elements of this category are functors $\mathcal{F} : \mathsf{Op}(X)^{\mathrm{op}} \to \mathsf{C}$ and morphisms are natural transformations $\phi : \mathcal{F} \to \mathcal{G}$.

Less abstractly, a C-valued presheaf $\mathcal{F}$ is the collection of data:

- For each open set $U \subseteq X$ an object $\mathcal{F}(U) \in \mathsf{C}$ called *sections of $\mathcal{F}$ over U*.
- For every inclusion of open sets $V \subseteq U$ a morphism $\mathrm{res}_{U,V} : \mathcal{F}(U) \to \mathcal{G}(U)$ in C called *restriction* such that
  - For every open set $U \subseteq X$, $\mathrm{res}_{U,U} = \mathbb{1}_U$.
  - For every inclusion of open sets $W \subseteq V \subseteq U$, $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

A morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ is a collection of morphisms

$$\{\phi(U) : \mathcal{F}(U) \to \mathcal{G}(U)\}_{U \in \mathsf{Op}(X)}$$

in C making the following diagram commute

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U) \\ {\scriptstyle \mathrm{res}_{U,V}}\downarrow & & \downarrow{\scriptstyle \mathrm{res}_{U,V}} \\ \mathcal{F}(V) & \xrightarrow[\phi(V)]{} & \mathcal{G}(V) \end{array}$$

Notation. When the context is clear, we will drop the subscript of $\mathsf{PSh}_X(\mathsf{C})$ and denote the category of C-valued presheaves over X as $\mathsf{PSh}(\mathsf{C})$.

Since $\mathsf{PSh}_X(\mathsf{C})$ is a functor category, if C has limits or colimits of a certain shape then $\mathsf{PSh}_X(\mathsf{C})$ also has them and they are computed pointwise. This means that limits and colimits can be computed *open set by open set*.

Proposition 10.1.1. Let X be a topological space. Let $\mathcal{I} \to \mathsf{PSh}_X(\mathsf{C}), i \mapsto \mathcal{F}_i$ be a diagram. If C has limits (resp. colimits) of shape $\mathcal{I}$ then $\lim_i \mathcal{F}_i$ (resp. $\operatorname{colim}_i \mathcal{F}_i$) exists in $\mathsf{PSh}_X(\mathsf{C})$. Moreover for every open set $U \subseteq X$

$$\left(\lim_i \mathcal{F}_i\right)(U) = \lim_i \left(\mathcal{F}_i(U)\right) \qquad \left(\operatorname*{colim}_i \mathcal{F}_i\right)(U) = \operatorname*{colim}_i \left(\mathcal{F}_i(U)\right)$$

In particular if C is an abelian category then kernels and cokernels exist in $\mathsf{PSh}_X(\mathsf{C})$ and are computed *open set by open set*, making $\mathsf{PSh}_X(\mathsf{C})$ an abelian category.

Example 10.1.1. Consider $X = \{*\}$ the one-point topological space. The category $\mathsf{Op}(X)$ has

$$\mathrm{Ob}(\mathsf{Op}(X)) = \{\varnothing, X\} \qquad \mathrm{Hom}(\mathsf{Op}(X)) = \{\varnothing \hookrightarrow X\}$$

Therefore every $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ consists of two objects $\mathcal{F}(\varnothing), \mathcal{F}(X) \in \mathsf{C}$ and a restriction morphism $\rho : \mathcal{F}(X) \to \mathcal{F}(\varnothing)$.
This makes $\mathsf{PSh}_{\{*\}}(\mathsf{C}) \cong \mathsf{Arr}(\mathsf{C})$.

10.1.1 Presheaves and continuous maps

Definition 10.1.2 (Direct image presheaf)**.** Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let $\mathcal{F}$ be a C-valued presheaf over X.
The **direct image presheaf** of $\mathcal{F}$ via f, $f_*\mathcal{F} \in \mathsf{PSh}_Y(\mathsf{C})$, is defined as

$$\forall V \in \mathsf{Op}(Y) \quad f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}V).$$

Given open subsets $W \subseteq V \subseteq Y$, the restriction maps $\mathrm{res}_{V,W}$ comes directly from the diagram

$$\begin{array}{ccc} f_*\mathcal{F}(V) & = & \mathcal{F}(f^{-1}V) \\ \vdots & & \downarrow{\scriptstyle \mathrm{res}_{f^{-1}V, f^{-1}W}} \\ f_*\mathcal{F}(W) & = & \mathcal{F}(f^{-1}W) \end{array}$$

Notice that this construction defines a functor $f_* : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{PSh}_Y(\mathsf{C})$.

Example 10.1.2 (Global section)**.** Let X be a topological space. Consider the one-point topological space $\{*\}$ and the unique continuous map $f : X \to \{*\}$. Let $\mathcal{F}$ be a C-valued presheaf over X.
Consider the direct image presheaf of $\mathcal{F}$ via f. Since this is a C-valued presheaf over $\{*\}$ by Example 10.1.1 we have that

$$f_*\mathcal{F} \cong (\mathcal{F}(\{*\}) \to \mathcal{F}(\varnothing))$$

Using the definition of the direct image presheaf we have

$$f_*\mathcal{F}(\varnothing) = \mathcal{F}(f^{-1}(\varnothing)) = \mathcal{F}(\varnothing)$$
$$f_*\mathcal{F}(\{*\}) = \mathcal{F}(f^{-1}(\{*\})) = \mathcal{F}(X)$$

Therefore we define the **global sections** of $\mathcal{F}$ as

$$\Gamma(X, \mathcal{F}) \coloneqq f_*\mathcal{F} \cong \mathcal{F}(X) \to \mathcal{F}(\varnothing)$$

This defines a functor $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Arr}(\mathsf{C})$.
In particular, if $\mathcal{F}(\varnothing)$ is a terminal object in C we can identify $\mathcal{F}(X) \to \mathcal{F}(\varnothing)$ with $\mathcal{F}(X)$, getting a functor $\Gamma(X, -) : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$.

Example 10.1.3 (Skyscraper presheaf). Let X be a topological space and $p \in X$. Consider the one-point topological space $\{*\}$ and the continuous map $\iota_p : \{*\} \to X, * \mapsto p$. Consider a morphism $\rho : S \to T$ in C, which by Example 10.1.1 corresponds to a presheaf $\mathcal{R} \in \mathsf{PSh}_{\{*\}}(\mathsf{C})$.
Consider the direct image presheaf of $\mathcal{R}$ via ι_p:

$$\forall U \in \mathsf{Op}(X) \quad \iota_{p,*}\mathcal{R}(U) = \mathcal{R}(\iota_p^{-1}(U)) = \begin{cases} \mathcal{R}(\{*\}) & p \in U \\ \mathcal{R}(\varnothing) & p \notin U \end{cases} = \begin{cases} S & p \in U \\ T & p \notin U \end{cases}$$

Note that if $V \subseteq U$ the restriction map is given by

$$\mathrm{res}_{U,V} = \begin{cases} \mathbb{1}_S & p \in V \\ \mathbb{1}_T & p \notin U \\ \rho & p \in U \wedge p \notin V \end{cases}$$

We define the **skyscraper presheaf** supported at ρ, $\iota_{p,*}\rho \in \mathsf{PSh}_X(\mathsf{C})$, as

$$\iota_{p,*}\underline{\rho}_{\mathrm{pre}} \coloneqq \iota_{p,*}\mathcal{R} \tag{10.1.1}$$

This defines a functor $\iota_{p,*} : \mathsf{Arr}(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$.
In particular if $T = \mathcal{R}(\varnothing)$ is a terminal object in C we can identify $\rho : S \to T$ with S, getting a functor $\iota_{p,*} : \mathsf{C} \to \mathsf{PSh}_X(\mathsf{C})$. In this situation we write $\iota_{p,*}\underline{S}_{\mathrm{pre}}$.

For sufficiently nice categories (those with direct limits), we have a *dual* construction.

Definition 10.1.3 (Inverse image presheaf). Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let C be a category with direct limits. Let $\mathcal{G}$ be a C-valued presheaf over Y.
The **inverse image presheaf** of $\mathcal{G}$, $f_{\mathrm{pre}}^{-1}\mathcal{G} \in \mathsf{PSh}_X(\mathsf{C})$, of $\mathcal{G}$ via f is the presheaf

$$\forall U \in \mathsf{Op}(X) \quad \left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)(U) \coloneqq \varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(Y)}} \mathcal{G}(V)$$

Notice that this defines a functor $f_{\mathrm{pre}}^{-1} : \mathsf{PSh}_Y(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$.

Remark 10.1.1. For $\mathsf{C} = \mathsf{Set}$ we can describe $f^{-1}\mathcal{G}(U)$ as

$$\left(f_{\mathrm{pre}}^{-1}\mathcal{G}\right)(U) = \{(s, V) \mid f(U) \subseteq V \in \mathsf{Op}(Y) \wedge s \in \mathcal{G}(V)\}$$

modulo the relation

$$(s, V) \sim (s', V') \iff \exists W \in \mathsf{Op}(Y) \mid W \subseteq V \cap V' \wedge f(U) \subseteq W \wedge s|_W = s'|_W$$

In the following examples, we assume that the category C has direct limits.

Example 10.1.4 (Constant presheaf). Let X be a topological space. Consider the one-point topological space $\{*\}$. Consider a morphism $\rho : S \to T$ in C. Note that by Example 10.1.1 this corresponds to a presheaf $\mathcal{R} \in \mathsf{PSh}_{\{*\}}(\mathsf{C})$.
Consider the inverse image presheaf of $\mathcal{R}$ via the map $f : X \to \{*\}$.

– If $U \in \mathsf{Op}(X)$ is not the empty set we have that

$$(f_{\mathrm{pre}}^{-1}\mathcal{R})(U) = \varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(\{*\})}} \mathcal{R}(V) = \varinjlim_{\substack{V \ni * \\ V \in \mathsf{Op}(\{*\})}} \mathcal{R}(V) = \mathcal{R}(\{*\}) = S$$

– If $U = \varnothing$ then we take the limit over the diagram $S \xrightarrow{\rho} T$, getting

$$(f_{\mathrm{pre}}^{-1}\mathcal{R})(\varnothing) = T$$

Therefore

$$\forall U \in \mathsf{Op}(X) \quad \left(f_{\mathrm{pre}}^{-1}\mathcal{R}\right)(U) = \begin{cases} S & U \neq \varnothing \\ T & U = \varnothing \end{cases}$$

Note that for $V \subseteq U$ open subsets of X the restriction maps are given by

$$\mathrm{res}_{U,V} = \begin{cases} \mathbb{1}_S & V \neq \varnothing \\ \mathbb{1}_T & U = \varnothing \\ \rho & V = \varnothing \wedge U \neq \varnothing \end{cases}$$

Therefore we can define the **constant presheaf** supported as ρ as

$$\underline{\rho}_{\mathrm{pre}}(U) \coloneqq f_{\mathrm{pre}}^{-1}\mathcal{R}$$

This defines a functor $\underline{\cdot}_{\mathrm{pre}} : \mathsf{Arr}(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$
Note that if $T = \mathcal{R}(\varnothing)$ is a terminal object in C we can identify $\rho : S \to T$ with S, getting a functor $\underline{\cdot}_{\mathrm{pre}} : \mathsf{C} \to \mathsf{PSh}_X(\mathsf{C})$. In this situation we write $\underline{S}_{\mathrm{pre}}$.

Example 10.1.5 (Stalk)**.** Let X be a topological space and $p \in X$. Consider the one-point topological space $\{*\}$ and the inclusion $f : \{*\} \hookrightarrow X, * \mapsto p$. Let $\mathcal{F}$ be a C-valued presheaf over X.
Consider the inverse image presheaf of $\mathcal{F}$ via f. Since this is a C-valued presheaf over $\{*\}$ by Example 10.1.1

$$f_{\mathrm{pre}}^{-1}\mathcal{F} \cong \left(f_{\mathrm{pre}}^{-1}\mathcal{F}(\{*\}) \to f_{\mathrm{pre}}^{-1}\mathcal{F}(\varnothing)\right)$$

Using the definition of the inverse image presheaf we have that

$$f_{\mathrm{pre}}^{-1}\mathcal{F}(\{*\}) = \varinjlim_{\substack{U \supseteq f(\{*\}) \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U) = \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U)$$

$$f_{\mathrm{pre}}^{-1}\mathcal{F}(\varnothing) = \varinjlim_{\substack{U \supseteq f(\varnothing) \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U) = \varinjlim_{U \in \mathsf{Op}(X)} \mathcal{F}(U) = \mathcal{F}(\varnothing)$$

Therefore

$$f_{\mathrm{pre}}^{-1}\mathcal{F} \cong \left(\varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U) \to \mathcal{F}(\varnothing) \right)$$

We define the **stalk** of $\mathcal{F}$ at p as

$$\mathcal{F}_p := f_{\text{pre}}^{-1}\mathcal{F}$$

This defines a functor $-_p : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Arr}(\mathsf{C})$.
In particular if $\mathcal{F}(\varnothing)$ is a terminal object in C we can identify

$$\left(\varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U) \to \mathcal{F}(\varnothing) \right) \cong \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathcal{F}(U)$$

getting a functor $-_p : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{C}$.

Lemma 10.1.1. Let $f : X \to Y$ be a continuous map between two topological space X, Y and $p \in X$. Let C be a category with direct limits. Let $\mathcal{G}$ be a C-valued presheaf over Y. Then $(f_{\text{pre}}^{-1}\mathcal{G})_p = \mathcal{G}_{f(p)}$.

Proof.

$$\begin{aligned}
(f_{\text{pre}}^{-1}\mathcal{G})_p &= \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} f_{\text{pre}}^{-1}\mathcal{G}(U) \to \varinjlim_{U \in \mathsf{Op}(X)} f_{\text{pre}}^{-1}\mathcal{G}(U) \\
&= \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \left(\varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(Y)}} \mathcal{G}(V) \right) \to \varinjlim_{U \in \mathsf{Op}(X)} \left(\varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(Y)}} \mathcal{G}(V) \right) \\
&= \varinjlim_{\substack{V \ni f(p) \\ V \in \mathsf{Op}(Y)}} \mathcal{G}(V) \to \varinjlim_{V \in \mathsf{Op}(Y)} \mathcal{G}(V) \\
&= \mathcal{G}_{f(p)}
\end{aligned}$$

□

Theorem 10.1.1. Let $f : X \to Y$ be a continuous function between two topological spaces X, Y. Let C be a category with direct limits. Then there exists an adjunction

$$f_{\text{pre}}^{-1} \dashv f_* : \mathsf{PSh}_X(\mathsf{C}) \underset{f_*}{\overset{f_{\text{pre}}^{-1}}{\leftrightarrows}} \mathsf{PSh}_Y(\mathsf{C})$$

Proof. We want to construct natural transformations

$$\begin{aligned}
\eta &: \mathbb{1}_{\mathsf{PSh}_Y(\mathsf{C})} \implies f_* \circ f_{\text{pre}}^{-1} \\
\varepsilon &: f_{\text{pre}}^{-1} \circ f_* \implies \mathbb{1}_{\mathsf{PSh}_X(\mathsf{C})}
\end{aligned}$$

making for all $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and $\mathcal{G} \in \mathsf{PSh}_Y(\mathsf{C})$ the following diagrams commute

$$\begin{array}{ccc}
f_{\text{pre}}^{-1}(\mathcal{G}) & \xrightarrow{\mathbb{1}_{\mathcal{G}}} & f_{\text{pre}}^{-1}(\mathcal{G}) \\
{\scriptstyle f_{\text{pre}}^{-1}(\eta_{\mathcal{G}})}\downarrow & \nearrow_{\varepsilon_{f_{\text{pre}}^{-1}(\mathcal{G})}} & \\
\left(f_{\text{pre}}^{-1} \circ f_* \circ f_{\text{pre}}^{-1}\right)(\mathcal{G}) & &
\end{array}
\qquad
\begin{array}{ccc}
f_*(\mathcal{G}) & \xrightarrow{\mathbb{1}_{\mathcal{F}}} & f_*(\mathcal{G}) \\
{\scriptstyle \eta_{f_*(\mathcal{F})}}\downarrow & \nearrow_{f_*(\varepsilon_{\mathcal{F}})} & \\
\left(f_* \circ f_{\text{pre}}^{-1} \circ f_*\right)(\mathcal{F}) & &
\end{array}$$

Note that, by definition of the direct image presheaf and inverse image presheaf we have, for $U \in \mathsf{Op}(X)$ and $V \in \mathsf{Op}(Y)$

$$\left(f_{\mathrm{pre}}^{-1} \circ f_*\right)(\mathcal{F})(U) = \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Op}(Y)}} (f_*\mathcal{F})(W) = \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Op}(Y)}} \mathcal{F}(f^{-1}(W))$$

$$\left(f_* \circ f_{\mathrm{pre}}^{-1}\right)(\mathcal{G})(V) = (f_{\mathrm{pre}}^{-1}\mathcal{G})(f^{-1}(V)) = \varinjlim_{\substack{W \supseteq (f \circ f^{-1})(V) \\ W \in \mathsf{Op}(Y)}} \mathcal{G}(W)$$

Since we have $(f \circ f^{-1})(V) \subseteq V$, we can define $\eta_{\mathcal{G}}(V)$ as the canonical injection into the direct limit

$$\eta_{\mathcal{G}}(V) : \mathcal{G}(V) \to \varinjlim_{\substack{W \supseteq (f \circ f^{-1})(V) \\ W \in \mathsf{Op}(Y)}} \mathcal{G}(W)$$

On the other hand, since for every $W \subseteq f(U)$ open we have $f^{-1}(W) \subseteq (f^{-1} \circ f)(U) \subseteq U$ there exists a restriction morphism

$$\mathrm{res}_{f^{-1}(W),U} : \mathcal{F}(f^{-1}(W)) \to \mathcal{F}(U)$$

Using this maps we obtain a morphism

$$\varepsilon_{\mathcal{F}}(U) : \varinjlim_{\substack{W \supseteq f(U) \\ W \in \mathsf{Op}(Y)}} \mathcal{F}(f^{-1}(W)) \to \mathcal{F}(U)$$

□

In particular, we get the following particular cases of Theorem 10.1.1 for a category C with direct limits.

Corollary 10.1.1. Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued presheaf over X and let $\rho : S \to T$ be a morphism in C. Then

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{\rho}_{\mathrm{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{Arr}(\mathsf{C})}(\rho, \Gamma(X, \mathcal{F}))$$

In particular, if $\mathcal{F}(\varnothing)$ and T are terminal objects in C we get

$$\mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\underline{S}_{\mathrm{pre}}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathsf{C}}(S, \Gamma(X, \mathcal{F}))$$

Corollary 10.1.2. Let X be a topological space and $p \in X$. Let $\mathcal{F}$ be a C-valued presheaf over X and let $\rho : S \to T$ be a morphism in C. Then

$$\mathrm{Hom}_{\mathsf{Arr}(\mathsf{C})}(\mathcal{F}_p, \rho) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{\rho}_{\mathrm{pre}})$$

In particular, if $\mathcal{F}(\varnothing)$ and T are terminal objects in C we get

$$\mathrm{Hom}_{\mathsf{C}}(\mathcal{F}_p, S) \simeq \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota_{p,*}\underline{S}_{\mathrm{pre}})$$

10.2 Sheaves

Write motivation on why sheaves capture more *geometric* information

We now turn our attention to a particular class of presheaves. Let's first tackle the case where the evaluation category is Set.

Definition 10.2.1. Let X be a topological space. The category of **sheaves of sets** over X is the full subcategory $\mathsf{Sh}_X(\mathsf{Set})$ of $\mathsf{PSh}_X(\mathsf{Set})$ whose objects verify, for every open set $U \subseteq X$ and every open cover $\{U_i\}_{i\in\mathcal{I}}$ of U

SH.1) **Identity**: Given two sections $s, t \in \mathscr{F}(U)$ such that for every $i \in \mathcal{I}$ $\quad s|_{U_i} = t|_{U_i}$ we have that $s = t$.

SH.2) **Gluability**: Given a collection of sections $\{s_i \mid s_i \in \mathscr{F}(U_i)\}_{i\in\mathcal{I}}$ that agree on pairwise intersections $(\forall i, j \in \mathcal{I} \quad s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j})$ there exists a section $s \in \mathscr{F}(U)$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = s_i$.

Presheaves that only satisfy axiom SH.1 are called **separated presheaves**.

Notation. From now on, to distinguish a presheaf from a sheaf we will denote the former by $\mathcal{F}, \mathcal{G}, \mathcal{H}, \ldots$ and the latter by $\mathscr{F}, \mathscr{G}, \mathscr{H}, \ldots$

Let's try to make Definition 10.2.1 work on a more general evaluation category. To do this, the first thing we need to do is to write axioms SH.1 and SH.2 in terms of universal properties.

SH.1) Consider the morphism

$$\begin{aligned} e : \mathscr{F}(U) &\to \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \\ s &\mapsto \{s|_{U_i}\}_{i\in\mathcal{I}} \end{aligned}$$

given by the restriction maps res_{U,U_i}.
The condition SH.1 is equivalent to e being injective, which (in Set) means that e is a monomorphism.

SH.2) Consider the morphisms

$$p, q : \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathscr{F}(U_i \cap U_j)$$

where p takes the section on U_i and sends it to the intersection and q takes the section on U_j and sends it to the intersection.
The condition SH.2 is then equivalent to saying that $\mathscr{F}$ is the equalizer of the diagram consisting of p and q.

Therefore a presheaf of sets $\mathscr{F}$ is a sheaf of sets if for every open set U and open cover $\{U_i\}_{i\in\mathcal{I}}$ of U the following diagram is an equalizer sequence

$$\mathscr{F}(U) \longrightarrow \prod_{i\in\mathcal{I}} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j)\in\mathcal{I}\times\mathcal{I}} \mathscr{F}(U_i \cap U_j)$$

Let's try to go now to a more general category C. First of all, note that the equalizer sequence makes sense if products indexed by $\mathcal{I}$ and $\mathcal{I} \times \mathcal{I}$ exist in C. Furthermore, since our categories are locally small categories applying the functor $\mathrm{Hom}_{\mathsf{C}}(-,-) : \mathsf{C} \times \mathsf{C} \to \mathsf{Set}$ by the Yoneda lemma we have that the diagram is an equalizer if and only if for every object $C \in \mathsf{C}$ the diagram

$$\mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U)) \longrightarrow \prod_{i \in \mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U_i)) \rightrightarrows \prod_{(i,j) \in \mathcal{I} \times \mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U_i \cap U_j))$$

is an equalizer diagram in Set.

Definition 10.2.2 (Sheaf)**.** Let X be a topological space. Let C be a category with products indexed by $\mathcal{I}$ and $\mathcal{I} \times \mathcal{I}$.
The category of C-valued **sheaves** over X is the full subcategory $\mathsf{Sh}_X(\mathsf{C})$ of the category of C-valued presheaves over X whose objects verify that, for every open subset $U \subseteq X$ and every open cover $\{U_i\}_{i \in \mathcal{I}}$ the diagram

$$\mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U)) \longrightarrow \prod_{i \in \mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U_i)) \rightrightarrows \prod_{(i,j) \in \mathcal{I} \times \mathcal{I}} \mathrm{Hom}_{\mathsf{C}}(C, \mathscr{F}(U_i \cap U_j))$$

is an equalizer sequence in Set.

Remark 10.2.1. From now on, every time we work with C-valued sheaves we will tacitly assume that C has sufficient products so that Definition 10.2.2 makes sense.

Remark 10.2.2. Note that the empty set $\varnothing \in \mathsf{Op}(X)$ is covered via the *empty cover* $\{U_i\}_{i \in \varnothing}$. Therefore for any sheaf $\mathscr{F}$ we have that $\mathscr{F}(\varnothing)$ is an element of the empty product, i.e: $\mathscr{F}(\varnothing)$ is a terminal object in C.
Therefore following Example 10.1.1 we have that, if $\{*\}$ is the one-point topological space,

$$\mathsf{Sh}_*(\mathsf{C}) \cong \mathsf{C}$$

Remark 10.2.3. If C has small limits then the equalizer sequence in Definition 10.2.2 can be written directly in C.
In this case, a C-valued presheaf is a sheaf if for every open set $U \subseteq X$ and open cover $\{U_i\}_{i \in \mathcal{I}}$ the following diagram

$$\mathscr{F}(U) \longrightarrow \prod_{i \in \mathcal{I}} \mathscr{F}(U_i) \rightrightarrows \prod_{(i,j) \in \mathcal{I} \times \mathcal{I}} \mathscr{F}(U_i \cap U_j)$$

is an equalizer sequence in C.

10.2.1 Sheaves of Set-like categories

Let's consider the case where our category C comes with a functor $F : \mathsf{C} \to \mathsf{Set}$.

Here, for every C-valued presheaf $\mathcal{F}$ we have an associated *underlying presheaf of sets* $F\mathcal{F}$.

We want to know in which categories we can check the sheaf conditions in the underlying presheaf of sets.

We have the following diagram

$$\begin{array}{ccccc}
\mathscr{F}(U) & \longrightarrow & \prod_{i\in I}\mathscr{F}(U_i) & \rightrightarrows & \prod_{(i,j)\in I\times I}\mathscr{F}(U_i\cap U_j) \\
\downarrow & & \downarrow & & \downarrow \\
(F\mathscr{F})(U) & \longrightarrow & \prod_{i\in I}(F\mathscr{F})(U_i) & \rightrightarrows & \prod_{(i,j)\in I\times I}(F\mathscr{F})(U_i\cap U_j)
\end{array}$$

If we suppose that the top row is an equalizer sequence in C ($\mathscr{F}$ is a C-valued sheaf) then F needs to preserve limits to make the bottom row an equalizer sequence in Set ($F\mathscr{F}$ is a sheaf of sets).

Similarly, if we suppose that the bottom row is an equalizer sequence in Set ($F\mathscr{F}$ is a sheaf of sets) then F needs to reflect limits to make the top row an equalizer sequence in C ($\mathscr{F}$ is a C-valued sheaf).

Note that this does not happen in general for all concrete categories. For example the functor $F : \mathsf{Top} \to \mathsf{Set}$ sending every topological space to its underlying set does not reflect limits.

Indeed, consider $X = (\mathbb{R}\times\mathbb{R}, \mathcal{T})$ the real plane with the usual product topology and $Y = (\mathbb{R}\times\mathbb{R}, \mathcal{T}_d)$ the real plane with the discrete topology. Let $Z = (\mathbb{R}, \mathcal{T})$ be the real line with the usual topology.

Consider the diagram in Top

$$\begin{array}{ccc}
 & Y & \\
{\scriptstyle p_1}\swarrow & & \searrow{\scriptstyle p_2} \\
Z & & Z
\end{array}$$

Note that this is not a limit in Top. Indeed the diagram

$$\begin{array}{ccccc}
 & & X & & \\
 & {\scriptstyle \pi_1}\swarrow & & \searrow{\scriptstyle \pi_2} & \\
 & & Y & & \\
 & {\scriptstyle p_1}\swarrow & & \searrow{\scriptstyle p_2} & \\
Z & & & & Z
\end{array}$$

cannot be completed since there does not exist a continuous map $X \to Y$.

However, applying the forgetful functor to the diagram we get

$$\begin{array}{ccc}
 & \mathbb{R}^2 & \\
\swarrow & & \searrow \\
\mathbb{R} & & \mathbb{R}
\end{array}$$

which is a limit in Set (it is the diagram of a product).

For other categories such as Ban (the category of Banach spaces) the forgetful functor $F : \mathsf{Ban} \to \mathsf{Set}$ does not preserve limits.

Luckily *algebraic categories* such as Grp, Ab, Ring, $\mathsf{Vect}_{\mathbb{K}}$, ... do preserve and reflect limits. Therefore in this categories we can *forget* the structure when checking for

sheaf conditions.

10.3 Sheafification

We have established the definition of presheaves and sheaves. It is clear from the definitions that every sheaf is a presheaf. This defines a "forgetful" functor:

$$\iota : \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$$

Since sheaves are the objects with "local-to-global" gluing properties, we often wish to associate to an arbitrary presheaf $\mathcal{F}$ the "best possible" sheaf that approximates it. In the language of category theory, this problem is framed as finding a left adjoint to the inclusion functor.

Definition 10.3.1 (Sheafification). Let X be a topological space and C a category. The **sheafification** (or associated sheaf) functor, denoted by $-^{\#} : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$, is the left adjoint to the inclusion functor $\iota : \mathsf{Sh}_X(\mathsf{C}) \hookrightarrow \mathsf{PSh}_X(\mathsf{C})$.
That is, for every presheaf $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ and every sheaf $\mathcal{G} \in \mathsf{Sh}_X(\mathsf{C})$, there is a natural bijection:

$$\mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathcal{F}^{\#}, \mathcal{G}) \cong \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\mathcal{F}, \iota\mathcal{G})$$

The unit of this adjunction gives a canonical morphism $\theta : \mathcal{F} \mapsto \iota(\mathcal{F}^{\#})$ in $\mathsf{PSh}_X(\mathsf{C})$.

The existence of the sheafification functor ... Therefore, we will first assume its existence and later show that it indeed exists for certain categories.

First of all, we can translate the concepts of direct image presheaf and inverse image presheaf to sheaves.

Definition 10.3.2. Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let $\mathscr{F}$ be a C-valued sheaf over X.
The **direct image sheaf** of $\mathscr{F}$ via f, $f_*\mathscr{F} \in \mathsf{Sh}_Y(\mathsf{C})$ is defined as

$$f_*\mathscr{F} = (f_*^{\mathrm{pre}} \iota\mathscr{F})^{\#}$$

where f_*^{pre} is the direct image presheaf of $\iota\mathscr{F} \in \mathsf{PSh}_X(\mathsf{C})$.
Therefore we have

$$\forall Y \in \mathsf{Op}(Y) \quad (f_*\mathscr{F})(V) = \left(f_*^{\mathrm{pre}} \iota\mathscr{F}\right)^{\#}(V) = (\iota\mathscr{F})^{\#}(f^{-1}V) = \mathscr{F}(f^{-1}V)$$

i.e: the direct image sheaf coincides with the direct image presheaf.
Note that this defines a functor $f_* : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{Sh}_Y(\mathsf{C})$.

We can also translate Example 10.1.2 and Example 10.1.3 to sheaves. Furthermore, since by Remark 10.2.2 $\mathscr{F}(\varnothing)$ is a terminal object in C the definitions simplify.

Definition 10.3.3 (Global section). Let X be a topological space. Let $\mathscr{F}$ be a C-valued sheaf over X.
Using Example 10.1.2 we can define the **global section** of $\mathscr{F}$ as

$$\Gamma(X, \mathscr{F}) := \mathscr{F}(X)$$

This induces a functor $\Gamma(X, -) : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$.

Definition 10.3.4 (Skyscraper sheaf)**.** Let X be a topological space and $p \in X$. Let C be a category with a terminal object $* \in \mathsf{C}$.
Using Example 10.1.3 we can define the **skyscraper sheaf** supported at $S \in \mathsf{C}$ as

$$\forall U \in \mathsf{Op}(X) \quad \iota_{p,*}\underline{S} = \begin{cases} S & p \in U \\ * & p \notin U \end{cases}$$

This induces a functor $\iota_{p,*} : \mathsf{C} \to \mathsf{Sh}_X(\mathsf{C})$.

Definition 10.3.5 (Inverse image sheaf)**.** Let $f : X \to Y$ be a continuous map between two topological spaces X, Y. Let C be a category with direct limits. Let $\mathscr{G}$ be a C-valued sheaf over Y.
The **inverse image sheaf** of $\mathscr{G}$, $f^{-1}\mathscr{G} \in \mathsf{Sh}_X(\mathsf{C})$, of $\mathscr{G}$ via f is the sheaf

$$f^{-1}\mathscr{G} \coloneqq \left(f^{-1}_{\mathrm{pre}}\iota\mathscr{G}\right)^{\#}$$

Therefore we have

$$\forall U \in \mathsf{Op}(X) \quad \left(f^{-1}\mathscr{G}\right)(U) = \left(f^{-1}_{\mathrm{pre}}\iota\mathscr{G}\right)^{\#}(U) = \left(\varinjlim_{\substack{V \supseteq f(U) \\ V \in \mathsf{Op}(Y)}} \iota\mathscr{G}(V)\right)^{\#}$$

In this occassion, unlike Definition 10.3.2, we are stuck with taking sheafification.
Notice that this defines a functor $f^{-1} : \mathsf{Sh}_Y(\mathsf{C}) \to \mathsf{Sh}_X(\mathsf{C})$.

Again, we can translate Example 10.1.4 and Example 10.1.5. We assume C is a category with direct limits as in the presheaf case.

Definition 10.3.6 (Constant sheaf)**.** Let X be a topological space. The **constant sheaf** supported at $S \in \mathsf{C}$ is

$$\underline{S} \coloneqq \left(\underline{S}_{\mathrm{pre}}\right)^{\#}$$

Note that this induces a functor $\underline{\cdot} : \mathsf{C} \to \mathsf{Sh}_X(\mathsf{C})$.

Definition 10.3.7 (Stalk)**.** Let X be a topological space and $p \in X$. Let $\mathscr{F}$ be a C-valued sheaf over X.
We define the **stalk** of $\mathscr{F}$ at p as

$$\mathscr{F}_p \coloneqq \varinjlim_{\substack{U \ni p \\ U \in \mathsf{Op}(X)}} \mathscr{F}(U)$$

Note that this induces a funtor $-_p : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$.

The adjunction of Theorem 10.1.1 also translates directly

Theorem 10.3.1. Let $f : X \to Y$ be a continuous function between two topological spaces X, Y. Let C be a category with direct limits. Then there exists an adjunction

$$f^{-1} \dashv f_* : \mathsf{PSh}_X(\mathsf{C}) \underset{f_*}{\overset{f^{-1}}{\leftrightarrows}} \mathsf{PSh}_Y(\mathsf{C})$$

Proof. Using the adjunctions $-^{\#} \dashv \iota$ and $f^{-1}_{\text{pre}} \dashv f_*$ we have

$$\begin{aligned}\mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(f^{-1}\mathscr{G}, \mathscr{F}) &= \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(f^{-1}\iota\mathscr{G}, \iota\mathscr{F}) \\ &= \mathrm{Hom}_{\mathsf{PSh}_X(\mathsf{C})}(\iota\mathscr{G}, f_*\iota\mathscr{F}) \\ &= \mathrm{Hom}_{\mathsf{Sh}_X(\mathsf{C})}(\mathscr{G}, f_*\mathscr{F})\end{aligned}$$

□

Therefore we have the analogous adjunctions of Corollary 10.1.2 and Corollary 10.1.2 for sheaves.

Proposition 10.3.1. Let X be a topological space and $p \in X$. For any C-valued presheaf $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{C})$ we have that $\mathcal{F}_p = (\mathcal{F}^{\#})_p$.

Proof. Note that the stalk functor $-_p : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{C}$ can be written as the $-^{\text{pre}}_p \circ \iota$ where $\iota : \mathsf{Sh}_X(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is the inclussion functor and $-^{\text{pre}}_p : \mathsf{PSh}_X(\mathsf{C}) \to \mathsf{Arr}(\mathsf{C})$ is the presheaf stalk functor.

Note that $\iota^{\text{pre}}_{p,*} : \mathsf{Arr}(\mathsf{C}) \to \mathsf{PSh}_X(\mathsf{C})$ is, by Corollary 10.1.2, right adjoint to $-^{\text{pre}}_p$. Furthermore, by Definition 10.3.4 we have that $\iota_{p,*} : \mathsf{C} \to \mathsf{Sh}_X(\mathsf{C})$ verifies $\iota_{p,*} = -^{\text{pre}}_p \circ \iota$.

Therefore if $F = (-^{\text{pre}}_p \circ \iota) \circ -^{\#}$ we have that its right adjoint is

$$R_F = \iota \circ \iota^{\text{pre}}_{p,*} = \iota_{p,*}$$

Therefore by *uniqueness* of the adjoint we have that $\mathcal{F}_p \cong \mathcal{F}^{\#}_p$. □

Proposition 10.3.2. Sheaf morphisms are determined by their induced maps on stalks

Proof. Let $\varphi, \psi : \mathscr{F} \to \mathscr{G}$ be two sheaf morphisms such that, for every point $p \in X$ $\varphi_x = \psi_x$. We want to show that, for every open subset $U \subseteq X$ the component maps $\varphi_U, \psi_U : \mathscr{F}(U) \to \mathscr{G}(U)$ are equal.

Let $U \subseteq X$ be an open subset and $s \in \mathscr{F}(U)$ be an arbitrary section. Consider $t_1 \coloneqq \varphi_U(s), t_2 \coloneqq \psi_U(s)$.

By the definition of stalk we have a commutative diagram

$$\begin{array}{ccc} \mathscr{F}(U) & \xrightarrow{\varphi_U} & \mathscr{G}(U) \\ \downarrow & & \downarrow \\ \mathscr{F}_p & \xrightarrow{\varphi_p} & \mathscr{G}_p \end{array}$$

Then for any $p \in U$ we have

$$\begin{aligned}(t_1)_p &= (\varphi_U(s))_p = \varphi_p(s_p) \\ (t_2)_p &= (\psi_U(s))_p = \psi_p(s_p)\end{aligned}$$

and therefore by hypothesis $(t_1)_p = (t_2)_p$ □

Existence of sheafification

We will study first the construction of the sheafification functor for the case where $\mathsf{C} = \mathsf{Set}$, and then prove that this construction can be carried in other categories.

Construction (Godement construction of sheafification)**.** Let $\mathcal{F} \in \mathsf{PSh}_X(\mathsf{Set})$.

For every open set $U \subseteq X$, let

$$\Lambda(\mathcal{F})(U) \coloneqq \prod_{p \in U} \mathcal{F}_p$$

Considering for $V \subseteq U$ the morphism

$$\begin{aligned} \mathrm{res}_{U,V} : \Lambda(\mathcal{F})(U) &\to \Lambda(\mathcal{F})(V) \\ \prod_{p \in U} \mathcal{F}_p &\mapsto \prod_{p \in V} \mathcal{F}_p \end{aligned}$$

as the restriction maps, we see that $\Lambda(\mathcal{F})$ is a Set-valued presheaf.

Let's check that $\Lambda(\mathcal{F})$ verifies the sheaf axioms. To do this, consider an open set $U \subseteq X$.

SH.1) Let $s, t \in \mathcal{F}(U)$ be two sections such that there exists an open covering $\{U_i\}_{i \in \mathcal{I}}$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = t|_{U_i}$.
Note that for any $x \in U$ since there exists $j \in \mathcal{I}$ such that $x \in U_j$ we have that $s_x = t_x$. Therefore, since this holds for all $x \in U$, we have that $s = t$.

SH.2) Let $\{U_i\}_{i \in \mathcal{I}}$ be an open covering of U and let $\{s_i \mid s_i \in \Lambda(\mathcal{F})(U_i)\}_{i \in \mathcal{I}}$ be a family of sections such that for any $i, j \in \mathcal{I}$ we have $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$.
Define a section $s \in \Lambda(\mathcal{F})(U)$ by choosing, for each $x \in U$, an index $k \in \mathcal{I}$ such that $x \in U_k$ and setting $s_x = (s_k)_x$. Note that this is well-defined by the compatibility condition of the family. Furthermore, for all $i \in \mathcal{I}$ by construction we have that $s|_{U_i} = s_i$.

Note that there is a presheaf morphism $\theta : \mathcal{F} \to \Lambda(\mathcal{F})$ defined for any $s \in \mathcal{F}(U)$ by sending s to its germ at every point:

$$\theta(U)(s) = (s_p)_{p \in U} \in \prod_{p \in U} \mathcal{F}_p$$

where s_p denotes the image of s in the stalk $\mathcal{F}_p$.

We define the sheafification $\mathcal{F}^\#$ of $\mathcal{F}$ as the sub-presheaf of $\Lambda(\mathcal{F})$ consisting of sections that are *locally compatible* with $\mathcal{F}$. Explicitly, for every open set $U \subseteq X$ an element $\sigma = (\sigma_p)_{p \in U} \in \Lambda(\mathcal{F})(U)$ belongs to $\mathcal{F}^\#(U)$ if and only if:

> For every $p \in U$, there exists an open neighborhood $V \subseteq U$ of p and a section $t \in \mathcal{F}(V)$ such that for all $q \in V$, $\sigma_q = t_q$ (as elements of $\mathcal{F}_q$).

We need to check that our definition of $\mathcal{F}^\#$ verifies the universal propery of sheafification. To do this, consider a sheaf $\mathcal{G}$ and a morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$. We wish to show that there exists a unique sheaf morphism $\psi : \mathcal{F}^\# \to \mathcal{G}$ such that $\phi \circ \theta = \psi$.

Let's check existence first by defining ψ explicitly. Let $U \subseteq X$ be an open set and $s \in \mathcal{F}^{\#}(U)$. By definition there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of U and sections $t_i \in \mathcal{F}(V_i)$ such that $\forall i \in \mathcal{I} \quad s|_{U_i} = \theta(t_i)$.

We want ψ to verify $\psi(s)|_{U_i} = \psi(s|_{U_i}) = \psi(\theta_{t_i})$. This forces us to define $\psi(\theta(t_i)) = \phi(t_i) \in \mathcal{G}(U_i)$.

Let's see that $\{\phi(t_i)\}_{i\in\mathcal{I}}$ can be glued in $\mathcal{G}$. Note that for all $y \in U_i \cap U_j$ we have $(t_i)_y = (t_j)_y$. Therefore using the induced stalk morphism $\phi_y : \mathcal{F}_y \to \mathcal{G}_y$ we have $\phi_y((t_i)_y) = \phi_y((t_j)_y)$. Therefore $(\phi(t_i))_y = (\phi(t_j))_y$ for all $y \in U_i \cap U_j$, meaning that $\phi(t_i)|_{U_i\cap U_j} = \phi(t_j)|_{U_i\cap U_j}$.

Applying SH.2 to this family we get that there is a unique section $t \in \mathcal{G}(U)$ such that $\forall i \in \mathcal{I}\, t|_{U_i} = \phi(t_i)$. Therefore we can define $\psi_U(s) = t$, which verifies the universal property by construction.

To check uniqueness, since sheaf morphisms are determined by their induced maps on stalks and for all points $x \in X$ we have $\theta_x = \mathbb{1}$, ψ verifies $\forall x \in X \quad \psi_x = \phi_x$ making it unique.

Now that we have seen how to construct the sheafification of a sheaf of sets, we will see that it also exists for *algebraic structures*.

Definition 10.3.8. A **type of algebraic structure** is a category C together with a functor $F : \mathsf{C} \to \mathsf{Set}$ verifying:

1. F is faithful.
2. C has limits and F commutes with them.
3. C has direct limits and F commutes with them.
4. F reflects isomorphisms.

Example 10.3.1. The following categories, together with their respective forgetful functors are types of algebraic structures:

- The category of pointed sets.
- The category of (abelian) groups.
- The category of monoids.
- The category of rings.
- The category of R-modules for a fixed ring R.
- The category of Lie algebras over a fixed field.

Remark 10.3.1. If (C, F) is a type of algebraic structure then C-valued sheaves are well defined.

We state some properties of this types of categories without proof.

Lemma 10.3.1 (Properties of types of algebraic structures). Let (C, F) be a type of algebraic structure. Then:

1. C has a terminal object 0 and $F(0) = \{*\}$.
2. C has products and $F\left(\prod A_i\right) = \prod F\left(A_i\right)$.
3. C has fibre products and $F(A \times_B C) = F(A) \times_{F(B)} F(C)$.
4. C has equalizers and if $E \to A$ is the equalizer of $a, b : A \to B$ then $F(E) \to F(A)$ is the equalizer of $F(a), F(b) : F(A) \to F(B)$.
5. $A \to B$ is a monomorphism if and only if $F(A) \to F(B)$ is injective.

6. If $F(a) : F(A) \to F(B)$ is surjective then a is an epimorphism.
7. Given $A_1 \to A_2 \to \cdots$, $\operatorname{colim} A_i$ exists and $F(\operatorname{colim} A_i) = \operatorname{colim} F(A_i)$. And more generally for any direct limit.

Remark 10.3.2. It follows from the definition of a type of algebraic structure and Lemma 10.3.1 that F both preserves and reflects limits.
Therefore by subsection 10.2.1 we have that $\mathscr{F}$ is a C-valued sheaf if and only if its underlying presheaf of sets $F(\mathscr{F})$ is a sheaf of sets.

Lemma 10.3.2. Let (C, F) be a type of algebraic structure. Let $f : A \to B$ and $g : C \to B$ be two morphisms in C. Then if $F(g)$ is injective and $\operatorname{im} F(f) \subseteq \operatorname{im} F(g)$ we have that f factors as $f = g \circ t$ for some morphism $t : A \to C$ in C.

Proposition 10.3.3. prop:sheafification-of-algebraic-structure] Let X be a topological space. Let (C, F) be a type of algebraic structure. Let $\mathcal{F}$ be a C-valued presheaf over X. Then there exists a C-valued sheaf $\mathcal{F}^\#$ and a morphism $\mathcal{F} \to \mathcal{F}^\#$ such that

1. The map $\mathcal{F} \to \mathcal{F}^\#$ identifies the underlying sheaf of sets of $\mathcal{F}^\#$ with the underlying sheafification of the underlying presheaf of sets of $\mathcal{F}$.
2. For any C-valued sheaf $\mathcal{G}$ and any morphism $\mathcal{F} \to \mathcal{G}$ there exists a unique factorization $\mathcal{F} \to \mathcal{F}^\# \to \mathcal{G}$.

Proof. Consider the *product sheaf* defined as, for any open subset $U \subseteq X$

$$\prod(\mathcal{F})(U) = \prod_{x \in U} \mathcal{F}_x$$

This is well-defined because C has products. Notice that we have a map $\mathcal{F} \to \prod(\mathcal{F})$ defined by $\mathcal{F}(U) \mapsto \prod(\mathcal{F})(U)$.

For any open set $U \subseteq X$ we define $\mathcal{F}^\#(U)$ as the fiber product in C of the diagram

$$\begin{array}{ccc} & & \prod(\mathcal{F})(U) \\ & & \downarrow \\ \prod_{x \in U} \mathcal{F}_x & \longrightarrow & \prod_{x \in U} \prod(\mathcal{F})_x \end{array}$$

We have that

$$F(\mathcal{F}^\#(U)) = F\left(\prod(\mathcal{F})(U)\right) \times_{F(\prod(\mathcal{F})_x)} \prod F(\mathcal{F}_x)$$

Therefore by **[stacks-project]** we have property 1.

For property 2. consider a sheaf $\mathcal{G}$ and a morphism $\mathcal{F} \to \mathcal{G}$. By the universal property of sheafification there exists a map of sheaves of sets $\psi_{\mathsf{Set}} : F(\mathcal{F}^\#) \to F(\mathcal{G})$. Therefore using Lemma 10.3.2 we have

$$\begin{array}{ccc} \mathcal{F}^\#(U) & \dashrightarrow & \mathcal{G}(U) \\ \downarrow & & \downarrow \\ \prod_{x \in U} \mathcal{F}_x & \longrightarrow & \prod_{x \in U} \mathcal{G}_x \end{array}$$

□

Chapter 11

Ringed spaces

11.1 Sheaves of modules

Definition 11.1.1. Let X be a topological space. Let $\mathcal{O}_X$ be a presheaf of rings over X. A **presheaf of $\mathcal{O}_X$-modules** is a presheaf of abelian groups $\mathcal{F}$ together with a map of presheaves of sets $\mathcal{O} \times \mathcal{F} \to \mathcal{F}$ such that for every open subset $U \subseteq X$ the map $\mathcal{O}(U) \times \mathcal{F}(U) \to \mathcal{F}(U)$ defines the structure of an $\mathcal{O}(U)$-module on $\mathcal{F}(U)$.
A **morphism** of presheaves of $\mathcal{O}_X$-modules $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves of abelian groups $\varphi : \mathcal{F} \to \mathcal{G}$ such that the following diagram commutes

$$\begin{array}{ccc} \mathcal{O}_X \times \mathcal{F} & \longrightarrow & \mathcal{F} \\ {\scriptstyle \mathbb{1}\times\varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ \mathcal{O}_X \times \mathcal{G} & \longrightarrow & \mathcal{G} \end{array}$$

We denote the category of presheaves of $\mathcal{O}_X$-modules over X as $\mathsf{PMod}(\mathcal{O}_X)$.

Notation. When the context is clear, we will drop the X subindex and say presheves of $\mathcal{O}$-modules and $\mathsf{PMod}(\mathcal{O})$.

Definition 11.1.2. Let X be a topological space. Let $\mathcal{O}_1 \to \mathcal{O}_2$ be a morphism of presheaves of rings. We define the **restriction** functor

$$-_{\mathcal{O}_1} : \mathsf{PMod}(\mathcal{O}_2) \to \mathsf{PMod}(\mathcal{O}_1)$$

by using for any presheaf of $\mathcal{O}_2$-modules $\mathcal{F}$ the composition map $\mathcal{O}_1 \times \mathcal{F} \to \mathcal{O}_2 \times \mathcal{F} \to \mathcal{F}$.

Definition 11.1.3. Let X be a topological space. For any presheaves of rings $\mathcal{O}_1, \mathcal{O}_2$ define the **change of rings** functor

$$\mathcal{O}_2 \otimes_{\mathrm{pre},\mathcal{O}_1} - : \mathsf{PMod}(\mathcal{O}_1) \to \mathsf{PMod}(\mathcal{O}_2)$$

as, for any presheaf of $\mathcal{O}_1$-modules $\mathcal{G}$ and open subset $U \subseteq X$

$$(\mathcal{O}_2 \otimes_{\mathrm{pre},\mathcal{O}_1} \mathcal{G})(U) \coloneqq \mathcal{O}_2(U) \otimes_{\mathcal{O}_1(U)} \mathcal{G}(U)$$

Lemma 11.1.1. For any $\mathcal{O}_1 \to \mathcal{O}_2$ morphism of presheaves of rings the restriction

functor and the change of rings functor are adjoint. In formula

$$\mathrm{Hom}_{\mathcal{O}_1}\left(\mathcal{G}, \mathcal{F}_{\mathcal{O}_1}\right) \cong \mathrm{Hom}_{\mathcal{O}_2}(\mathcal{O}_2 \oplus_{\mathrm{pre},\mathcal{O}_1} \mathcal{G}, \mathcal{F})$$

Proof. This follows directly from the adjuntion of the corresponding functors in Ring. □

Definition 11.1.4. Let X be a topological space. Let $\mathscr{O}_X$ be a sheaf of rings. The category of **sheaves of $\mathscr{O}_X$-modules** $\mathsf{Mod}(\mathscr{O}_X)$ is the full subcategory of $\mathsf{PMod}(\iota\mathscr{O}_X)$ consisting of presheaves of $\iota\mathscr{O}_X$-modules where the the underlying presheaves of abelian modules are sheaves.

Since, at the end of the day, presheaves of $\mathcal{O}$-modules and sheaves of $\mathscr{O}$-modules are presheaves and sheaves of abelian groups respectively, all the constructions we have seen in the past chapters apply.

For example, in the case of presheaves of $\mathcal{O}$-modules we have.

Lemma 11.1.2. Let X be a topological space and $x \in X$. Let $\mathcal{O}$ be a presheaf of rings over X. Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$-modules.
The canonical map $\mathcal{O}_x \times \mathcal{F}_x \to \mathcal{F}_x$ coming from the map $\mathcal{O} \times \mathcal{F} \to \mathcal{F}$ defines an $\mathcal{O}_x$-module structure on the abelian group $\mathcal{F}_x$.

Lemma 11.1.3. Let X be a topological space and $x \in X$. Let $\mathcal{O} \to \mathcal{O}'$ be a morphism of presheaves of rings over X. Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$-modules over X.
A $\mathcal{O}'_x$-modules we have that

$$\mathcal{F}_x \otimes_{\mathcal{O}_x} \mathcal{O}'_x \cong (\mathcal{F} \oplus_{\mathrm{pre},\mathcal{O}} \mathcal{O}')_x$$

The case of sheafification is a little bit more tricky. We have an analogous of **??**.

Proposition 11.1.1. Let X be a topological space. Let $\mathcal{O}$ be a presheaf of rings over X and $\mathcal{O}^\#$ be its sheafification. Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$-modules. Let $\mathcal{F}^\#$ be the sheafification of $\mathcal{F}$ as a presheaf of abelian groups.
There exists a map of sheaves of sets $\mathcal{O}^\# \times \mathcal{F}^\# \to \mathcal{F}^\#$ making the diagram

$$\begin{array}{ccc} \mathcal{O} \times \mathcal{F} & \longrightarrow & \mathcal{F} \\ \downarrow & & \downarrow \\ \mathcal{O}^\# \times \mathcal{F}^\# & \longrightarrow & \mathcal{F}^\# \end{array}$$

commute and which makes $\mathcal{F}^\#$ into a sheaf of $\mathcal{O}^\#$-modules. Furthermore, for any sheaf of $\mathcal{O}^\#$-modules $\mathscr{G}$, all morphisms of presheaves of $\mathcal{O}$-modules $\mathcal{F} \to \mathscr{G}$ factor uniquely as $\mathcal{F} \to \mathcal{F}^\# \to \mathscr{G}$ where $\mathcal{F}^\# \to \mathscr{G}$ is a morphism of sheaves of $\mathcal{O}^\#$-modules.
This defines a sheafification functor

$$-^\# : \mathsf{PMod}(\mathcal{O}) \to \mathsf{Mod}(\mathcal{O}^\#)$$

Lemma 11.1.4. Let X be a topological space. Let $\mathcal{O}$ be a presheaf of rings over X.
The functors $-^\# : \mathsf{PMod}(\mathcal{O}) \to \mathsf{Mod}(\mathcal{O}^\#)$ and $\iota : \mathsf{Mod}(\mathcal{O}^\#) \to \mathsf{PMod}(\mathcal{O})$ are adjoint.
In formula

$$\mathrm{Hom}_{\mathsf{PMod}(\mathcal{O})}(\mathcal{F}, \iota\mathscr{G}) \cong \mathrm{Hom}_{\mathsf{Mod}(\mathcal{O}^\#)}(\mathcal{F}^\#, \mathscr{G})$$

Using this adjuntion we can *translate* Definition 11.1.2 and Definition 11.1.3.

Definition 11.1.5. Let X be a topological space. Let $\mathscr{O}_1 \to \mathscr{O}_2$ be a morphism of sheaves of rings. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_2$-modules.
We define the **restriction** of $\mathscr{F}$ to $\mathscr{O}_1$ as

$$\mathscr{F}_{\mathscr{O}_1} := \left(\iota\left(\mathscr{F}_{\mathscr{O}_1}\right)\right)^{\#} = \mathscr{F}_{\mathscr{O}_1}$$

where $\iota(\mathscr{F}_{\mathscr{O}_1})$ is the restriction as a presheaf of $\iota\mathscr{O}_2$-modules.
I.e: the restriction of sheaves of modules coincides with the restriction of presheaves of modules. This induces a functor

$$-_{\mathscr{O}_1} : \mathsf{Mod}(\mathscr{O}_2) \to \mathsf{Mod}(\mathscr{O}_1)$$

Definition 11.1.6. Let X be a topological space. Let $\mathscr{O}_1, \mathscr{O}_2$ be sheaves of rings over X. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_2$-modules.
We define the **change of basis** of $\mathscr{F}$ into $\mathscr{O}_2$ as

$$\mathscr{O}_2 \otimes_{\mathscr{O}_1} \mathscr{F} := \left(\iota\mathscr{O}_2 \otimes_{\mathrm{pre},\iota\mathscr{O}_1} \iota\mathscr{F}\right)^{\#}$$

This induces a functor

$$\mathscr{O}_2 \otimes_{\mathscr{O}_1} - : \mathsf{Mod}(\mathscr{O}_1) \to \mathsf{Mod}(\mathscr{O}_2)$$

We also have an analogous of Lemma 11.1.1

Lemma 11.1.5. Let X be a topological space. Let $\mathscr{O}_1 \to \mathscr{O}_2$ be a morphism of sheaves of rings over X.
The functors $-_{\mathscr{O}_1}$ and $\mathscr{O}_2 \otimes_{\mathscr{O}_1} -$ are adjoint. In formula, if $\mathscr{F}$ is a sheaf of $\mathscr{O}_2$-modules and $\mathscr{G}$ is a sheaf of $\mathscr{O}_1$-modules we have

$$\mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_1)}\left(\mathscr{G}, \mathscr{F}_{\mathscr{O}_1}\right) \cong \mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_2)}\left(\mathscr{O}_2 \otimes_{\mathscr{O}_1} \mathscr{G}, \mathscr{F}\right)$$

Lemma 11.1.6. Let X be a topological space and $x \in X$. Let $\mathscr{O} \to \mathscr{O}'$ be a morphism of sheaves of rings over X. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}$-modules.
As $\mathscr{O}'_x$-modules we have

$$\mathscr{F}_x \otimes_{\mathscr{O}_x} \mathscr{O}'_x \cong (\mathscr{F} \otimes_{\mathscr{O}} \mathscr{O}')_x$$

Theorem 11.1.1. $\mathsf{Mod}(\mathscr{O})_X$ is an abelian category

Proof. In Proposition 10.1.1 we have seen that limits and colimits of presheaves can be computed *open set by open set*. Therefore since the category of modules over a ring is abelian, the category $\mathsf{PMod}(\mathscr{O}_X)$ inherits this property.

Let's see how we can see that $\mathsf{Mod}(\mathscr{O}_X)$ is also abelian. It is clear that $\mathsf{Mod}(\mathscr{O}_X)$ is an additive category. Therefore, we only need to check that kernels and cokernels exist in $\mathsf{Mod}(\mathscr{O}_X)$.

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves of $\mathscr{O}$-modules. We have an associated morphism of presheaves of $\iota\mathscr{O}$-modules $\iota\varphi : \iota\mathscr{F} \to \iota\mathscr{G}$. Using that $\mathsf{PMod}(\iota\mathscr{O}_X)$ is an abelian category the morphism $\iota\varphi$ admits a kernel $\ker_{\mathrm{pre}}\iota\varphi$ and cokernel $\mathrm{coker}_{\mathrm{pre}}\iota\varphi$ in $\mathsf{PMod}(\iota\mathscr{O}_X)$. Therefore we can define the kernel and cokernel of φ using the

sheafification functor

$$\ker \varphi := \left(\ker_{\text{pre}} \iota\varphi\right)^{\#} = \ker_{\text{pre}} \left(\iota\varphi\right)^{\#} = \ker_{\text{pre}} \varphi$$
$$\operatorname{coker} \varphi := \left(\operatorname{coker}_{\text{pre}} \iota\varphi\right)^{\#}$$

where the equalities in $\ker \varphi$ come from the fact that right adjoints preserve limits. □

11.2 Ringed spaces

Definition 11.2.1 (Ringed space). A **ringed space** is a pair $(X, \mathscr{O}_X)$ consisting of a topological space X and a sheaf of rings $\mathscr{O}_X$ over X.
A **morphism** of ringed spaces $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a pair $(f, f^{\#})$ where $f : X \to Y$ is a continuous map and $f^{\#} : \mathscr{O}_y \to f_*\mathscr{O}_X$ is a morphism of sheaves.

11.2.1 Ringed spaces and continuous functions

We wish to translate what we did for sheaves in subsection 10.1.1 to sheaves of $\mathscr{O}_X$-modules.

The direct image is direct.

Definition 11.2.2. Let $(f, f^{\#}) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of ringed spaces.
Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules.
We define the **direct image** of $\mathscr{F}$ via f as the sheaf of $\mathscr{O}_Y$-modules which has the direct image sheaf $f_*\mathscr{F}$ as sheaf of abelian groups and with module structure given by the map $f^{\#} : \mathscr{O}_Y \to f_*\mathscr{O}_X$.
This defines a functor

$$f_* : \mathsf{Mod}(\mathscr{O}_X) \to \mathsf{Mod}(\mathscr{O}_Y)$$

However, as in the case for the inverse image for sheaves, the inverse image of a sheaf of $\mathscr{O}_X$-modules needs more work.

Explain why ⊗

Definition 11.2.3. Let $(f, f^{\#}) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of ringed spaces.
Let $\mathscr{G}$ be an $\Gamma(X, \mathscr{O}_X)$-module, where
We define the **inverse image** of $\mathscr{G}$ via f to be the sheaf of $\mathscr{O}_X$-modules

$$f^*\mathscr{G} := \mathscr{O}_X \otimes_{f^{-1}\mathscr{O}_Y} f^{-1}\mathscr{G}$$

where f^{-1} is the inverse image sheaf.
This defines a functor

$$f^* : \mathsf{Mod}(\mathscr{O}_Y) \to \mathsf{Mod}(\mathscr{O}_X)$$

Theorem 11.2.1. Let $(f, f^{\#}) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of ringed spaces.
The functors f_* and f^* are adjoint.
In formula, if $\mathscr{F}$ is a sheaf of $\mathscr{O}_X$-modules and $\mathscr{G}$ is a sheaf of $\mathscr{O}_Y$-modules we have

$$\operatorname{Hom}_{\mathsf{Mod}(\mathscr{O}_X)}\left(f^*\mathscr{G}, \mathscr{F}\right) \cong \operatorname{Hom}_{\mathsf{Mod}(\mathscr{O}_Y)}\left(\mathscr{G}, f_*\mathscr{F}\right)$$

Proof.

$$\begin{aligned}\mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_X)}\left(f^*\mathscr{G},\mathscr{F}\right) &\cong \mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_X)}\left(\mathscr{O}_X \otimes_{f^{-1}\mathscr{O}_Y} f^{-1}\mathscr{G},\mathscr{F}\right)\\ &\cong \mathrm{Hom}_{\mathsf{Mod}(f^{-1}\mathscr{O}_Y)}\left(f^{-1}\mathscr{G},\mathscr{F}_{f^{-1}\mathscr{O}_Y}\right)\\ &\cong \mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_Y)}\left(\mathscr{G},f_*\mathscr{F}\right)\end{aligned}$$

□

With the adjunction between f_* and f^* out of the way, let's now ...

Let $(X,\mathscr{O}_X)$ be a ringed space and $p \in X$.

Consider the one-point topological space $\{*\}$. Recall that by Remark 10.2.2 we have that $\mathsf{Sh}_{\{*\}}(\mathsf{Ring}) \cong \mathsf{Ring}$. Therefore to make $\{*\}$ into a ringed space we only need to *accompany* it by a ring.

We are interested in the direct and inverse images of the morphisms of ringed spaces induced by the maps

$$\pi : X \to \{*\} \qquad \iota_p : \{*\} \to X$$

Note that

$$\begin{aligned}\pi^\# &: \mathscr{O}_\pi \to \pi_*\mathscr{O}_X = \Gamma(X,\mathscr{O}_X)\\ \iota_p^\# &: \mathscr{O}_X \to \iota_{p,*}\mathscr{O}_{\iota_p} = \mathscr{O}_{\iota_p,p}\end{aligned}$$

Therefore we choose $\mathscr{O}_\pi = \Gamma(X,\mathscr{O}_X)$ and $\mathscr{O}_{\iota_p} = \mathscr{O}_{X,p}$.

Definition 11.2.4 (Global sections). Let $(X,\mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules. Consider the morphism of ringed spaces

$$(\pi,\pi^*) : (X,\mathscr{O}_X) \to (\{*\},\Gamma(X,\mathscr{O}_X))$$

We define the **global sections** of $\mathscr{F}$ as the direct image of $\mathscr{F}$ via π:

$$\Gamma(X,\mathscr{F}) \coloneqq \pi_*\mathscr{F}$$

Note that this defines a functor

$$\Gamma(X,-) : \mathsf{Mod}(\mathscr{O}_X) \to \mathsf{Mod}(\Gamma(X,\mathscr{O}_X))$$

Definition 11.2.5 ("Constant" sheaf). Let $(X,\mathscr{O}_X)$ be a ringed space. Let M be a $\Gamma(X,\mathscr{O}_X)$ module. Consider the morphism of ringed spaces

$$(\pi,\pi^\#) : (X,\mathscr{O}_X) \to (\{*\},\Gamma(X,\mathscr{O}_X))$$

We define the **"constant" sheaf** supported at M as inverse image of M via π:

$$\mathscr{O}_X \otimes_{\Gamma(X,\mathscr{O}_X)} M \coloneqq \pi^*M = \mathscr{O}_X \otimes_{\pi^{-1}\Gamma(X,\mathscr{O}_X)} \pi^{-1}M = \mathscr{O}_X \otimes_{\underline{\Gamma(X,\mathscr{O}_X)}} \underline{M}$$

Note that this defines a functor

$$\mathscr{O}_X \otimes_{\Gamma(X,\mathscr{O}_X)} - : \mathsf{Mod}(\Gamma(X,\mathscr{O}_X)) \to \mathsf{Mod}(\mathscr{O}_X)$$

Corollary 11.2.1. Let $(X, \mathscr{O}_X)$ be a ringed space. The global sections functor and constant sheaf functor of sheaves of $\mathscr{O}_X$-modules are adjoint.
In formula, if $\mathscr{F} \in \mathsf{Mod}(\mathscr{O}_X)$ and $M \in \mathsf{Mod}(\Gamma(X, \mathscr{O}_X))$

$$\mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_X)}(\mathscr{O}_X \otimes_{\Gamma(X,\mathscr{O}_X)} M, \mathscr{F}) \cong \mathrm{Hom}_{\mathsf{Mod}(\Gamma(X,\mathscr{O}_X))}(M, \Gamma(X, \mathscr{F}))$$

Definition 11.2.6 (Skyscraper sheaf). Let $(X, \mathscr{O}_X)$ be a ringed space with $p \in X$. Let M be a $\mathscr{O}_{X,p}$-module. Consider the morphism of ringed spaces

$$(\iota_p, \iota_p^\#) : (\{*\}, \mathscr{O}_{X,p}) \to (X, \mathscr{O}_X))$$

We define the **skyscraper sheaf** supported at M as the direct image of M via ι_p:

$$\iota_{p,*}\underline{M} \coloneqq \iota_{p,*}M$$

Note that this defines a functor

$$\iota_{p,*} : \mathsf{Mod}(\mathscr{O}_{X,p}) \to \mathsf{Mod}(\mathscr{O}_X)$$

Definition 11.2.7 (Stalk). Let $(X, \mathscr{O}_X)$ be a ringed space with $p \in X$. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules. Consider the morphism of ringed spaces

$$(\iota_p, \iota_p^\#) : (\{*\}, \mathscr{O}_{X,p}) \to (X, \mathscr{O}_X))$$

We define the **stalk** of $\mathscr{F}$ at p as inverse image of $\mathscr{F}$ via ι_p:

$$\mathscr{F}_p \coloneqq \iota_p^*\mathscr{F} = \mathscr{O}_{X,p} \otimes_{\iota_p^{-1}\mathscr{O}_X} \iota_p^{-1}\mathscr{F} = \mathscr{O}_{X,p} \otimes_{\mathscr{O}_{X,p}} \mathscr{F}_p = \mathscr{F}_p$$

Note that this defines a functor

$$-_p : \mathsf{Mod}(\mathscr{O}_X) \to \mathsf{Mod}(\mathscr{O}_{X,p})$$

Corollary 11.2.2. Let $(X, \mathscr{O}_X)$ be a ringed space with $p \in X$. The skyscraper functor and stalk functor of sheaves of $\mathscr{O}_X$-modules are adjoint.
In formula, if $\mathscr{F} \in \mathsf{Mod}(\mathscr{O}_X)$ and $M \in \mathsf{Mod}(\mathscr{O}_{X,p})$

$$\mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_{X,p})}(\mathscr{F}_p, M) \cong \mathrm{Hom}_{\mathsf{Mod}(\mathscr{O}_X)}(\mathscr{F}, \iota_{p,*}\underline{M})$$

Lemma 11.2.1. Let $(f, f^\#) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of ringed spaces and $p \in X$. Let $\mathscr{G}$ be a sheaf of $\mathscr{O}_Y$-modules.
As $\mathscr{O}_{X,p}$ modules we have

$$(f^*\mathscr{G})_p = \mathscr{G}_{f(p)} \otimes_{\mathscr{O}_{Y,f(p)}} \mathscr{O}_{X,p}$$

11.3 Vector bundles

Definition 11.3.1 (Topological vector bundle). Let X be a topological space. Let $k \in \{\mathbb{R}, \mathbb{C}\}$ be either the field of real or complex numbers.
A **topological k-vector bundle** over X (denoted by $E \xrightarrow{\pi} X$) consists of

- A topological space E.
- A continuous map $\pi : E \to X$.
- For every point $x \in X$ the structure of a finite-dimensional k-vector space

over $E_x \coloneqq \pi^{-1}(\{x\})$.

That admits a *local trivialization*, i.e: there exists an open cover $\{U_i\}_{i\in\mathcal{I}}$ of X and for each $i \in \mathcal{I}$ an $n_i \in \mathbb{N}$ and a homeomorphism

$$\phi_i : U_i \times k^{n_i} \xrightarrow{\sim} \pi^{-1}(U_i)$$

satisfying

1. $\pi \circ \phi_i = \mathrm{pr}_1$, meaning that $\phi_i(\{x\} \times k^{n_i}) \subseteq \pi^{-1}(\{x\})$.
2. For all $x \in X$ the map $\phi_i(x) : k^{n_i} \to E_x$ is linear.

If all the numbers n_i are equal to some $n \in \mathbb{N}$ we say that the vector bundle has rank n.
A **morphism** of topological k-vector bundles from $E_1 \xrightarrow{\pi_1} X$ to $E_2 \xrightarrow{\pi_2} X$ is a continuous map $f : E_1 \to E_2$ such that $\pi_2 \circ f = \pi_1$ and for every $x \in X$ the map $f|_x : (E_1)_x \to (E_2)_x$ is linear.
We denote by $\mathsf{Vect}_k(X)$ the category of topological k-vector bundles over X.

Notation. A topological k-vector bundle of rank n is:

- A **real topological vector bundle** if $k = \mathbb{R}$.
- A **complex topological vector bundle** if $k = \mathbb{C}$.
- A **line bundle** if $n = 1$.

Example 11.3.1 (Trivial topological vector bundle)**.** Let X be a topological space and $n \in \mathbb{N}$. The product topological space $X \times k^n \xrightarrow{\mathrm{pr}_1} X$ is a topological k-vector bundle of rank n called the **trivial** topological vector bundle over X.

Remark 11.3.1. Let $E \xrightarrow{\pi} X$ be a topological k-vector bundle of rank n with local trivialization

$$\{\phi_i : U_i \times k^n \xrightarrow{\sim} \pi^{-1}(U_i)\}_{i\in\mathcal{I}}$$

For each $i, j \in \mathcal{I}$ the composite function

$$\phi_i^{-1} \circ \phi_j : (U_i \cap U_j) \times k^n \to (U_i \cap U_j) \times k^n$$

is well-defined and satisfies

$$\left(\phi_i^{-1} \circ \phi_j\right)(x, y) = (x, g_{i,j}(x)y)$$

for some $\mathrm{GL}_n(k)$-valued function

$$g_{i,j} : U_i \cap U_j \to \mathrm{GL}_n(k)$$

We call these functions $g_{i,j}$ the **transition functions** of the bundle.
Note that this functions satisfy $g_{i,i}(x) = \mathrm{Id}_n$ for all $x \in U_i$ and, for all open subsets U_i, U_j, U_k such that the bundle trivializes and $U_i \cap U_j \cap U_k \neq \varnothing$

$$\forall x \in U_i \cap U_j \cap U_k \quad g_{i,j}(x)g_{j,k}(x)g_{k,i}(x) = \mathrm{Id}_n$$

This is called a *Čech cocycle*.

Lemma 11.3.1. Let (E, X, π, k^n) be a fiber bundle with a $\mathrm{GL}_n(k)$ cocycle acting in the standard way on the fiber k^n. Then there exists a topological k-vector bundle with transition functions determined by the cocycle.

Example 11.3.2.
Examples of vector bundles: Cylinder, Möbius, ...

Definition 11.3.2 (Section of a topological vector bundle). Let $E \xrightarrow{\pi} X$ be a topological k-vector bundle.
For any open set $U \subseteq X$ a **section** of $E \xrightarrow{\pi} X$ over U is a continuous map $\sigma : U \to E$ such that $\pi \circ \sigma = \mathbb{1}_U$.
The set of all sections of $E \xrightarrow{\pi} X$ over U is denoted by $\Gamma(U, E \xrightarrow{\pi} X)$.

Remark 11.3.2. The set $\Gamma(U, E \xrightarrow{\pi} X)$ admits the structure of a k-vector space.

Definition 11.3.3. Let $E \xrightarrow{\pi} X$ be a topological k-vector bundle. The **sheaf of sections** of $E \xrightarrow{\pi} X$ is

$$\begin{aligned} \mathcal{E} : \mathsf{Op}^{\mathrm{op}}(X) &\to \mathsf{Vect}_k \\ U &\mapsto \Gamma(U, E \xrightarrow{\pi} X) \end{aligned}$$

Proposition 11.3.1. Let $E \xrightarrow{\pi} X$ be a topological k-vector bundle. Let $C_X(k)$ be the sheaf of continuous k-valued functions over X.
Then the sheaf of sections $\mathcal{E}$ is a sheaf of $C_X(k)$-modules.

Proof. Let $U \subseteq X$ be an open subset. Let $f \in \mathscr{O}_X(U)$ and $\sigma \in \mathcal{E}(U)$.

We define the action $f \cdot \sigma$ pointwise: for every $x \in U$

$$(f \cdot \sigma)(x) := f(x) \cdot \sigma(x)$$

where the operation in the right is scalar multiplication in E_x.

Let's verify that $f \cdot \sigma$ is a valid section of $E \xrightarrow{\pi} X$.

1. **Linearity:** Since $(f \cdot \sigma)(x)$ is a scalar multiple of a vector in E_x, it remains in the fiber $\pi^{-1}(\{x\})$. Thus $\pi \circ (f \cdot \sigma) = \mathbb{1}_U$.
2. **Continuity:** Since the bundle admits local trivializations, scalar multiplication $k \times E \to E$ is a continuous map. Because f and σ are both continuous, their pointwise product $f \cdot \sigma$ is continuous.

The module axioms follow directly from the vector space structure on E_p. □

Definition 11.3.4 (Locally free sheaf). A sheaf of $\mathscr{O}_X$-modules $\mathscr{F}$ is **locally free** of rank n if there exist an open cover $\{U_i\}_{i \in I}$ of X such that for every $i \in I$ there is an isomorphism of $\mathscr{O}_X|_{U_i}$-modules

$$\mathscr{F}|_{U_i} \cong \bigoplus_{j=1}^{n} \mathscr{O}_X|_{U_i} =: (\mathscr{O}_X|_{U_i})^{\oplus n}$$

We denote by $\mathsf{LocFree}_n(\mathscr{O}_X)$ the full subcategory of $\mathsf{Mod}(\mathscr{O}_X)$ consisting of locally free sheaves of $\mathscr{O}_X$-modules.

Theorem 11.3.1. Let $k \in \{\mathbb{R}, \mathbb{C}\}$. There is an equivalence of categories between the category of topological k-vector bundles of rank n over X and the category of locally free sheaves of $C_X(k)$-modules of rank n.
Under this correspondence

- The trivial topological k-vector bundle $X \times k^n$ corresponds to $(C_X(k))^{\oplus n}$.
- A section of $E \xrightarrow{\pi} X$ corresponds to a global section of $C_X(k)$.

Proof. In Proposition 11.3.1 we have seen that $\mathcal{E}$ defines a functor

$$\mathsf{Vect}_n(X) \longrightarrow \mathsf{LocFree}_n(C_X(k))$$

Therefore we only need to construct the inverse correspondence.

Let $\mathcal{E}$ be a locally free sheaf of $C_X(k)$-modules. Then there exists an open covering $\mathcal{U} = \{U_i\}_{i \in \mathcal{I}}$ of X such that for every $U_i \in \mathcal{U}$ there is an isomorphism of sheaves of $C_X(k)$-modules

$$\tau|_i : \mathcal{E}|_{U_i} \xrightarrow{\sim} \left(C_X(k)|_{U_i}\right)^{\oplus n}$$

Therefore on $U_i \cap U_j$ we have an isomorphism of locally free sheaves of $C_X(k)$-modules

$$\tau_j \circ \tau_i^{-1} : \left(C_X(k)|_{U_i \cap U_j}\right)^{\oplus n} \xrightarrow{\sim} \left(C_X(k)|_{U_i \cap U_j}\right)^{\oplus n}$$

Note that this isomorphism is given by an $n \times n$ matrix $M_{i,j}$ consisting of elements of $C_X(k)|_{U_i \cap U_j}$ that is invertible at every point.

This matrices are not arbitrary. Indeed they satisfy the *cocycle condition*:

$$M_{i,j} \cdot M_{j,k} \cdot M_{k,i} = \mathrm{Id}_n$$

Therefore we can assign to $\mathcal{E}$ the vector bundle of rank n, trivial on the open sets U_i of the covering, and whose transitions are the matrics $M_{i,j}$. Less abstractly, E is obtained by gluing $U_i \times \mathbb{R}^n$ via the identification

$$U_i \cap U_j \times \mathbb{R}^n \subset U_i \times \mathbb{R}^n \xrightarrow{\sim} U_i \times U_j \times \mathbb{R}^n \subset U_j \times \mathbb{R}^n$$

given by $\mathbb{1} \times M_{i,j}$.

This defines a functor

$$\mathsf{LocFree}_n(C_X(k)) \longrightarrow \mathsf{Vect}_n(X)$$

that by construction is *inverse* to the previous functor. □

Definition 11.3.5 (Smooth vector bundle). Let M be a smooth manifold. Let $k \in \{\mathbb{R}, \mathbb{C}\}$ be either the field of real or complex numbers.

A **smooth k-vector bundle** is a topological k-vector bundle $E \xrightarrow{\pi} M$ such that

1. E is a smooth manifold.
2. The projection map $\pi : E \to M$ is smooth.
3. The local trivialization maps $\phi_i : U_i \times k^{n_i} \xrightarrow{\sim} \pi^{-1}(U_i)$ are diffeomorphism.

A **morphism** of smooth k-vector bundles from $E_1 \xrightarrow{\pi_1} M$ to $E_2 \xrightarrow{\pi_2} M$ is morphism of topological vector bundles $f : E_1 \to E_2$ such that f is smooth.
We denote by $\mathsf{Vect}_k^\infty(X)$ the category of smooth k-vector bundles over X.

Definition 11.3.6 (Holomorphic vector bundle)**.** Let M be a complex manifold.

A **holomorphic vector bundle** is a topological $\mathbb{C}$-vector bundle $E \xrightarrow{\pi} M$ such that

1. E is a complex manifold.
2. The projection map $\pi : E \to M$ is an holomorphism.
3. The local trivialization maps $\phi_i : U_i \times \mathbb{C}^{n_i} \xrightarrow{\sim} \pi^{-1}(U_i)$ are biholomorphisms[a].

A **morphism** of holomorphic vector bundles from $E_1 \xrightarrow{\pi_1} M$ to $E_2 \xrightarrow{\pi_2} M$ is morphism of topological vector bundles $f : E_1 \to E_2$ such that f is holomorphic. We denote by $\mathsf{Vect}^{\omega}_{\mathbb{C}}(X)$ the category of holomorphic vector bundles over X.

[a] A map $f : \mathbb{C} \to \mathbb{C}$ is biholomorphic is it is holomorphic, bijective and its inverse is holomorphic.

Remark 11.3.3. Theorem 11.3.1 can be proved analogously to the case of smooth and holomorphic vector bundles.

PART IV

Sheaf cohomology

Chapter 12

Cohomology of sheaves

Let $(X, \mathscr{O}_X)$ be a ringed space.

We want to define the cohomology of X with coefficients in a sheaf of $\mathscr{O}_X$-modules as the right derived functor of the global sections functor

$$\Gamma(X, -) : \mathsf{Mod}(\mathscr{O}_X) \to \mathsf{Mod}(\Gamma(X, \mathscr{O}_X))$$

To do this, we need to check the following:

1. $\mathsf{Mod}(\mathscr{O}_X)$ and $\mathsf{Mod}(\Gamma(X, \mathscr{O}_X))$ are abelian categories.
2. The functor $\Gamma(X, -)$ is left exact.
3. $\mathsf{Mod}(\mathscr{O})_X$ has enough injectives.

In Theorem 11.1.1 we have proved that the category of sheaves of $\mathscr{O}_X$-modules is abelian. Furthermore, since $\Gamma(X, \mathscr{O}_X)$ is a ring, $\mathsf{Mod}(\Gamma(X, \mathscr{O}_X))$ is also abelian.

Recall that by Corollary 11.2.1 $\Gamma(X, -)$ is right adjoint. Therefore by Theorem 5.3.4 $\Gamma(X, -)$ is left exact.

It only remains to show that $\mathsf{Mod}(\mathscr{O}_X)$ has enough injectives.

Lemma 12.0.1. Let $(X, \mathscr{O}_X)$ be a ringed space. The category $\mathsf{Mod}(\mathscr{O}_X)$ of sheaves of $\mathscr{O}_X$-modules over X has enough injectives.

Proof. We will use the following claim

> The category $\mathsf{Mod}(R)$ of modules over a ring R has enough injectives.

For a proof, see [Alu09, Corollary VIII.6.4].

Let $\mathscr{F}$ be an $\mathscr{O}_X$-module. Let $\mathscr{F}_p \in \mathsf{Mod}(\mathscr{O}_{X,p})$ be that stalk of $\mathscr{F}$ at a point $p \in X$.

Since $\mathsf{Mod}(\mathscr{O}_{X,p})$ has enough injectives we can choose an object I_p and a monomorphism

$$i_p : \mathscr{F}_p \rightarrowtail I_p$$

Consider the sheaf

$$\mathscr{I} := \bigoplus_{p \in X} \iota_{p,*} \underline{I_p}$$

where $\iota_{p,*}I_p$ is the skyscraper sheaf. More explicitly, for every open set $U \subseteq X$ the sheaf $\mathscr{I}$ is defined as

$$\mathscr{I}(U) = \bigoplus_{p \in U} I_p$$

Recall that by Corollary 11.2.2 the skyscraper functor is left-adjoint to the stalk functor, which is exact. Therefore by Proposition 8.1.2 $\mathscr{I}$ is an injective object.

Furthermore, there exists a monomorphism $\phi : \mathscr{F} \to \mathscr{I}$ defined as, for any open set $U \subseteq X$

$$\begin{aligned} \phi(U) : \mathscr{F}(U) &\to \mathscr{I}(U) \\ s &\mapsto \bigoplus_{p \in U} j_p(s_p) \end{aligned}$$

□

Having seen this, we now can finally define sheaf cohomology.

Definition 12.0.1 (Sheaf Cohomology). Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules over a topological space X. The i-th **cohomology** of X with coefficients in $\mathscr{F}$ as the i-th right derived functor of the global sections functor:

$$\mathrm{H}^i(X, \mathscr{F}) \coloneqq R^i\Gamma(X, \mathscr{F})$$

Therefore, given a sheaf of $\mathscr{O}_X$-modules $\mathscr{F}$ to compute its cohomology we *only* need to construct an injective (or $\Gamma(X, -)$ acyclic by Theorem 9.2.3) resolution of $\mathscr{F}$

$$0 \longrightarrow \mathscr{F}^{(0)} \longrightarrow \mathscr{F}^{(1)} \longrightarrow \cdots$$

and compute the cohomology of the $\Gamma(X, \mathscr{O}_X)$-modules

$$0 \longrightarrow \Gamma(X, \mathscr{F}^{(0)}) \longrightarrow \Gamma(X, \mathscr{F}^{(1)}) \longrightarrow \cdots$$

Since injective resolutions are usually hard to find (note how Lemma 12.0.1 doesn't give an *easy* construction) we will mostly work with $\Gamma(X, -)$ acyclic resolutions.

12.1 The Godement resolution

Definition 12.1.1 (Flasque sheaf). A sheaf of $\mathscr{O}_X$-modules $\mathscr{F}$ is **flasque** (or **flabby**) if for any inclussion of open subsets $V \subseteq U$ the restriction map $\mathrm{res}_{U,V} : \mathscr{F}(U) \to \mathscr{F}(V)$ is surjective.

Proposition 12.1.1. Flasque sheaves are $\Gamma(X, -)$ acyclic.

Proof. Let $\mathscr{F}$ be a flasque sheaf. Since $\mathsf{Mod}(\mathscr{O}_X)$ has enough injectives we have a short exact sequence

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{I} \longrightarrow \mathscr{G} \longrightarrow 0$$

where $\mathscr{I}$ is an injective object.

We will first show that, for any open subset $U \subseteq X$, the map $\mathscr{I}(U) \twoheadrightarrow \mathscr{G}(U)$ is surjective. Let $s \in \mathscr{G}(U)$. Let $V, W \subseteq U$ be two open subsets and $\tau_V \in \mathscr{I}(V), \tau_W \in \mathscr{I}(W)$ be two sections lifting s to $\mathscr{I}$. Consider

$$\tau_V|_{V\cap U} - \tau_W|_{V\cap W} \in \mathscr{F}(V \cap W)$$

Since $\mathscr{F}$ is flasque there exists a section $\chi_V \in \mathscr{F}(V)$ such that $\chi_V|_{V\cap W} = \tau_V - \tau_W$. Therefore if $\tau'_V = \tau_V - \chi_V$ we have $\tau'_V|_{V\cap W} = \tau_W|_{V\cap W}$, i.e: there exists a section $\tau \in \mathscr{I}(V \cup W)$ such that $\tau|_V = \tau'|_V$ and $\tau'|_W = \tau|_W$. Clearly τ is sent to $\sigma|_{V\cap W}$.

Consider now a pair (W, τ) which is maximal (for the obvious order relation), where W is an open subset of U and τ is a section of $\mathscr{I}$ lifting σ to W. Since σ lifts locally on $\mathscr{I}$ and using the previous result we see that $W = U$.

But then the long exact sequence in cohomology of our sequence and the fact that for $i > 0 \quad \mathrm{H}^i(X, \mathscr{I}) = 0$ imply that $\mathrm{H}^1(X, \mathscr{F}) = 0$ and

$$\forall k - 1 \geq 1 \quad \mathrm{H}^k(X, \mathscr{F}) = \mathrm{H}^{k-1}(X, \mathscr{G})$$

Furthermore surjectivity of the maps $\mathscr{I}(U) \twoheadrightarrow \mathscr{G}(U)$ implies that $\mathscr{G}$ is flasque.

It follows by induction that

$$\forall k > 0 \quad \mathrm{H}^k(X, \mathscr{F}) = 0$$

□

Using flasque sheaves we can construct a canonical (and functorial) resolution of a sheaf of $\mathscr{O}_X$-modules to compute its cohomology.

Construction (Godement resolution)**.** Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules. The **Godement resolution** of $\mathscr{F}$ is constructed by considering the inclussion of $\mathscr{F}$ into the *Godement sheaf* $\mathscr{F}_{\mathrm{God}}$

$$U \mapsto \bigoplus_{p\in U} \mathscr{F}_p =: \mathscr{F}_{\mathrm{God}}(U)$$

Note that $\mathscr{F}_{\mathrm{God}}$ is flasque.

We then inject the quotient $\mathscr{F}_{\mathrm{God}} / \mathscr{F}$ into $\left(\mathscr{F}_{\mathrm{God}} / \mathscr{F}\right)_{\mathrm{God}}$ and so on.

Example 12.1.1. Let's try to compute sheaf cohomology using the Godement resolution.
Let $X = \{p, q\}$ with the *Sierpiński* topology $\mathcal{T} = \{\varnothing, \{p\}, X\}$. Consider the constant sheaf supported at the integers $\mathscr{F} = \underline{\mathbb{Z}}$.
To compute the cohomology of $\mathscr{F}$, we are going to construct the complex of global sections:

$$0 \longrightarrow \Gamma(X, \mathscr{F}_{\mathrm{God}}) \xrightarrow{d^0} \Gamma\left(X, \left(\mathscr{F}_{\mathrm{God}} / \mathscr{F}\right)_{\mathrm{God}}\right) \xrightarrow{d^1} \cdots$$

Note that the stalks of $\mathscr{F}$ are:

$$\mathscr{F}_p = \mathscr{F}(\{p\}) = \mathbb{Z}, \qquad \mathscr{F}_q = \mathscr{F}(X) = \mathbb{Z}.$$

Consider the Godement sheaf associated to $\mathscr{F}$:

$$\mathscr{F}_{\text{God}}(U) = \bigoplus_{x \in U} \mathscr{F}_x.$$

We have that:

$$\Gamma(X, \mathscr{F}_{\text{God}}) = \mathscr{F}_{\text{God}}(X) = \mathscr{F}_p \oplus \mathscr{F}_q = \mathbb{Z} \oplus \mathbb{Z}.$$

This means that we have a map d^{-1} as follows:

$$d^{-1} : \underbrace{\Gamma(X, \mathscr{F})}_{\mathbb{Z}} \to \underbrace{\Gamma(X, \mathscr{F}_{\text{God}})}_{\mathbb{Z}\oplus\mathbb{Z}}, \quad d^{-1}(k) = (k, k).$$

Let $\mathscr{Q}^1 = {}^{\mathscr{F}_{\text{God}}}\big/_{\mathscr{F}}$ be the cokernel sheaf. If we compute its stalks, we have that:

$$(\mathscr{Q}^1)_p = {}^{\mathscr{F}_{\text{God},p}}\big/_{\mathscr{F}_p} = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0,$$
$$(\mathscr{Q}^1)_q = {}^{\mathscr{F}_{\text{God},q}}\big/_{\mathscr{F}_q} = {}^{\mathbb{Z}\oplus\mathbb{Z}}\big/_{\langle (k,k) \rangle} \cong \mathbb{Z}.$$

Therefore, the global sections of the Godement sheaf of $\mathscr{Q}^1$ are:

$$\Gamma\left(X, (\mathscr{Q}^1)_{\text{God}}\right) = (\mathscr{Q}^1)_p \oplus (\mathscr{Q}^1)_q = 0 \oplus \mathbb{Z} = \mathbb{Z}.$$

This produces a differential map:

$$d^0 : \underbrace{\Gamma(X, \mathscr{F}_{\text{God}})}_{\mathbb{Z}\oplus\mathbb{Z}} \to \underbrace{\Gamma\left(X, (\mathscr{Q}^1)_{\text{God}}\right)}_{\mathbb{Z}}, \quad d^0(m, n) = m - n.$$

Let $\mathscr{Q}^2$ be the cokernel of the map $\mathscr{F}_{\text{God}} \to (\mathscr{Q}^1)_{\text{God}}$. If we compute its stalks, we have:

$$(\mathscr{Q}^2)_p = {}^{0}\big/_{0} = 0,$$
$$(\mathscr{Q}^2)_q = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0.$$

Therefore, we have that:

$$\Gamma\left(X, (\mathscr{Q}^2)_{\text{God}}\right) = 0 \oplus 0 = 0.$$

This means that our resolution of global sections is:

$$0 \longrightarrow \mathbb{Z} \oplus \mathbb{Z} \xrightarrow[(m,n)\mapsto m-n]{d^0} \mathbb{Z} \longrightarrow 0$$

Therefore, the cohomology groups of X with coefficients in $\underline{\mathbb{Z}}$ are:

$$\mathrm{H}^0(X, \underline{\mathbb{Z}}) = \ker(d^0) = \{(m, n) \in \mathbb{Z} \oplus \mathbb{Z} \mid m - n = 0\} \cong \mathbb{Z},$$
$$\mathrm{H}^1(X, \underline{\mathbb{Z}}) = \operatorname{coker}(d^0) = {}^{\mathbb{Z}}\big/_{\operatorname{im}(d^0)} = {}^{\mathbb{Z}}\big/_{\mathbb{Z}} = 0,$$
$$\mathrm{H}^i(X, \underline{\mathbb{Z}}) = 0 \quad \forall i > 1.$$

Notice that if the Godement resolution makes computing sheaf cohomology difficult on a *simple* space, it renders the task nearly impossible on more *complicated* spaces.

To address this, the upcoming chapters will explore how we can select different,

more efficient resolutions to compute cohomology on sufficiently nice topological spaces.

Chapter 13

de Rham Cohomology

Definition 13.0.1 (Fine sheaf). A sheaf of $\mathscr{O}_X$-modules $\mathscr{F}$ is **fine** if for every open cover $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ of X there exists a partition of unity $\{f_i\}_{i\in\mathcal{I}}$ subordinate to this covering.

Proposition 13.0.1. Fine sheaves are $\Gamma(X, -)$ acyclic.

Proof. Let $\mathscr{F}$ be a fine sheaf. Using the Godement resolution we have that $\mathscr{F}$ admits a flasque resolution $\mathscr{F}[0] \to \mathscr{I}^\bullet$.

Using Proposition 12.1.1 by Theorem 9.2.3 we have that

$$\mathrm{H}^k(X, \mathscr{F}) = \ker\left(\Gamma(X, \mathscr{I}^k) \to \Gamma(X, \mathscr{I}^{k+1})\right) \Big/ \operatorname{im}\left(\Gamma(X, \mathscr{I}^{k-1}) \to \Gamma(X, \mathscr{I}^k)\right)$$

Let $\alpha \in \ker\left(\Gamma(X, \mathscr{I}^k) \to \Gamma(X, \mathscr{I}^{k+1})\right)$. Since $\mathscr{I}^\bullet$ is locally exact in degree greater than 0 we have that α comes from $\mathscr{I}^{k-1}$, i.e: there exists an open cover $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ of X such that $\alpha|_{U_i} = d\beta_i$ with $\beta_i \in \mathscr{I}^{k-1}(U_i)$.

Consider a partition of unity $\{f_i\}_{i\in\mathcal{I}}$ subordinate to $\mathscr{U}$ and let

$$\beta := \sum_{i\in\mathcal{I}} f_i\beta_i$$

where the sum is locally finite. Note that $f_i\beta_i$ is a section of $\mathscr{I}^{k-1}(X)$ with values in $\mathscr{I}_p^{k-1}(X)$:

$$(f_i\beta_i)_p = \begin{cases} 0 & p \notin U_i \\ f_i\beta_i & p \in U_i \end{cases}$$

Since f_i has support in U_i this defines a section in $\mathscr{I}^{k-1}(U_i)$.

We have that $d\beta = \alpha$ since $\alpha = \sum_{i\in\mathcal{I}} f_i\alpha|_{U_i}$. Therefore $\ker\left(\Gamma(X, \mathscr{I}^k) \to \Gamma(X, \mathscr{I}^{k+1})\right) = \operatorname{im}\left(\Gamma(X, \mathscr{I}^{k-1}) \to \Gamma(X, \mathscr{I}^k)\right)$, which means

$$\forall k > 0 \quad \mathrm{H}^k(X, \mathscr{F}) = 0$$

□

Definition 13.0.2. Let M be a smooth manifold of dimension m.
The **bundle of k-forms** $\Lambda^k(\mathrm{T}^* M)$ is the smooth vector bundle defined with:

- *Total space*: $\Lambda^k(\mathrm{T}^* M) = \coprod_{p\in M} \Lambda^k(\mathrm{T}^*_p M)$.
- *Projection map*: $\pi : \Lambda^k(\mathrm{T}^* M) \to M,\ \pi(p,\omega) = p$.
- *Fiber space*: For every $p \in M$ the fiber $E_p = \pi^{-1}(\{p\}) = \Lambda^k(\mathrm{T}^*_p M)$ is a real vector space of dimension $\binom{m}{k}$.

To define the local trivialization, consider a local chart $(U, \{x^i\}_{i=1}^m)$ of M. Then the differentials $\{\mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}\}_{1\le i_1<\cdots<i_k\le m}$ form a basis for each fiber with base point in U and the map $\phi^{-1} : \pi^{-1}(U) \to U \times \mathbb{R}^{\binom{m}{k}}$ is given by sending a form ω to its coefficients in this base.

Notation. Let $I = (i_1, \dots, i_k)$ be a multi-index. We will write

$$\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \mathrm{d}x^{i_k}$$

In this way the basis of the local trivialization is $\{\mathrm{d}x^I\}_I$.

Definition 13.0.3 (Smooth k-form)**.** Let M be a smooth manifold of dimension m. The **sheaf of k-forms** $\Omega^k(M)$ over M is the sheaf of sections of the bundle of k-forms $\Lambda^k(\mathrm{T}^* M)$.
More concretely, a k-form over an open set U is a map $\omega : U \to \Lambda^k(\mathrm{T}^* M)$ such that $\pi \circ \omega = \mathbb{1}_U$.

Example 13.0.1. Let M be a smooth manifold of dimension m.
Note that $\Lambda^0(\mathrm{T}^* M) = M \times \mathbb{R}$ is the trivial vector bundle. Therefore a 0-form is a map $\omega : M \to M \times \mathbb{R}$ such that $\pi \circ \omega = \mathbb{1}_M$, i.e: ω is a smooth function.
This means that $\Omega^0(M) \cong C^\infty(M)$.

Example 13.0.2. Let M be a smooth manifold of dimension m.
Suppose that $k > m$. Since the fibers of the bundle of k-forms $\Lambda^k(\mathrm{T}^* M)$ have dimension $\binom{m}{k}$ this means that at least one index i in the multi-index I will be repeated. Therefore since $\mathrm{d}x^i \wedge \mathrm{d}x^i = 0$ we have that $\mathrm{d}x^I = 0$, i.e: $\Lambda^k(\mathrm{T}^* M) = 0$.
In particular $\Omega^k(M) = 0$ for all $k > m$.

Definition 13.0.4 (Exterior derivative)**.** Let ω be a k-form over a smooth manifold M. If $(U, \{x^i\}_{i=1}^m)$ is a local chart then there exist functions $\alpha_I \in C^\infty(M)$ such that

$$\omega = \sum_I \alpha_I \mathrm{d}x^I$$

where $I = (i_1, \dots, i_k)$ is a multi-index and $\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \mathrm{d}x^{i_k}$.
We can define the **exterior derivative** of ω as the $k+1$-form

$$\mathrm{d}\omega = \sum_I \mathrm{d}\alpha_I \wedge \mathrm{d}x^I = \sum_I \left(\sum_{j=1}^m \frac{\partial \alpha_I}{\partial x^j} \mathrm{d}x^j \right) \wedge \mathrm{d}x^I$$

Lemma 13.0.1. Let M be a smooth manifold of dimension m.
Then for any k-form the exterior derivative verifies $\mathrm{d}^2\omega = 0$, i.e: the sequence

$$\Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \Omega^2(M) \xrightarrow{\mathrm{d}} \cdots$$

is a complex of sheaves of $C^\infty(M)$-modules.

Proof. It suffices to show that for any smooth function $f \in C^\infty(M) \cong \Omega^0(M)$ we have

$\mathrm{d}^2 f = 0$. Indeed, if $\mathrm{d}^2 f = 0$ then for any *basic* form $f\,\mathrm{d}x^I$ we have

$$\mathrm{d}(f\,\mathrm{d}x^I) = \mathrm{d}(\mathrm{d}f \wedge \mathrm{d}x^I) = \mathrm{d}^2 f \wedge \mathrm{d}x^I - \mathrm{d}f \wedge \mathrm{d}^2 x^I = 0$$

Let $f \in C^\infty(M)$ be a smooth function. Its exterior derivative is

$$\mathrm{d}f = \sum_{i=1}^{m} \frac{\partial f}{\partial x^i} \mathrm{d}x^i$$

Differentiating again we get

$$\mathrm{d}^2 f = \sum_{j=1}^{m} \frac{\partial}{\partial x^j} \left(\sum_{i=1}^{m} \frac{\partial f}{\partial x^i} \right) \wedge \mathrm{d}x^j = \sum_{i,j=1}^{m} \frac{\partial^2 f}{\partial x^i \partial x^j} \mathrm{d}x^i \wedge \mathrm{d}x^j$$

By Clairaut's theorem we have that $\frac{\partial^2 f}{\partial x^i \partial x^j} = \frac{\partial^2 f}{\partial x^j \partial x^i}$ and by the antisymmetry of the wedge product $\mathrm{d}x^i \wedge \mathrm{d}x^j = -\mathrm{d}x^j \wedge \mathrm{d}x^j$. Therefore all the terms cancel leaving

$$\mathrm{d}^2 f = 0$$

□

Definition 13.0.5 (de Rham complex). Let M be a smooth manifold of dimension m. The **de Rham complex** of M is the complex of sheaves of $C^\infty(M)$-modules

$$\Omega^\bullet(M) \coloneqq \Omega^0(M) \xrightarrow{\mathrm{d}} \Omega^1(M) \xrightarrow{\mathrm{d}} \cdots \xrightarrow{\mathrm{d}} \Omega^m(M) \xrightarrow{\mathrm{d}} 0$$

where $\Omega^k(M)$ is the sheaf of k-forms over M and d is the exterior derivative.

Lemma 13.0.2 (Poincaré). Let $U \subseteq \mathbb{R}^n$ be a contractible open set and $k \geq 1$. If $\omega \in \Omega^k(U)$ is a k-form such that $\mathrm{d}\omega = 0$ then there exists a $(k-1)$-form $\eta \in \Omega^{k-1}(U)$ such that $\mathrm{d}\eta = \omega$.

Lemma 13.0.3. The de Rham complex is exact.

Proof. For $k \geq 1$ consider the relevant part of the complex

$$\Omega^{k-1} \xrightarrow{\mathrm{d}^{k-1}} \Omega^k \xrightarrow{\mathrm{d}^k} \Omega^{k+1}$$

We need to show that for any $p \in M$ the complex

$$\left(\Omega^{k-1}\right)_p \xrightarrow{\mathrm{d}_p^{k-1}} \left(\Omega^k\right)_p \xrightarrow{\mathrm{d}_p^k} \left(\Omega^{k+1}\right)_p$$

is exact, i.e: $\operatorname{im} \mathrm{d}_p^{k-1} = \ker \mathrm{d}_p^k$.

For the $[\subseteq]$ inclusion, let $\mathrm{d}\eta \in \operatorname{im} \mathrm{d}^{k-1} \subseteq \Omega^k$. By Lemma 13.0.1 we have that $\mathrm{d}(\mathrm{d}\eta) = 0$, i.e: $\mathrm{d}\eta \in \ker \mathrm{d}^k$.

For the $[\supseteq]$ inclusion, let $[\omega]_p \in \ker \mathrm{d}_p^k$, i.e: $[\omega]_p \in \left(\Omega^k\right)_p$ verifies $\mathrm{d}_p^k [\omega]_p = 0$. By definition there exists an open neighbourhood U of p and a k-form ω such that $\mathrm{d}^k \omega|_U = 0$.

Since M is a smooth manifold of dimension m we can choose a local chart (V, ψ) around p such that $V \subseteq U$ and $\psi(V)$ is an open ball in $\mathbb{R}^m$. Since open balls are

contractible, by Poincaré's lemma there exists a $(k-1)$-form $\eta \in \Omega^{k-1}(V)$ such that $\mathrm{d}^{k-1}\eta = \omega|_V$.

Therefore there exists $[\eta]_p \in \left(\Omega^{k-1}\right)_p$ such that $\mathrm{d}_p^{k-1}[\eta]_p = [\omega]_p$, i.e: $[\omega]_p \in \operatorname{im} d_p^{k-1}$. □

Lemma 13.0.4. Let M be a smooth paracompact manifold. Then for all k the sheaf of k-forms $\Omega^k(M)$ is fine.

Proof. Let $\mathscr{U} = \{U_i\}_{i\in\mathcal{I}}$ be an arbitrary open cover of M.

Since M is paracompact there exists a partition of unity $\{f_i\}_{i\in\mathcal{I}} \subset C^\infty(M)$ subordinate to $\mathscr{U}$. This means that

$$\begin{aligned} \forall i \in \mathcal{I} \quad & \operatorname{supp} f_i \subseteq U_i \\ \forall p \in M \quad & \sum_{i\in\mathcal{I}} f_i(p) = 1 \end{aligned}$$

and this last sum is finite.

For all indices $i \in \mathcal{I}$ consider the map

$$\begin{aligned} \eta_i : \Omega^k(M) &\to \Omega^k(M) \\ \omega &\mapsto f_i \cdot \omega \end{aligned}$$

Note that since $\operatorname{supp} f_i \subseteq U_i$ we have $\operatorname{supp} \eta_i \subseteq U_i$. Furthermore for any k-form ω

$$\sum_{i\in\mathcal{I}} \eta_i(\omega) = \sum_{i\in\mathcal{I}} f_i \cdot \omega = \left(\sum_{i\in\mathcal{I}} f_i\right) \cdot \omega = 1 \cdot \omega = \omega$$

Therefore $\sum_{i\in\mathcal{I}} \eta_i = \mathbb{1}$, making $\{\eta_i\}_{i\in\mathcal{I}}$ a partition of unity subordinate to $\mathscr{U}$. □

Theorem 13.0.1. Let M be a smooth paracompact manifold. Then

$$\mathrm{H}^\bullet(M, \underline{\mathbb{R}}) \cong \mathrm{H}^\bullet_{\mathrm{dR}}(M, \mathbb{R})$$

Proof. Consider the resolution

$$0 \longrightarrow \underline{\mathbb{R}} \longrightarrow \Omega^\bullet(M)$$

where $\Omega^\bullet$ is the de Rham complex of M.

By Lemma 13.0.3 we know that $\Omega^\bullet(M)$ is exact, therefore we only need to show that this resolution is exact at $\Omega^0(M)$, i.e: that for all $p \in M$

$$0 \longrightarrow \underline{\mathbb{R}}_p \xrightarrow{\iota_p} \Omega^0_p \xrightarrow{\mathrm{d}_p} \Omega^1_p$$

is a short exact sequence, i.e: $\operatorname{im} \iota_p = \ker \mathrm{d}_p$.

- $\subseteq$ Let $[c]_p \in \underline{\mathbb{R}}_p$. By definition this is a locally constant function $c : U \to \mathbb{R}$ on some open neighbourhood U of p.
  Since c is locally constant we have that $\mathrm{d}c|_U = 0$, i.e $\mathrm{d}_p[c]_p = 0$.

$\supseteq$ Let $[f]_p \in \Omega^0_p$ such that $\mathrm{d}_p[f]_p = 0$. This means that there exists an open neighbourhood U of p and a function $f \in \Omega^0(V) \cong C^\infty(M)$ such that $\mathrm{d}f|_V = 0$. Without loss of generality we may assume that U is connected. Therefore we have that $f|_U$ is a constant function, i.e: $[f]_p \in \underline{\mathbb{R}}_p$.

Therefore since by Lemma 13.0.4 the resolution is made of fine sheaves and fine sheaves are $\Gamma(X, -)$ acyclic by definition of sheaf cohomology

$$\mathrm{H}^\bullet(M, \underline{\mathbb{R}}) = \mathrm{H}^\bullet_{\mathrm{dR}}(M, \mathbb{R})$$

□

Corollary 13.0.1. Let M be a smooth paracompact manifold of dimension m. Then for all $k > m$

$$\mathrm{H}^k(M, \underline{\mathbb{R}}) = 0$$

Example 13.0.3 (The cohomology of S^1)**.** Consider the circle S^1 as a smooth manifold of dimension 1.
We are going to compute its sheaf cohomology with coefficients in $\underline{\mathbb{R}}$, $\mathrm{H}^\bullet(S^1, \underline{\mathbb{R}})$, using the de Rham complex

$$\Omega^0(S^1) \xrightarrow{\mathrm{d}^0} \Omega^1(S^1) \xrightarrow{\mathrm{d}^1} 0$$

On the one hand

$$\mathrm{H}^0(S^1, \underline{\mathbb{R}}) = \ker d^0 = \{f \in \Omega^0(S^1) \cong C^\infty(S^1) \mid \mathrm{d}f = 0\}$$

Since S^1 is connected, the only functions with 0 derivative are the constant functions. Therefore

$$\mathrm{H}^0(S^1, \underline{\mathbb{R}}) = \mathbb{R}$$

On the other

$$\mathrm{H}^1(S^1, \underline{\mathbb{R}}) = {\ker d^1}/{\operatorname{im} d^0} = {\{\omega \in \Omega^1(S^1) \mid \mathrm{d}\omega = 0\}}/{\{\mathrm{d}f \mid f \in \Omega^0(S^1) \cong C^\infty(S^1)\}}$$

Note that since $\Omega^2(M) = 0$ all 1-forms $\omega \in \Omega^1(S^1)$ verify $\mathrm{d}\omega = 0$. Therefore

$$\{\omega \in \Omega^1(S^1) \mid \mathrm{d}\omega = 0\} = \Omega^1(S^1)$$

Consider now the *integration* map

$$\begin{aligned} I : \Omega^1(S^1) &\to \mathbb{R} \\ \omega &\mapsto \int_{S^1} \omega \end{aligned}$$

Let $\mathrm{d}\theta$ be the differential of the angle. Since $I(\mathrm{d}\theta) = 2\pi$ we have that I is surjective. Furthermore by Stoke's theorem we have that $I(\omega) = 0$ if and only if $w = \mathrm{d}f$ for some $f \in C^\infty(S^1)$.
Therefore by the first isomorphism theorem we have that

$$\mathrm{H}^1(S^1, \underline{\mathbb{R}}) = {\Omega^1(S^1)}/{\ker I} \cong \operatorname{im} I = \mathbb{R}$$

In summary, we have that

$$\mathrm{H}^k(S^1, \underline{\mathbb{R}}) = \begin{cases} \mathbb{R} & k \in \{0, 1\} \\ 0 & k > 1 \end{cases}$$

Chapter 14

Čech Cohomology

Definition 14.0.1 (Čech complex). Let $(X, \mathscr{O}_X)$ be a ringed space and let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules.
Consider an open covering $\mathcal{U} = \{U_i\}_{i\in\mathbb{N}}$ and denote

$$U_{i_0,\dots,i_p} \coloneqq U_{i_0} \cap \cdots \cap U_{i_p}$$

For all $p \geq 0$ set

$$\check{C}^p(\mathcal{U}, \mathscr{F}) = \prod_{(i_0,\dots,i_p)} \mathscr{F}(U_{i_0,\dots,i_p})$$

and for all $s \in \check{C}^p(\mathcal{U}, \mathscr{F})$ denote by $s_{i_0,\dots,i_p}$ its value in $\mathscr{F}(U_{i_0,\dots,i_p})$.
Define the *differential maps* $d^p : \check{C}^p(\mathcal{U}, \mathscr{F}) \to \check{C}^{p+1}(\mathcal{U}, \mathscr{F})$ as

$$d^p(s)_{i_0,\dots,i_p} \coloneqq \sum_{j=0}^{p+1} (-1)^j s_{i_0,\dots,\hat{i_0},\dots,i_{p+1}}|_{U_{i_0,\dots,i_{p+1}}}$$

A straightforward calculation shows that $d^{p+1} \circ d^p = 0$, making $\check{C}^\bullet(\mathcal{U}, \mathscr{F})$ a complex. We call this the **Čech complex** associated to $\mathscr{F}$ and $\mathcal{U}$.
Note that we can define a *natural* map $\varepsilon : \mathscr{F} \to \check{C}(\mathcal{U}, \mathscr{F})$ by sending $s \in \mathscr{F}(U)$ to $\varepsilon(s)_i \coloneqq s|_{U\cap U_i}$.

Proposition 14.0.1. Let $(X, \mathscr{O}_X)$ be a ringed space and let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules.
Let $\mathcal{U}$ be an open covering of X.
Then the Čech complex associated to $\mathscr{F}$ and $\mathcal{U}$ is a resolution of $\mathscr{F}$.

Proof. □

Definition 14.0.2 (Čech cohomology). Let $(X, \mathscr{O}_X)$ be a ringed space and let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules. Let $\mathcal{U}$ be an open covering of X.
The **Čech cohomology** associated to $\mathscr{F}$ and the covering $\mathcal{U}$ is

$$\check{\mathrm{H}}^\bullet(\mathcal{U}, \mathscr{F}) \coloneqq \mathrm{H}^\bullet(\check{C}^\bullet(\mathcal{U}, \mathscr{F}))$$

Theorem 14.0.1. Let $(X, \mathscr{O}_X)$ be a ringed space, where X is Hausdorff and paracompact. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules.

Then for all open coverings $\mathcal{U}$

$$\check{H}^\bullet(\mathcal{U}, \mathscr{F}) \cong H^\bullet(X, \mathscr{F})$$

Proof. [God98, Theorem 5.10.1] □

Chapter 15

Singular cohomology

Definition 15.0.1 (Standard simplex). The **standard n-simplex** Δ^n is the subset of $\mathbb{R}^{n+1}$ defined by

$$\Delta^n := \left\{ (t_0, \dots, t_{n+1}) \in [0,1]^{n+1} \mid \sum_{i=0}^{n+1} t_i = 1 \right\}$$

Definition 15.0.2 (Simplex). Let X be a topological space. A **singular n-simplex** in X is a continuous map

$$\sigma : \Delta^n \to X$$

where Δ^n is the standard n-simplex.

Definition 15.0.3 (Singular chains). The group of **singular n-chains** with coefficients in an abelian group G, denoted $C_n(X, G)$, is the free abelian group generated by the set of all singular n-simplices in X.
We have *boundary* maps $\delta_n : C_n^{\text{sing}}(X, G) \to C_{n-1}^{\text{sing}}(X, G)$ defined as

$$\forall \sigma \in C_n^{\text{sing}}(X, G) \quad \delta_n(\sigma) := \sum_{i=0}^{n} (-1)^i \sigma|_{[v_0, \dots, \hat{v_i}, \dots, v_n]}$$

A straightforwards calculations shows that $\delta_{n-1} \circ \delta_n = 0$. Therefore we have a **singular chain** complex

$$C_\bullet^{\text{sing}}(X, G) = \cdots \longrightarrow C_{n+1}^{\text{sing}} \xrightarrow{\delta_{n+1}} C_n^{\text{sing}} \xrightarrow{\delta_n} C_{n-1} \xrightarrow{\delta_{n-1}} \cdots \xrightarrow{\delta_1} C_0^{\text{sing}}$$

Definition 15.0.4 (Singular cochains). The group of **singular n-cochains** with coefficients in an abelian group G, denoted $C^n_{\text{sing}}(X, G)$, is the dual of the group of singular n-chains:

$$C^n_{\text{sing}}(X, G) := \text{Hom}_{\mathsf{Ab}}(C_n^{\text{sing}}(X, G), G)$$

We have *coboundary* maps $d^n : C^n_{\text{sing}}(X, G) \to C^{n+1}_{\text{sing}}(X, G)$ defined as

$$\forall \phi \in C^n_{\text{sing}}(X, G), \sigma \in C_{n+1}^{\text{sing}}(X, G) \quad (d^n \phi)(\sigma) := \phi(\delta^n(\sigma)) = \sum_{i=0}^{n+1} (-1)^i \phi(\sigma|_{[v_0, \dots, \hat{v_i}, \dots, v_{n+1}]})$$

As before, it is easy to check that $d^{n+1} \circ d^n = 0$, so we have a **singular cochain** complex

$$C^\bullet_{\text{sing}}(X,G) = C^0_{\text{sing}}(X,G) \xrightarrow{d^0} C^1_{\text{sing}}(X,G) \xrightarrow{d^1} \cdots$$

Definition 15.0.5 (Singular cohomology). Let X be a topological space and G an abelian group. The **singular cohomology** of X with coefficients in G is the cohomology of the singular cochain complex:

$$\mathrm{H}^\bullet_{\text{sing}}(X,G) := \mathrm{H}^\bullet(C^\bullet_{\text{sing}}(X,G))$$

Lemma 15.0.1. Let X be a contractible space. Then for all abelian group G and $n > 0$

$$\mathrm{H}^n_{\text{sing}}(X,G) = 0$$

Theorem 15.0.1. Theorem of small chains

Theorem 15.0.2. Let $(X, \mathscr{O}_X)$ be a ringed space where X is locally contractible[a]. Let G be a commutative ring.
Then

$$\mathrm{H}^\bullet_{\text{sing}}(X,G) \cong \mathrm{H}^\bullet(X,\underline{G})$$

[a] A topological space X is locally contractible if any open subset $U \subseteq X$ can be covered by contractible open sets.

Proof. Consider the sheaf $\mathscr{C}^q_{\text{sing}}(-,G)$ associated to the presheaf

$$U \mapsto C^q_{\text{sing}}(U,G)$$

The differential $d : C^q_{\text{sing}}(U,G) \to C^{q+1}_{\text{sing}}(U,G)$ gives rise to a differential

$$d : \mathscr{C}^q_{\text{sing}} \to \mathscr{C}^{q+1}_{\text{sing}}$$

This complex is a resolution of $\underline{G}$.

Indeed, the complex $C^{>0}_{\text{sing}}(U,G)$ is exact on each contractible open set U using that the cohomology of the complex is by definition singular cohomology and Lemma 15.0.1.

Since X is locally contractible every point $p \in X$ has a neighbourhood basis consisting of contractible open sets. The stalk of the complex is the direct limit over this neighbourhoods, so the complex of presheaves $C^\bullet_{\text{sing}}(-,G)$ is exact at the level of stalks, making the complex of sheaves $\mathscr{C}^\bullet_{\text{sing}}(-,G)$ is exact.

Furthermore since X is locally pathwise connected we have

$$\ker\left(d^0 : \mathscr{C}^0_{\text{sing}}(-,G) \to \mathscr{C}^1_{\text{sing}}(-,G)\right) = \underline{G}$$

Note that the presheaf $C_{\text{sing}}(-,G)$ is flasque. Indeed for any restriction of open sets $V \subseteq U$ the restricion map is surjective because any singular cochain on V can be extended to one in U by assigning it to 0 to any simplex in U not contained in V. Furthermore we have

$$\mathscr{C}^q_{\text{sing}}(U,G) = {}^{C^q_{\text{sing}}(U,G)}\big/_{C^q_{\text{sing}}(U,G)_0}$$

were $C^q_{\text{sing}}(U, G)_0$ is the set of singular q-cochains $\alpha \in C^q_{\text{sing}}(U, G)$ such that there exists a covering $\mathcal{V}$ of U with $\alpha|_{C_q(V,G)} = 0$ for all $V \in \mathcal{V}$.

Therefore by Theorem 9.2.3 we have

$$\mathrm{H}^q(X, G) = \mathrm{H}^q(\Gamma(\mathscr{C}^\bullet_{\text{sing}}(X, G)))$$

It remains to show that the complex

$$\Gamma(\mathscr{C}^\bullet_{\text{sing}}(X, G)) = {}^{C^\bullet_{\text{sing}}(X, G)}\big/_{C^\bullet_{\text{sing}}(X, G)_0}$$

is quasi-isomorphic to the complex $C^\bullet_{\text{sing}}(X, G)$, which is Theorem 15.0.1. □

Appendix A

Algebra

Definition A.0.1 (Algebra)**.**

Definition A.0.2 (Local ring)**.**

Definition A.0.3 (Module)**.**

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